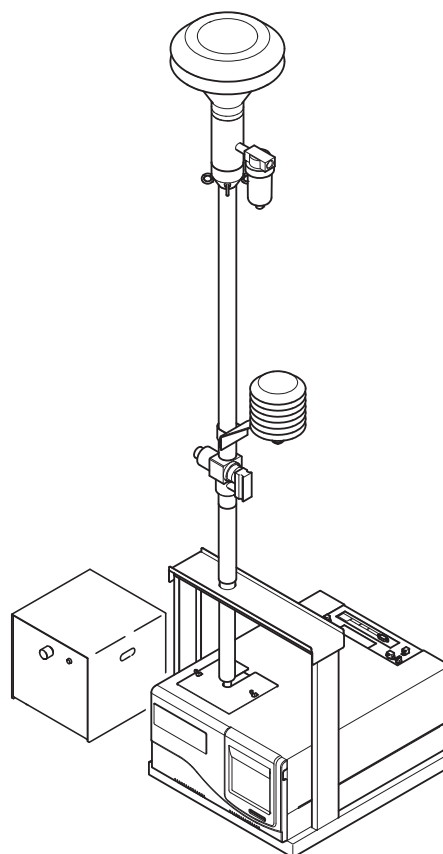




PM2.5 MONITOR

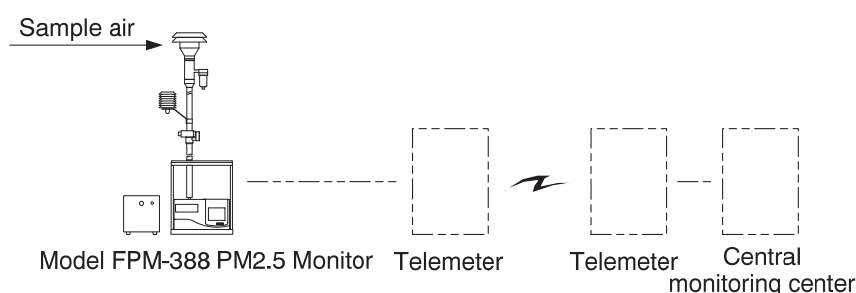
MODEL FPM-388



- Please keep this instruction manual close at hand of the persons who are in charge of the operation of this product.
- Before operating this product, please read this instruction manual carefully for its correct handling.

Introduction

- (a) Thank you for your purchase of our product. The Model FPM-388 PM2.5 Monitor (hereafter called the monitor or the product) can measure the concentration of fine particulate matter (PM2.5) in the air continuously.
- (b) The measurement method is beta (hereafter called “ β ”) ray absorption method. With β -ray absorption method, PM2.5 in the air is collected by filter and β -ray is irradiated and the transmitted beam intensity is measured to obtain the mass concentration of fine particulate matter. The measured data and various I/O signals can be connected to a central monitoring center via a telemeter. The figure shown below shows an example of measurement system including this product.



Example of Measurement System

- (c) An abnormal measured value may be indicated or output by the following causes. Build a system such that related facilities are not damaged.
- Any problem of the product such as deterioration or damage of the detecting section or inappropriate insulation of cables.
 - Improper setting of operating conditions or calibration operation.
 - Electrical interference such as noise in the vicinity or improper grounding.
 - Other unpredictable phenomena
- (d) Since important items are described in “Safety Information,” read the contents carefully.
- (e) The product should be handled by persons who have received proper training. In addition, for technical services such as repairs, ask a specialist to do who is qualified for the technical certification system in our company or a person who has technical skills equivalent to that certification system.

Safety Information

(1) Meaning of markings

The signal terminology and symbols related to warnings in the instruction manual are defined below. The alert symbol mark (: General caution mark) indicates the possibility of hazard or damage and also means "Refer to the instruction manual."

WARNING:

Indicates the degree of hazard which can lead to death or serious injury if you fail to operate the product properly.

Serious injury means an injury such as loss of sight, burns (high temperature or low temperature), electric shock, bone fracture and poisoning, and the aftereffects of the injury remains or the injury requires hospitalization or long periods of outpatient treatment.

CAUTION:

Indicates the degree of hazard/loss which can result in injury or property damage if you fail to operate the product properly.

Injury means an injury not requiring hospitalization or long periods of outpatient treatment and refers to burns or electric shock. Property damage refers to widespread damage to the home, household goods and livestock, pets, equipment, materials, etc. (damage to other than the product itself).

[IMPORTANT] Indicates important matters other than  **WARNING** and  **CAUTION**. They are the matters such as preventing damage to the product main body, preventing data destruction, preventing wasting time, maintaining performance, and observing regulations.

[NOTE] Indicates comments, reasons, background information, a case example and other items to help the reader understand the meaning.

> > Indicates reference items.

①, ②, ③ Indicates item numbers such as the ones used in operations.

(2) Safety compliance items

WARNING

Radiant ray

- β -ray source in the measurement unit is the radioactive material of 10MBq or less sealed radiation source (14C). For the handling, observe laws and regulations in the country where the product is used.
Since β -ray source is a radioactive material, never disassemble the upper measurement unit. Damage of radiation source window or removal of upper measurement unit may cause radiation exposure.

- Do not move bare skin of hand or face etc. close within 250mm range from the radiation source window at the position where β -ray source in the radiation source window is seen. At the cleaning of upper measurement unit, be sure to wear latex gloves (for kitchen use) and do not use a sharp item (screwdriver, tweezers, etc.) that may damage the film surface of radiation source window.
- When the measurement unit including β -ray source is replaced or the product is discarded, ask the sales company to pick it up.
- If the radiation source window is damaged, carry out the emergency treatment shown below, and then contact the sales company immediately.
 - (a) Put the paper whose thickness is three name cards worth to the radiation source window and fix it with tape.
 - (b) If a part of β -ray source may be attached to the tip of tools or periphery, wipe it off with a piece of cloth (waste cloth) sufficiently. Put the used piece of cloth in a reclosable poly bag and put it in a cardboard box to store it strictly.

Fall Injury

- When you work at high place such as roof of monitoring station for installation and maintenance, wear safety belt, etc. to prevent falling. In addition, to prevent injury, wear a helmet, life jacket, safety shoes, etc.

Hazardous Gasses

- Do not use the product in an area where explosive gas or flammable gas exists. Using the product in any of these areas can cause explosion or fire.

Lithium battery

- For lithium battery, do not carry out charging, short-circuit, disassembly, deformation, heating, or putting it in fire.
- When you store or discard a lithium battery, put the tape on the terminal to insulate it. Contact with other metal or battery may cause heating, explosion, or ignition, etc.

Electric Shock

- Do not touch the terminals inside the product while power is applied. Touching the terminals can cause electric shock.
- When you remove the upper cover, be sure to turn off the power (press "O") preliminarily
- The ground terminal must be grounded. If the ground terminal is not grounded, electric shock can result if a problem occurs in the power system.



Lightning protection

- If surge voltage by thunder may intrude from connection cable with external such as power supply, take a measure. Otherwise, a failure or ignition may be caused.

Disassembly and Modification

- Do not disassemble or modify the sections of the product that are not described in the instruction manual. The product can be damaged.

Warning Label Lost

- If any warning label affixed to this product becomes too difficult to read or lost, please order a new one through your local sales agent or our sales office and affix it to its original position.
-

(3) Notes on use of the instruction manual

Important items such as “Safety compliance items” are described in this manual. Handle the manual as follows:

- (a) The instruction manual is required not only at the start of operation but also required when maintenance is performed or in case a failure occurs. Please keep the manual at hand all the time so that the operator who actually operates the product can read the manual at any time.
- (b) If the manual is lost or too smeared to read, please order a new copy through your local sales agent or directly from our sales office.
- (c) For easier comprehension, some of the diagrams used in the manual or on product labels may be modified with part of their shapes or displays omitted or they may be described in abstract form. In addition, numbers etc. shown on the screen example are just examples for such cases.
- (d) The contents of the manual may be changed without prior notice for reasons such as to improve performance.
- (e) Intellectual property right of the manual belongs to DKK-TOA. All or part of the manual must not be reproduced without permission.

Warranty

(1) Warranty Coverage

DKK-TOA Corporation (DKK-TOA) warrants its products against defective material or workmanship for the warranty period.

- (a) The warranty period is one year from the date of delivery to the original user. If the date of delivery cannot be specified, the warranty period is 24 months from the month following the date of manufacture shown on the product nameplate.
- (b) Specific written agreements with DKK-TOA, if any, shall take precedence over this warranty.
- (c) The limitation of warranty described herein may not apply where applicable laws do not allow such limitation.

(2) Limited Warranty

This warranty does not cover the cases listed below.

- (a) Direct or indirect failure or damage caused by the use of the product for a purpose or in a manner not prescribed by the specifications or the instruction manual for the product.
- (b) Direct or indirect failure or damage caused by force majeure, including but not limited to an act of God, natural disaster such as earthquake, storm and flood damage, and lightning, fire, accident, abnormal voltage, salt damage, gas damage, labor unrest, acts of war (declared or undeclared), terrorism, civil strife, or acts of any governmental jurisdiction.
- (c) Failure or damage caused by any repair or modification not authorized by DKK-TOA.
- (d) Failure or damage caused by the transport, moving, or dropping of the product after the purchase that is not attributable to DKK-TOA.
- (e) Electrodes and consumables (The warranty period for each part has priority when the period is shorter than that for the main unit of the product. If the customer requires any part after more than six months from the date of manufacture, consult DKK-TOA or its distributor.)
- (f) Failure or damage caused by the use of consumables, parts, or software not supplied by DKK-TOA.
- (g) Malfunctions or damage caused by the use of connecting equipment not supplied by DKK-TOA
- (h) Loss of data, settings, programs, or software stored on the product not attributable to DKK-TOA.
- (i) Any product other than DKK-TOA's, if specified by the purchaser or user, that incorporates, or is incorporated into or combined with DKK-TOA's products (*1). In such cases, this warranty covers DKK-TOA's products only.
- (j) Any product not under proper maintenance in accordance with the instruction manual furnished by DKK-TOA.
- (k) Products without a nameplate (excluding products proved to have been delivered by DKK-TOA).
- (l) If installation, piping, or wiring is not performed in accordance with the delivery specification specified by DKK-TOA

EXCEPT AS EXPRESSLY SET FORTH IN THE PRECEDING SENTENCES, DKK-TOA MAKES NO WARRANTY OF ANY KIND WHATSOEVER WITH RESPECT TO ANY PRODUCT. DKK-TOA EXPRESSLY DISCLAIMS ANY WARRANTY IMPLIED BY LAW, INCLUDING BUT NOT LIMITED TO ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

LIMITATION OF REMEDIES: In the event that a defect is discovered within the warranty period, DKK-TOA or its authorized distributor will, at its option, repair or replace the defective product or its part, or will refund the purchase price of the product. **THIS IS THE EXCLUSIVE REMEDY FOR ANY BREACH OF WARRANTY.**

LIMITATION OF DAMAGES: IN NO EVENT SHALL DKK-TOA BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES OF ANY KIND FOR BREACH OF ANY WARRANTY, NEGLIGENCE, ON THE BASIS OF STRICT LIABILITY, OR OTHERWISE.

(3) Others

- (a) Maintenance parts (*2) for product will normally be supplied for five years (*3) from the date manufacturing and sales are discontinued.
- (b) The cause of any malfunction or damage shall be determined by a DKK-TOA technician.
- (c) For repairs, contact a local distributor in your country or state.

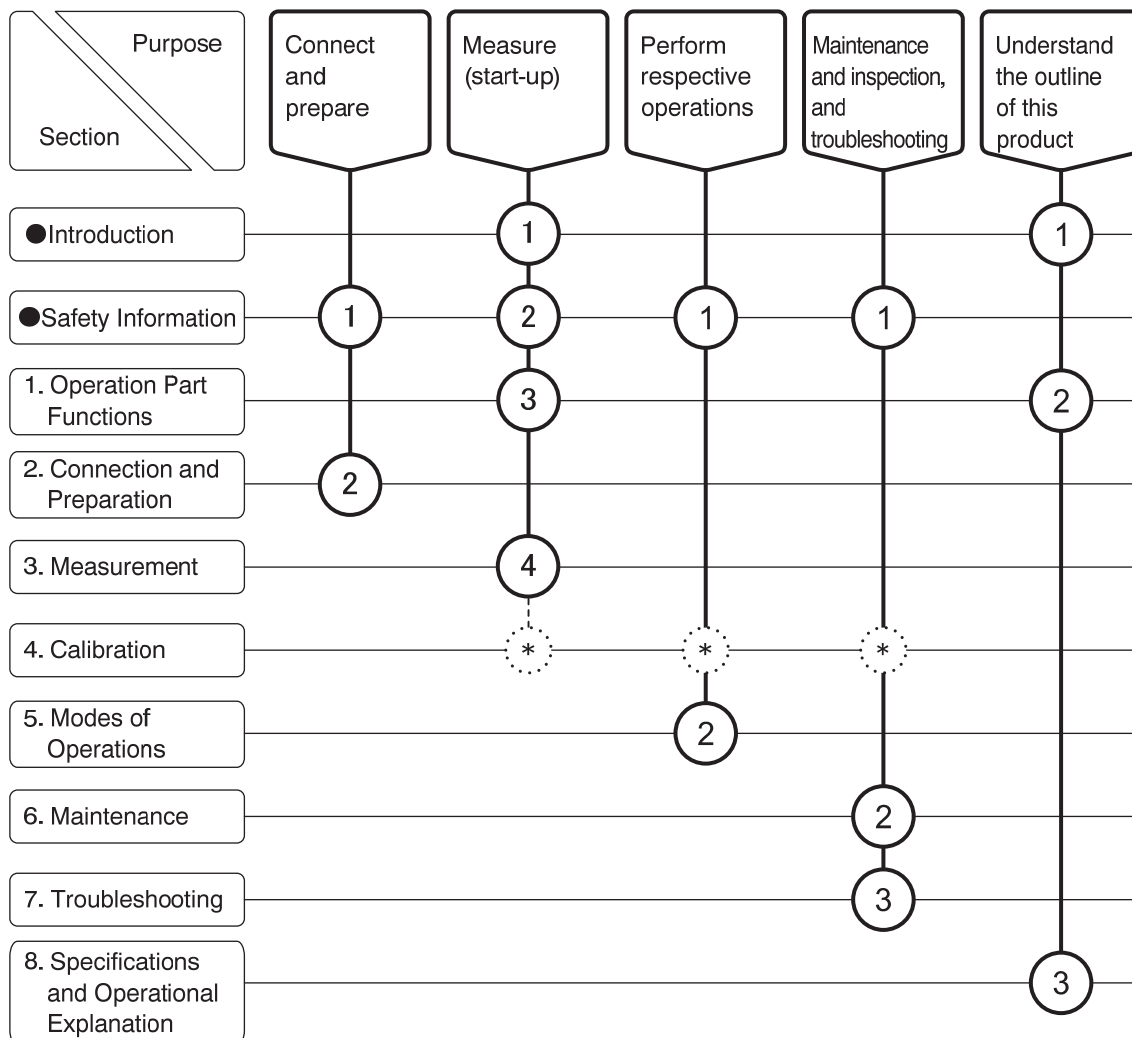
*1: Warranties for products from other companies must be maintained by the user.

*2: Maintenance parts refers to parts that are required to maintain operation of the product.

*3: This five-year period is subject to availability of parts or their replacement.

Reading Guide

Refer to the necessary sections of this instruction manual depending on your purposes such as understanding the outline of this product or starting the product as shown below. The numbers in circles indicate sections to be referred to in sequential order.



*: When necessary, operate according to the contents.

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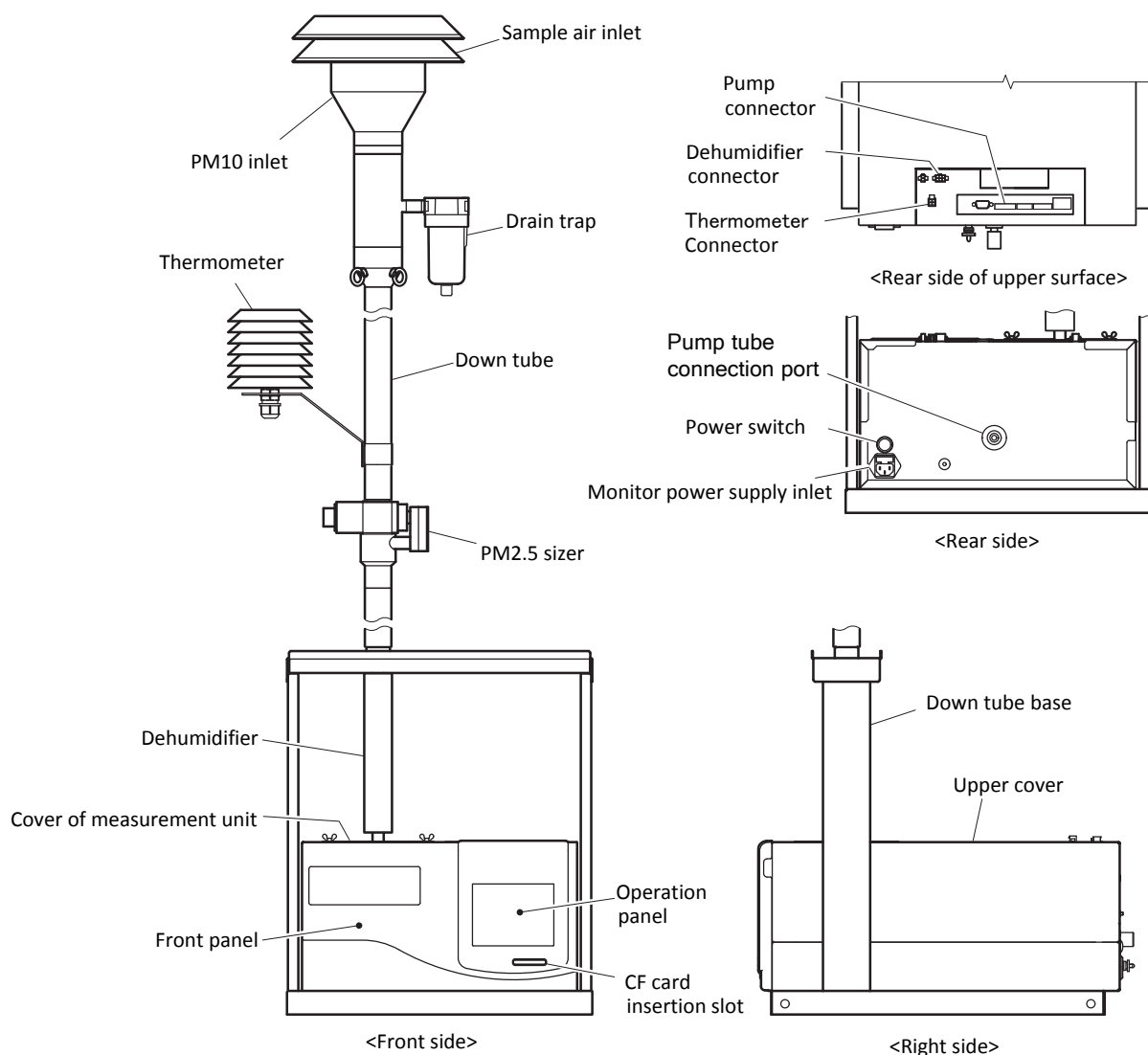
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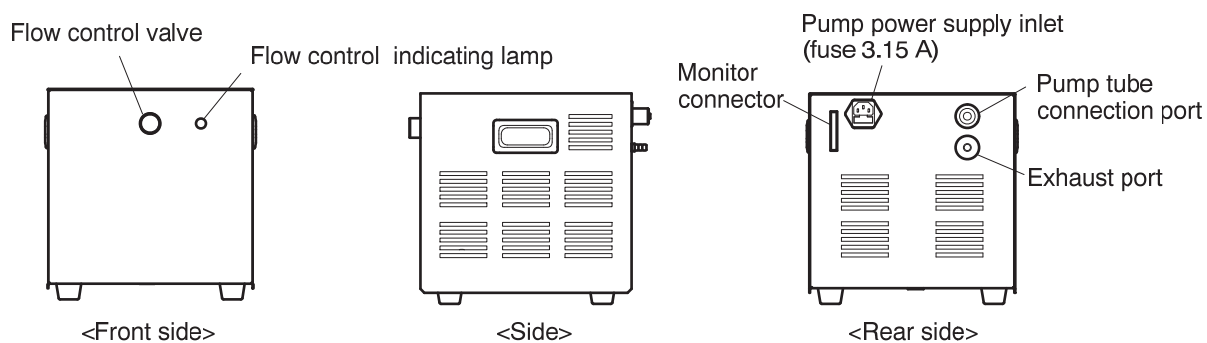
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1. Operation Part Functions

(1) Names of main parts



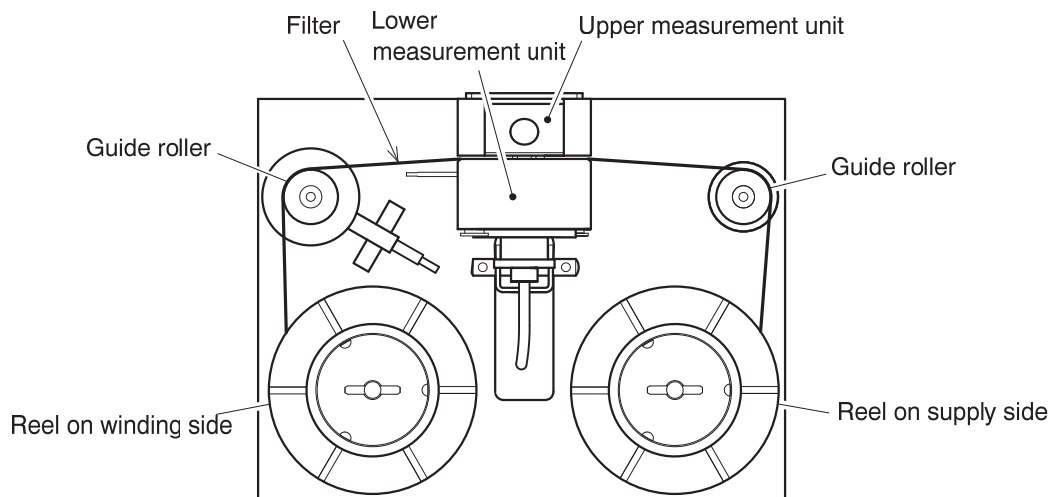
Names of Main Parts



Names of Pump Unit Each Part

(2) Names of measurement part

The measurement part can be seen by removing the front panel. The front panel can be removed by putting your finger on the cutout part on the upper part of side surface and pulling it out. >> ② Figure “Attachment and Removal of Front Panel” in 3.1 “Procedure for Starting the Automatic Measurement”

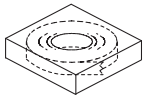


Names of Measurement Part

(3) Accessories

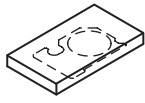
- (a) The table below shows the general accessories. These may be different depending on the specifications, or may be changed without notice.. Some parts are embedded in measurement part.
- (b) For spare parts and replacement parts, refer to the relevant item. >> 6.2 “Spare Parts and Replacement Parts”

Accessories List

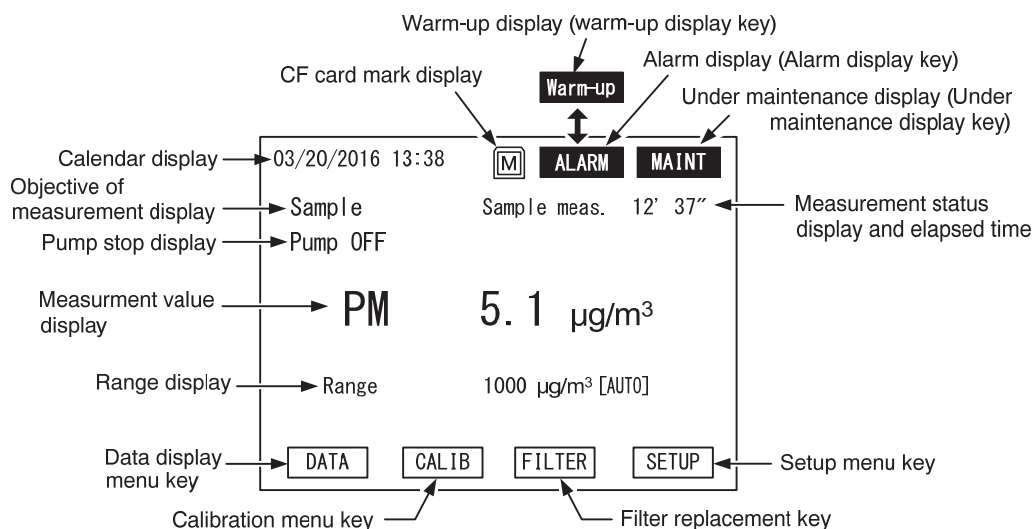
Item No.	Code No.	Part name	Sketch	Q'ty	Remark
1	—	Instruction manual		1	
2	—	Inspection result		1	
3	7008500K	PTFE type filter AP-50		1	Already attached to measurement part Name in this document: Filter
4	136A970	Filter winding core for AP-50		1	Already attached to measurement part Name in this document: Winding core

(To be continued)

(Continued from previous page)

Item No.	Code No.	Part name	Sketch	Q'ty	Remark
5	7219670K	Calibration film assembly		1	Name in this document: Calibration film
6	6958270K	CF card assembly		1	Name in this document: CF card

(4) Names and functions of home screen



Example of Home Screen Display

- (a) Current date and time, set average value range, current measured value, and status of monitor, etc. can be checked on "Home screen".
- (b) If the key on the screen (within frame line enclosed by square) is pressed (touched), "Home screen" switches to other menu screen, etc. >> 5.1(1) "Map of home screen and home-related screens"

Main Display Functions

Display	Functions, etc.
Calendar display	Indicates current year (dominical year), month and day, and time.
CF card mark display	Indicates that the CF card is inserted.
Alarm display	Indicates that PM2.5 alarm 1 signal (abbreviation "alarm 1 signal") or PM2.5 alarm 2 signal (abbreviation "alarm 2 signal") is being output.
Warm-up display	- Indicates that warm-up is carried out after the power is turned on. - Indicates the status before temperature in monitor, temperature of measurement unit, sample temperature, or sample flow rate enters normal range. - If they do not enter normal range in 60 minutes, this display switches to alarm display.

(To be continued)

(Continued from previous page)

Display	Functions, etc.
Under maintenance display	Indicates that PM2.5 maintenance is being carried out.
Objective of measurement display	- Indicates the current objective to be measured. “Sample” Indicates the sample air is being measured. “Film” Indicates the calibration by calibration film is being carried out. - Not displayed when the measurement status is “Stop”.
Measurement status display	- Indicates the current measurement status and elapsed time. “Cell Open” Measurement unit is being opened. “Cell Close” Measurement unit is being closed. “Filter Tape Fed” Filter paper is being fed. “Wait” Standby “Sample meas.” Indicates that the usual measurement is being carried out. “Stop” Measurement is stopped due to generation of alarm.
Pump stop display	- Current pump operation status. “Pump OFF” Pump OFF is being executed. - Not displayed when the measurement status is “Stop”.
Measurement value display	- PM2.5 mass concentration of last 1-hour period. - Unit is “μg/m³” - Decimal point position is fixed (“±XXXX.X”). - If the measurement value is positive, “+” is not displayed. “- - - - -” indicates that the measured value could not be obtained. - Blinking of measured value display: indicates that the measured value exceeded the range of average value.
Range display	- Indicates the current PM2.5 average value range per hour (abbreviation “average value range”). [AUTO]: indicates the automatic switching of range.

Main Keys Functions

Names of keys	Notation in text	Functions, etc.
Data display menu key		Pressing the key switches the screen to “Data display menu” screen.
Calibration menu key		- Pressing the key switches the screen to “Calibration menu” screen. - Highlight display (): indicates that the internal calibration (by calibration film) is being carried out.
Filter replacement key		Pressing the key switches the screen to “Filter replacement” screen.
Setup menu key		Pressing the key switches the screen to “Setup menu” screen.
Alarm display key		Pressing the key switches the screen to “Ongoing alarm” screen.
Warm-up display key		Pressing the key switches the screen to “Warm-up alarm” screen.
Under maintenance display key		Pressing the key switches the screen to “Maintenance clear” screen.

(To be continued)

(Continued from previous page)

Names of keys	Notation in text	Functions, etc.
CF card mark display key		Pressing the key switches the screen to “CF card removal execution” screen. > > 5.3(4) “Removal of CF card”
ALL CLEAR key		Pressing the key clears all alarm information, etc.
BACK key		Pressing the key returns to the previous screen.
YES key		Pressing the key goes to the explanation status on the screen.
NO key		Pressing the key does not go to the explanation status on the screen, but returns to the previous screen.
HOME key		Pressing the key returns to the “Home” screen.
CF key		Pressing the key switches the screen to “CF card” screen.
Log Data key		Pressing the key switches the screen to “Log data” screen.
FEED key		Pressing the key feeds 1 spot of filter.

2. Connection and Preparation

2.1 Installation Location and Installation

(1) Installation location conditions

Use the monitor in a location which conforms to the specifications and satisfies the following conditions. In addition, fulfill the standard of a region where the monitor is installed.



Hazardous Gasses

- Do not use the product in an area where explosive gas or flammable gas exists. Using the product in any of these areas can cause explosion or fire.
-

(a) A location where the ventilation is sufficiently performed all the time.

- [IMPORTANT]** • Be sure to place the monitor at a sufficiently ventilated measurement room. If the place cannot be sufficiently ventilated, connect a tube to discharge ports of all dry monitors and wet monitors in the measurement room and lead them to the external of measurement room (separate them from sample air collection point as much as possible).
-

(b) A location where salty wind and dust can be avoided.

- [IMPORTANT]** • Place the monitor at a place where there is less dust. The dust negatively affects lifetime and measurement accuracy of monitor. If there is much dust, it is recommended to clean up the installation location periodically.
-

(c) A location where the product is not exposed to water drops or chemicals.

(d) A location where temperature and humidity are within the specification range. A location where the temperature rarely changes avoiding direct sunlight and abnormal high temperature in summer season.

(e) A location where power supply voltage and frequency rarely fluctuate. A location where there is no large electromagnetic induction facility or equipment that generates spark etc. in the near distance. Measures for lightning protection are required for some regions.

(f) A location where vibration is rarely generated. A location where the measures for controlling vibration are taken depending on the circumstances.

(g) A location where the monitor can be installed horizontally.

(h) A location where installation and maintenance work can be performed safely.

(2) Monitor installation

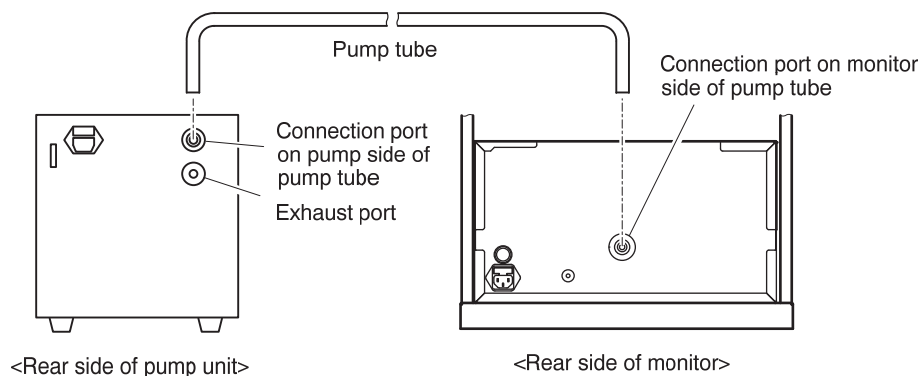
(a) Fix the inlet part that is installed outdoors securely.

(b) Install so that the upper surface of monitor becomes even.

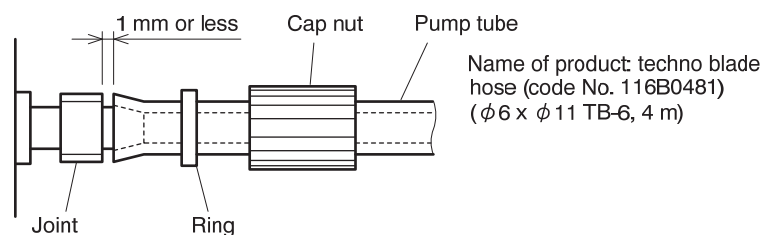
(c) Ensure the maintenance space around the monitor. Space for wiring, piping, or leading in the cables is also required.

2.2 Connection and Preparation of Pump Unit

(1) Connection of pump tube



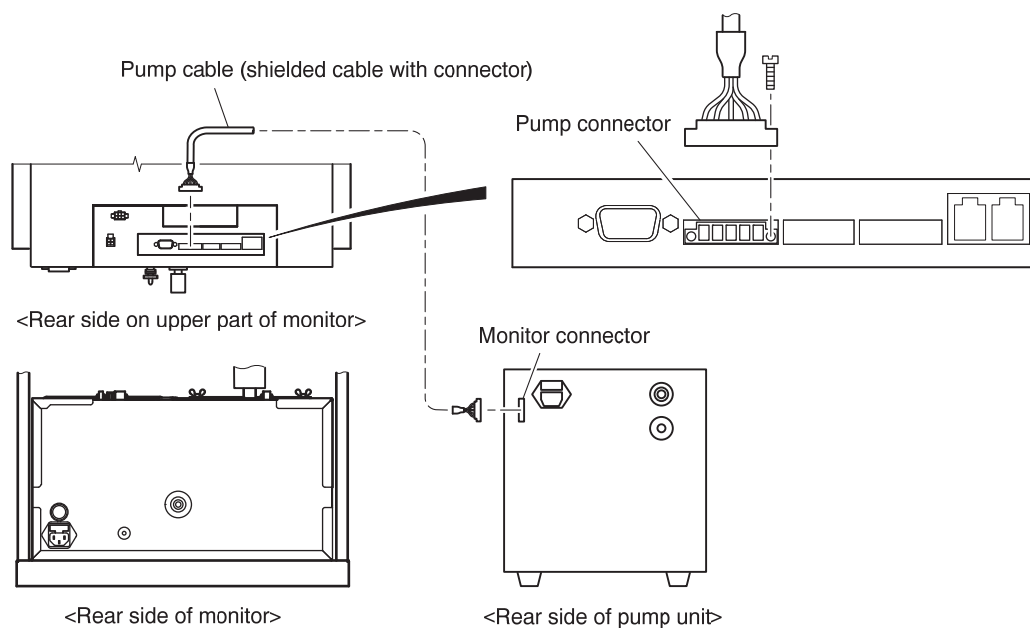
- (a) Connect the attached pump tube to the connection port on pump side of pump tube on the rear surface of pump unit. Next, connect the other end to the connection port on monitor side of pump tube on the rear surface of monitor.
- (b) Connect the pump tube in the following procedure.



- ① Pull out the cap nut and ring from the connection port.
- ② Embed cap nut and ring in the pump tube as shown in the figure.
- ③ Insert the tip of pump tube in the joint securely.
 - If the tip of pump tube is moistened with pure water, it can be easily inserted in the joint.
- ④ Screw the cap nut securely by hand.
- ⑤ Tighten the cap nut by hand additionally in approximately one hour after the operation starts.
 - If it is tightened securely by using a tool, the resin cap nut, etc. may be damaged.
- (c) When the pump tube is replaced with a new one, use a techno blade hose (code No. 116B0481).

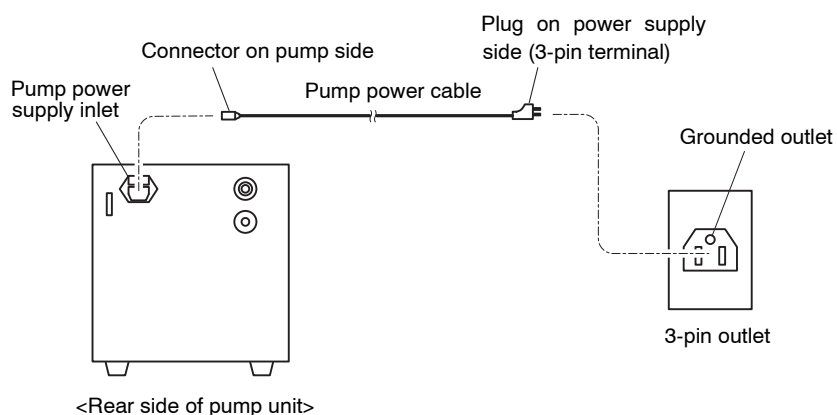
(2) Connection of pump cable

- (a) Connect the pump connector on the rear side on upper part of monitor and the monitor connector on the rear surface of pump unit using the attached pump cable.
- (b) For safety, carry out this connection before connecting the pump power cable to the outlet.



Connection of Pump Cable

(3) Pump power cable and grounding preparation



Pump Power Cable and Preparation for Grounding

- ① **Prepare the power cable.** Prepare the attached pump power cable.

【IMPORTANT】 • The attached pump power cable is dedicated to this pump. Do not use it for other electric device. Do not use the power cable of other device for this pump.

- ② **Prepare 3-pin outlet.** Prepare the 3-pin outlet that can be grounded by type D grounding work (grounding resistance value 100Ω or less).

- ③ **Check the power supply voltage of outlet.** Check the power supply of used 3-pin outlet complies with the specified power supply (power supply voltage is shown in delivery specification).

-
- 【IMPORTANT】**
- If the power supply at high voltage exceeding the specified range is supplied, the product may be damaged.
 - Connect the power supply that does not contain electric noise.
-

- ④ **Reset the pump power cable to the status before connection.** Connect the connector on pump side of pump power cable to the pump power supply inlet so that the plug on power supply side can be inserted in 3-pin outlet. However, do not insert it actually.
- Carry out the insertion of plug in the outlet in the following 3.1 “Procedure for Starting the Automatic Measurement”.

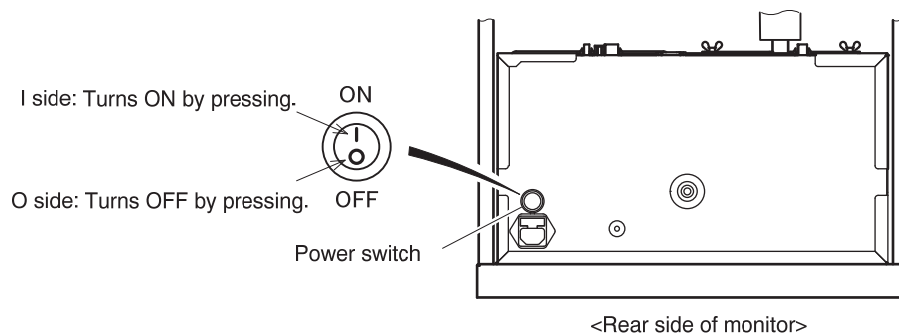
-
- 【IMPORTANT】**
- For safety, do not connect the pump power cable to the outlet at this phase.
-

2.3 Power Cable and Grounding Preparation

- ① **Prepare the power cable.** If the attached power cable cannot be used, prepare the power cable complying with the installation location.

【IMPORTANT】 • The attached power cable is dedicated to this monitor. Do not use it for other electric device. If the attached power cable cannot be used, use the cable either equaling or surpassing the standard.

- ② **Check the power switch is turned off.** Check the power switch on the rear surface of monitor is turned off (press “O” side).

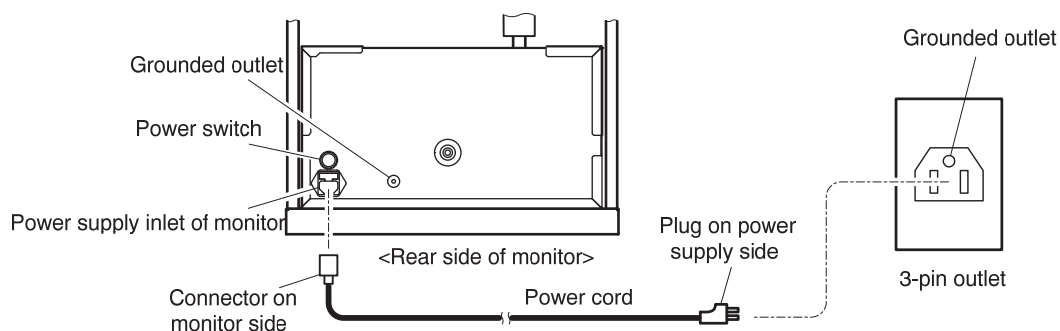


Power Switch

- ③ **Check the power supply of outlet.** Check the power supply of used outlet complies with the specified power supply of this monitor.

【IMPORTANT】 • If the power supply at high voltage exceeding the specified range is supplied, the product may be damaged.
• Connect the power supply that does not contain electric noise.

- ④ **Connect the monitor side of power cable.** Connect the connector on monitor side of power cable to the power supply inlet of monitor.
- ⑤ **Ensure the situation in which grounding can be carried out** Ensure the situation in which type D grounding work (grounding resistance value 100Ω or less) can be carried out in one of the following methods. Carry out the connection to 3-pin outlet etc. in the following 3.1 “Procedure for Starting the Automatic Measurement”.



Power Cable Preparation

- (i) **Grounding with 3-pin outlet** This is the method for connecting the plug on power supply side of power cable to 3-pin outlet.

- (ii) Grounding with grounding terminal This is the method for grounding the grounding terminal on the rear surface of monitor. In this case, do not connect the power cable to the grounding pin at 3-pin outlet.

 WARNING

Electric Shock

- Be sure to carry out grounding. Otherwise, electric shock can result if a problem occurs in the power system.

-
- ⑥ **Enable the connection of power cable.** Reset the plug on power supply side of power cable to the status before the insertion in outlet. Carry out the insertion in outlet in the following 3.1 “Procedure for Starting the Automatic Measurement”.

-
- 【IMPORTANT】** • For safety, do not connect the pump power cable to the outlet at this phase.
-

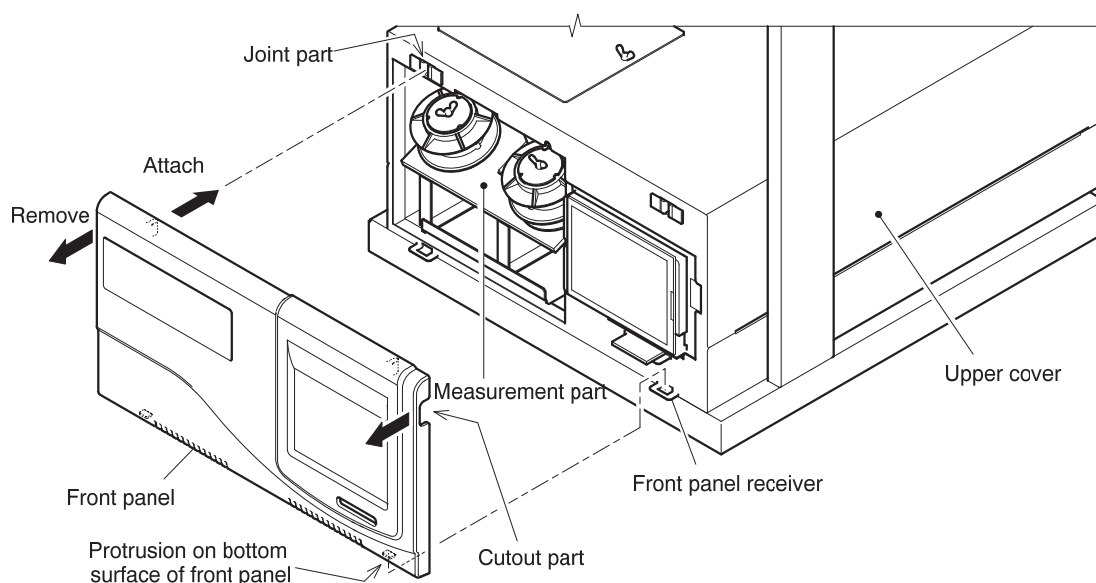
3. Measurement

3.1 Procedure for Starting the Automatic Measurement

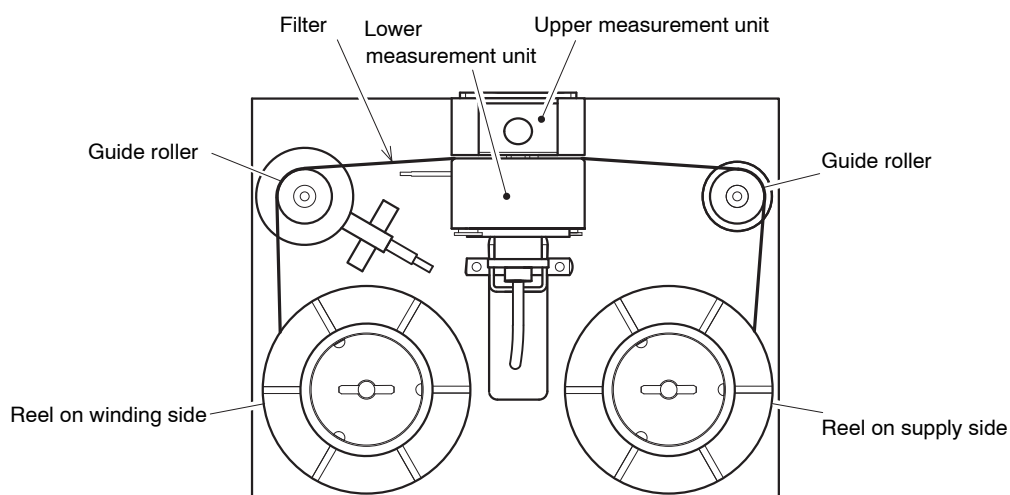
Set the monitor to the usual automatic measurement status in the following procedure. To use in the step “① Make a note about data display”, make a copy of “Engineering Service Inspection Sheet” at the end of the book preliminarily.

① **Check the connection and preparation.** Check that monitor installation, connection and preparation of pump unit, grounding and preparation of power cable, etc. are complete.
> > 2. “Connection and Preparation”

② **Remove the front panel.** Put your fingers on cutout part on both sides of front panel and pull the front panel.



Attachment and Removal of Front Panel



Measurement Part

③ **Check that the filter is loaded.** Check that the filter reeled out from the reel on supply side passes through between upper measurement unit and lower measurement unit and it is rewound by the reel on rewinding side.

- If the filter is not loaded, load it. >> 6.7 “Replacement of Filter”

④ **Attach the front panel.** Place the protrusion on bottom surface of front panel on the front panel receiver and push the upper part of front panel against the measurement unit to join them.

>> Figure “Attachment and Removal of Front Panel” in “② Remove the front panel”

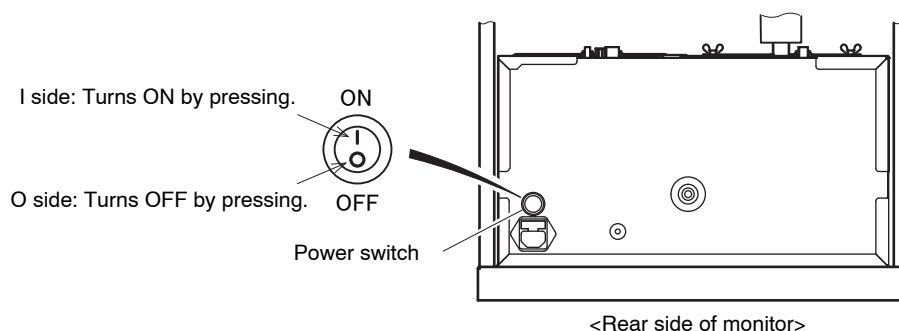
⑤ **Connect the power cable to the outlet.** Connect the power cable of pump unit and the power cable of monitor to the outlets. If the grounding terminal on the rear surface of monitor is used, ground it by type D grounding work (grounding resistance value 100Ω or less). >> 2.2(3) “Pump power cable and grounding preparation”, 2.3 “Power Cable and Grounding Preparation”

⚠ WARNING

Electric Shock

- Be sure to carry out grounding. Otherwise, electric shock can result if a problem occurs in the power system.
- Do not touch the terminals inside the product while power is applied. Touching the terminals can cause electric shock.

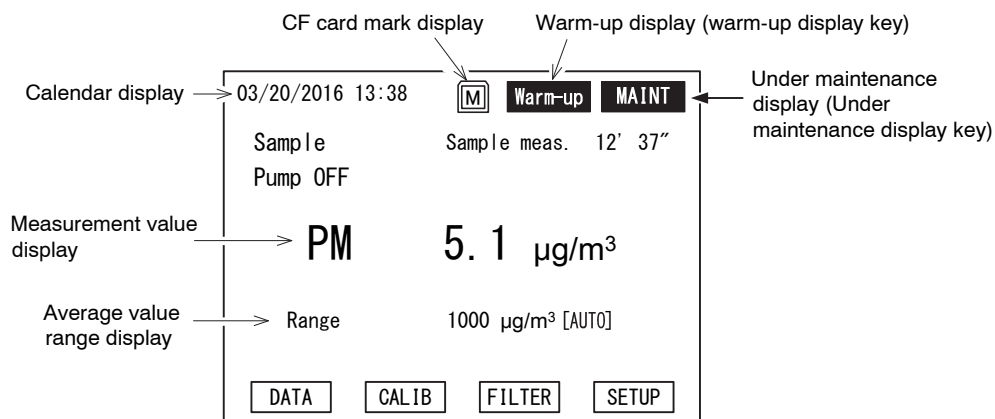
⑥ **Turn on the power.** Turn on the power (press “I” side).



Power Switch

⑦ **Check the display.** After the screen lights up fully once, check that “Home” screen is displayed.

- The monitor starts the warm-up and **Warm-up** and **MAINT** are displayed in the upper right of the screen.
-  indicates that a CF card is inserted.



“Home” Screen at Start-up

⑧ Wait for the measurement to be stabilized. Wait until **Warm-up** and **MAINT** disappear.

- If temperature of monitor and sample flow rate enter the range of set value, the measurement enters stable measurement status and **Warm-up** and **MAINT** disappear. Maximum 60 minutes are required to enter the range.
- If the monitor does not enter the stable measurement status in 60 minutes, the monitor forcibly terminates the warm-up status and **ALARM** appears instead of **Warm-up**. The monitor enters measurement status keeping the alarm generation status. >> 7.1 “Alarm Symbols and Measures”
- To ensure more stable measurement, it is recommended to keep this status for the following time after **Warm-up** and **MAINT** disappear.

At start-up or after long-term operation stop Approx. 24 hours

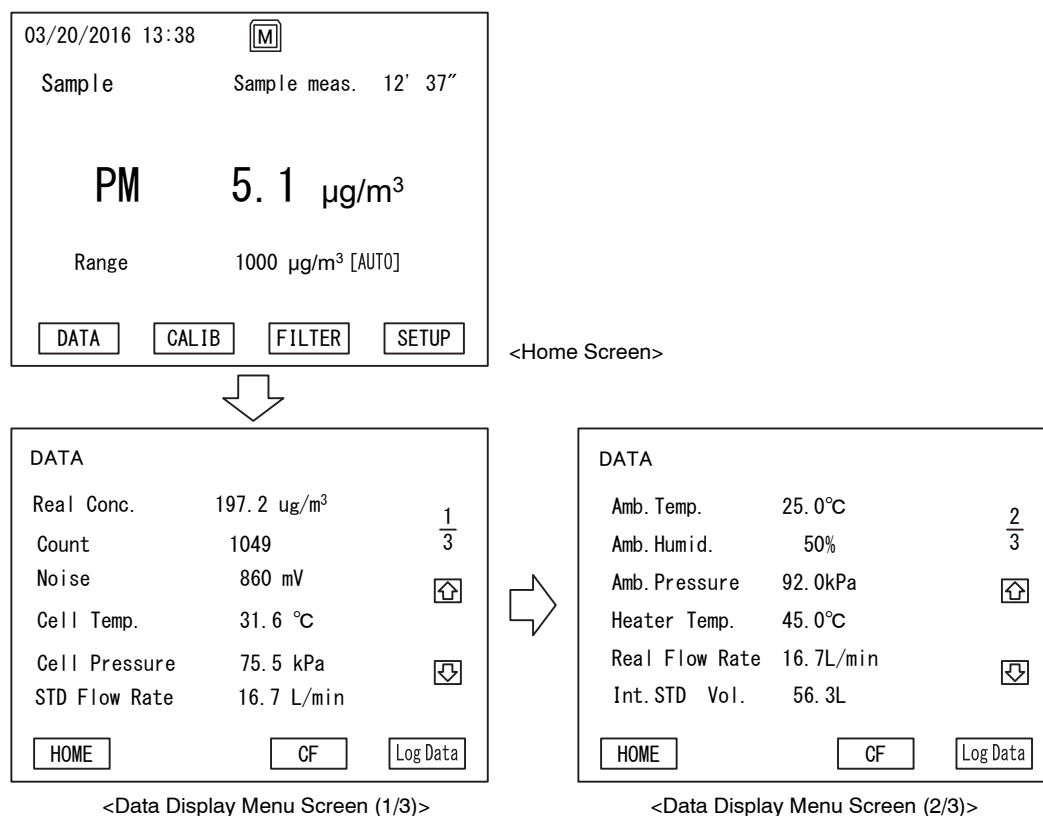
After short operation stop Approx. 3 hours

- Carry out “⑩ Check the flow rate” and subsequent operations after the monitor enters the stable measurement status.

⑨ Check date and time. Check that date and time in the upper left of the screen are correct. If they have lag, make an adjustment. >> 5.5(2) “Change of date and time”

⑩ Check the flow rate.

- Check that the flow control indicating lamp of pump unit is turned off.
- Press **DATA** on “Home” screen. “Data Display Menu” screen (1/3) appears.



From “Home” Screen to “Data Display Menu” Screen (2/3)

- ③ Press once on “Data Display Menu” screen (1/3). “Data Display Menu” screen (2/3) appears.
- ④ Check that the value of “Actual flow rate of sample (STD Flow Rate)” is within the following range on “Data Display Menu” screen (2/3).
- Range of actual flow rate of sample $16.7 \pm 0.3 \text{ L/min}$

⑪ **Make a note about data display.** Press or on “Data Display Menu” screen (1/3 to 3/3) to switch the “Data Display Menu” screen. It is recommended to make a note of values shown on “1/3” to “3/3” in the copied “Engineering Service Inspection Sheet”, which was prepared before starting this operation, and save it.

- The data on this screen is mainly the value as of factory shipment status. You can reference this data when an error is generated.

⑫ **Perform the internal calibration.** The monitor has been calibrated when it was shipped from the factory, however, it is recommended to calibrate it with calibration film for confirmation.

>> 4.2 “Internal Calibration”

The monitor enters the usual measurement status. For the content recorded in a CF card, refer to 3.2 (3) “File of measurement data folder”.

3.2 Data in CF Card

(1) Data collection by CF card

(a) The log data recorded in the monitor can be written (copied) in a CF card.

(b) Measurement data collection cycle It is recommended to collect the log data in the monitor every 10 days to transfer it to PC by inserting a CF card.

- The number of log data that can be recorded by a monitor varies depending on the type of data file as shown below. >> 5.3(1) “Essential point of data display menu”

1-hour average value of PM2.5 Max. 75 days data

Accuracy management information Max. 10 days data
(temperature, pressure, and flow rate etc. in the monitor)

1-minute value, which is the basis of accuracy management information Max. 3 days data

[NOTE] • The accuracy management information is the generic name of internal data required for managing the accuracy of PM2.5 measurement such as sample flow rate, noise voltage, air temperature, pressure of measurement unit.

- The data to be saved in the backup folder is the same data.

[IMPORTANT] • The data of 1-hour average value 76 days before is deleted by overwriting. It is necessary to collect the data by CF card in at least 75-day cycle.

(c) Damage of data due to disturbance It is recommended to insert a CF card at every collection to avoid the effects due to disturbance.

[IMPORTANT] • The data in the inserted CF card may be damaged by disturbance of thunder, etc.

(d) Removal of CF card Before pulling out a CF card, be sure to carry out the operation shown in 5.3(4) “Removal of CF card”.

[IMPORTANT] • If you pull out the CF card forcibly without carrying out “Removal of CF card”, the data of CF card may not be read out properly.

(e) Genuine CF card of our company For the CF card, be sure to use our company's genuine product (code No. 6958270K).

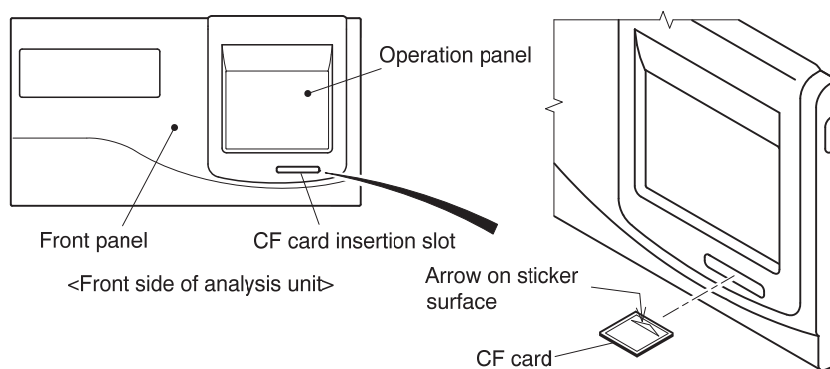
[IMPORTANT] • If a CF card other than our company's genuine product is used, the data may not be written properly or the data may not be recognized.

(f) Referential item For the following operation, refer to the items shown in 5.3 “Operation of Data Display Menu”.

- Removal of CF card >> 5.3(4) “Removal of CF card”
- Backup by CF card >> 5.3(5) “Backup by writing in CF card”

① **Insert a CF card** Face the sticker surface of CF card on which an arrow is printed upward and insert the card in the CF card insertion slot in the arrow direction slowly until the card stops.

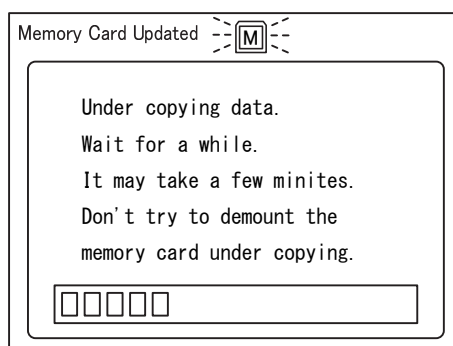
- A CF card can be inserted every time regardless of power ON/OFF of monitor and operation status, etc.



CF Card Insertion

- If a CF card is inserted,  (CF card mark) is displayed on the screen. If a CF card is inserted when the power supply of monitor is turned off,  mark will be displayed on the screen after the power supply will be turned on.

② **Check that the writing is started.** Check that “CF Card Writing Execution” screen is displayed and lighting  mark starts blinking (during writing).



CF Card Writing Execution Screen

- If a CF card is inserted, “CF Card Writing Execution” screen is automatically displayed and measurement data and log data, which are not recorded in the CF card, are copied from CPU of measurement part to the CF card.
- If there is no measured data or there is only data which has been collected in the CF card, “CF Card Writing Execution” screen is not displayed.
- The writing time varies depending on the amount of cumulative data, etc.

【IMPORTANT】 • While the data is written ( mark blinks), do not turn off the power. The data may not be read properly.

- [NOTE]
- The data that is automatically written by inserting a CF card is saved in the “measurement data folder (data)”. On the other hand, if you carry out 5.3(5) “Backup by writing in CF card” by operating the screen, the same data can be saved in the “backup folder (Backup)”.
 - The data for 10 years or more can be collected in a CF card, however, if the free capacity of memory of CF card becomes 1% or less, an alarm (“A310 CF card memory shortage”) is displayed (>> 7.1 “Alarm Symbols and Measures”). Moreover, if the free capacity becomes 1KB or less, the copy to CF card is disabled.

[In case of shortage of memory]

If “A310 CF card memory shortage” is displayed, carry out the operation as shown below.

- ① Press BACK on the error screen.
- ② Pull out the CF card by the operation of 5.3 (4) “Removal of CF card” on “CF card” screen.
- ③ Delete all measurement data saved in the measurement data folder (data) of CF card on PC.
- ④ Carry out the operations from step “① Insert a CF card” again.

[If writing does not proceed]

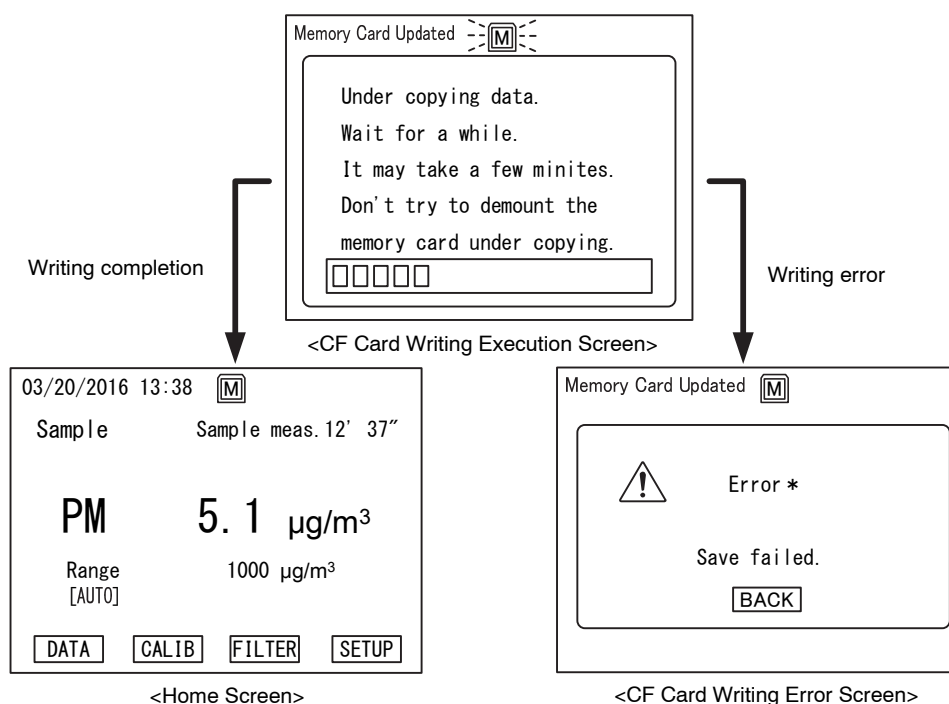
If the “CF Card Writing Execution” screen is not displayed although there must be measured data, or the operation does not proceed in the process of “CF Card Writing Execution” screen, carry out the operation as shown below.

- ① Turn off the power of monitor and pull out the CF card during it, and the turn on the power again.
- ② Format (initialize) the CF card by the file system FAT16(FAT) of PC.

-
- [IMPORTANT]**
- Do not format a CF card by the file system FAT32 of PC. Otherwise, reading and writing of CF card will be disabled.
 - If a CF card is formatted, every data in the CF card is deleted.
-

- ③ Carry out the operation from step “① Insert a CF card” again.

- ③ Check that the writing is completed** Check that “Home” screen is displayed and the CF card mark display (M) is not blinking.

**Writing Completion or Writing Error Screen**

[In case of writing error]

If “CF Card Writing Error” screen is displayed instead of “Home” screen, carry out the operation as shown below.

- ① Press BACK screen on “CF Card Writing Error” screen.
 - ② On the displayed “CF card” screen, carry out the operation shown in 5.3 (4) “Removal of CF card” to pull out the CF card.
 - ③ Carry out the operation from the step “① Insert a CF card” again.
 - ④ **Save the backup data.** Write the data in the backup folder of CF card too if necessary.
>> 5.3 (5) “Backup by writing CF card”
 - ⑤ **Pull out the CF card.** >> 5.3 (4) “Removal of CF card”
-
- [IMPORTANT]** • If you pull out the CF card forcibly without carrying out “Removing the CF card”, the data of CF card may not be read out properly.
-
- ⑥ **Transfer the data in CF card to PC.** Connect the CF card to PC and transfer the data to the PC.

(2) CF card folder

- (a) There is no folder in a CF card when it is shipped from factory. If a new CF card is inserted, measurement data folder, parameter folder, backup folder, and communication log folder is automatically generated in the station number folder of CF card as shown in Table “Types of CF Card Folders”.
- (b) If the data of CF card is collected in a PC, the content can be checked using the functions of spreadsheet software (Excel*), etc. Do not operate any folders other than measurement data folder and backup folder. (*: Excel is a registered trademark of Microsoft Corporation in US.)

[IMPORTANT] • Do not change the data in parameter folder and communication log folder on the PC. Otherwise, it will be obstacle to writing of measurement data, etc.

Types of CF Card Folders

Folder name (actual folder name)		Contents
Station number folder (□□□□□□)	Measurement data folder (data)	• Data measured by monitor, etc. is recorded.
	Parameter folder (param)	• Parameters of main CPU in the display unit of monitor are recorded. • Parameters may not be recorded depending on the specifications.
	Backup folder (back up)	• If 5.3 (5) “Backup by writing in CF card” is carried out, the data recorded in the main CPU of display unit is copied.
	Communication log folder (log)	• Communication log of main CPU of display unit and I/O board and RS-232C communication log of main CPU of display unit and external device are recorded. • The logs may not be recorded depending on the specifications.

[Note] • For the actual name of CF card, upper cases and lower cases are not distinguished. For the station number folder, 6-digit number of station number is the folder name.

(3) File of measurement data folder

- (a) The measurement data, etc. is recorded in measurement data folder and backup folder as log data as shown in Table “Data and File Name of Measurement Data Folder”.
- (b) For some file names of measurement data folder, starting “year” of measurement data (yyyy: Dominical year 4 digits, <0000 to 9999>) or “year and month” (yy: lower 2 digits of Dominical year and mm: month <01 to 12>) are added. If the file is copied to the backup folder, the numbers are not added.
- (c) If date and time are changed, the time of measurement data or log data may be redundant or have lag.

Data and File Name of Measurement Data Folder

Data of measurement data folder (or backup folder)		File name		Contents
		Measurement data folder	Backup folder	
PM2.5 measurement data	1-hour average value data (concentration)	K7HAyyyy.CSV	K7HA.CSV	Data integrated every hour
	Daily average value data (concentration)	K7HDyyyy.CSV	K7HD.CSV	Data averaging 1-hour average value per 1 to 24 hours
Accuracy management information	1-day value data (temperature, pressure, flow rate etc.)	K7Hyyyy.CSV	K7H.CSV	Data averaging instantaneous value of temperature, pressure, flow rate, etc. every second per 1 hour
	1-day value data (temperature, pressure, flow rate etc.)	K7Dyyyy.CSV	K7D.CSV	Data averaging 1-hour value data (temperature, pressure, flow rate etc.) shown above per 1 day
	Noise voltage instantaneous value data	K7NVyymm.CSV	K7NV.CSV	1 minute value of instantaneous value of noise voltage of every second per 1 minute
Event data	Zero deviation	K7ZERO.CSV		Log data when zero deviation is updated
	Span factor	K7SPAN.CSV		Log data when span factor is updated
	Alarm	K7ALM.CSV		Log data when an alarm is generated
	Event	K7IVT.CSV		Log data of power on and maintenance work, etc.

- (d) The format (recording format) of each file generated in measurement data folder and backup folder is shown below.

File Format and Contents

Data and file name	File format and contents																											
1-hour average value data (concentration) (K7HAyyyy.CSV)	YYYYMMDDHH, xxxxxxxx, xxxxxxxx, µg/m3, ****, xxxxxxxx, Year, month, day, and hourConcentration in standard statusActual concentrationUnitAttributeNoise voltage (Table 1)(mV) xxxxxxx, xxxxxxxx Integrated standard flow rate (mV)Integrated actual flow rate (L) - Concentration in standard status Concentration dividing the mass of collected PM2.5 by collected cubic content in “standard status (20°C, 1 atmosphere (101.3kPa))” - Actual concentration Concentration dividing the collection mass by collected cubic content of “temperature, pressure (actual status) at that point” - Priority order of attribute If multiple status of attribute symbol (>> Table 2) is generated at the same time, one status with higher priority order is described. - Attribute when time is changed If the time is changed crossing the hour, the attribute symbol “ADJ” is described for 1-hour average values of both of before time change and after time change. However, in case of time change without crossing the hour, the attribute symbol is not described. 1 minute after filter is fed For 1-hour average value of noise voltage, integrated standard flow rate, and integrated actual flow rate, the value of 1 minute after filter is fed and filter is transferred should not be calculated. Table 1. Symbol and Status of Attribute <table><tr><th>Priority order</th><th>Symbol</th><th>Status</th></tr><tr><td>1</td><td>NON</td><td>Measurement was not carried out.</td></tr><tr><td>2</td><td>POFF</td><td>Power supply was turned off.</td></tr><tr><td>3</td><td>ADJ</td><td>There was transmission during maintenance or time change.</td></tr><tr><td>4</td><td>AL2</td><td>Alarm 2 was generated.</td></tr><tr><td>5</td><td>AL1</td><td>Alarm 1 was generated.</td></tr><tr><td>6</td><td>CAL</td><td>Calibration was carried out.</td></tr><tr><td>7</td><td>RST</td><td>Measurement was started by internal reset.</td></tr><tr><td>8</td><td>□□□□ (blank)</td><td>Measurement was started by external reset.</td></tr></table>	Priority order	Symbol	Status	1	NON	Measurement was not carried out.	2	POFF	Power supply was turned off.	3	ADJ	There was transmission during maintenance or time change.	4	AL2	Alarm 2 was generated.	5	AL1	Alarm 1 was generated.	6	CAL	Calibration was carried out.	7	RST	Measurement was started by internal reset.	8	□□□□ (blank)	Measurement was started by external reset.
Priority order	Symbol	Status																										
1	NON	Measurement was not carried out.																										
2	POFF	Power supply was turned off.																										
3	ADJ	There was transmission during maintenance or time change.																										
4	AL2	Alarm 2 was generated.																										
5	AL1	Alarm 1 was generated.																										
6	CAL	Calibration was carried out.																										
7	RST	Measurement was started by internal reset.																										
8	□□□□ (blank)	Measurement was started by external reset.																										
1-hour value data (temperature, pressure, flow rate, etc.) (K7Hyyyy.CSV)	YYYYMMDDHH, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, Year, month, day, and hourβ-ray countMeasurement unit temperature (°C)Measurement unit pressure (kPa)Standard flow rate (L / min)Temperature in gauge (°C) xxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, Outdoor air humidity (%)Heater temperature (°C)Outdoor air temperature (°C)Ambient pressure (kPa)Actual flow rate (L / min)Measurement unit target temperature (°C)Maximum noise value xxxxxxx, xxxxxxxx, xxxxxxxx Minimum noise valueMaximum β-ray count valueMinimum β-ray count value 1 minute after the filter is fed 1-hour value of β-ray count, standard flow rate, actual flow rate, maximum noise value, minimum noise value, maximum β-ray count value, and minimum β-ray count value, the value for 1 minute after the filter is fed and after the filter is transferred should not be calculated.																											

(To be continued)

(Continued from previous page)

Data and file name	File format and contents									
1-day value data (temperature, pressure, flow rate, etc.) (K7Dyyyy.CSV)	YYYYMMDDHH, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, Year, month, day, and hour β-ray count Measurement unit temperature (°C) Measurement unit pressure (kPa) Standard flow rate (L / min) Temperature in gauge (°C) xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx Outdoor air humidity (%) Heater temperature (°C) Outdoor air temperature (°C) Ambient pressure (kPa) Actual flow rate (L / min) • 1 minute after a filter is fed For 1-day value of β-ray count, standard flow rate, and actual flow rate, the value for 1 minute after the filter is fed and after the filter is transferred should not be calculated.									
Noise voltage instantaneous value data (K7NVyymm.CSV)	YYYYMMDDHHmm, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx Year, month, day, hour, and minute Noise voltage (mV) β-ray count Outdoor air temperature (°C) Ambient pressure (kPa) • Insertion of CF card If a CF card is not inserted, the noise voltage instantaneous value data is not saved.									
Zero deviation log data (K7ZERO.CSV)	YYYY/MM/DD, HH:mm:00, ****, xxxxxxxx Year, month, and day Hour, minute, and second Calibration status (Table 2) Zero deviation Table 2 Symbol and Status of Calibration Status <table><tr><th>Priority order</th><th>Symbol</th><th>Status</th></tr><tr><td>1</td><td>CAL</td><td>Internal calibration (calibration by calibration film) was carried out.</td></tr><tr><td>2</td><td>(NULL)</td><td>The calibration that forcibly changed the values on screen was carried out by external calibration, flow rate calibration, pressure / temperature / humidity calibration, etc.</td></tr></table>	Priority order	Symbol	Status	1	CAL	Internal calibration (calibration by calibration film) was carried out.	2	(NULL)	The calibration that forcibly changed the values on screen was carried out by external calibration, flow rate calibration, pressure / temperature / humidity calibration, etc.
Priority order	Symbol	Status								
1	CAL	Internal calibration (calibration by calibration film) was carried out.								
2	(NULL)	The calibration that forcibly changed the values on screen was carried out by external calibration, flow rate calibration, pressure / temperature / humidity calibration, etc.								
Span factor log data (K7SPAN.CSV)	YYYY/MM/DD, HH:mm:00, ***, xxxxxxxx, xxxxxxxx Year, month, and day Hour, minute, and second Calibration status (Table 2) Span factor Calibration film measured value • Calibration status >> Table 2 “Symbol and Status of Calibration Status”									
Alarm log data (K7ALM.CSV)	YYYY/MM/DD, HH:mm:00, A***, ***, xxxxxxxx, xxxxxxxx, Year, month, and day Hour, minute, and second Alarm number Alarm classification β-ray count Measurement unit temperature (°C) xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx, Measurement unit pressure (kPa) Standard flow rate (L / min) Temperature in gauge (°C) Outdoor air humidity (%) Heater temperature (°C) Outdoor air temperature (°C) xxxxxxxx, xxxxxxxx, xxxxxxxx, xxxxxxxx Ambient pressure (kPa) Actual flow rate (L / min) Noise voltage (mV) Span factor • For alarm number and alarm classification, refer to 7.1 “Alarm Symbols and Measures”.									

(To be continued)

(Continued from previous page)

Data and file name	File format and contents		
Event log data (K7IVT.CSV)	YYYY/MM/DD, HH:mm:00, I***, xxxx Year, month, and day Hour, minute, Event Event and second number item		
	Event Number and Item		
	Event No.	Event item	Contents
	I901	Power OFF	The power is turned OFF.
	I902	Power ON	The power is turned ON.
	I903	Cancel warm-up	Warm-up is cancelled after power ON.
	I920	Writing of parameter	☐ YES is pressed on “Write Parameter” screen.
	I921	Update of program	☐ YES is pressed on “Program Update” screen.
	I922 I923	Initialization of CPU	☐ YES is pressed on “Initialization” screen.
	I924	Log clear	☐ YES is pressed on “Log Data Clear” screen.
	I925	Removal of CF card	CF card is pulled out without carrying out CF card removal operation.
	I926	Writing of CF card	☐ YES is pressed on “Memory Card Update” screen.
	I927	Change of time	☐ OK is pressed on “Date” and “Time” popup screen.
	I975	Replacement of filter	☐ YES is pressed on “Filter Replacement Confirmation” screen.
	I976	Calibration of flow rate	☐ Exec.Calibration of flow rate calibration operation is pressed. However, the number of event histories should be one until ☐ BACK on “Flow Rate Calibration” screen is pressed.
	I977	Pump OFF	☐ YES is pressed on “Pump Off Confirmation” screen.
	I978	Flow rate correction factor	“Flow rate correction factor” is changed on the calibration menu “Flow Rate Calibration” screen. > > 4.3 “Calibration of Flow Rate by Standard Flowmeter”
	I979	Sequence start	☐ YES is pressed on “Sample Sequence Start Confirmation” screen.
	I981	Basic calibration of flow rate	Event for service > > 5.5(6) “Service menu”
	I982	Filter transfer	Transfer of filter is executed.
• This table includes the event item for engineering service.			

3.3 Measurement Stop and Storage

(1) Measurement stop

To stop the measurement, turn off the power. >> Figure “Power Switch” in step “⑥” of 3.1 “Procedure for Starting the Automatic Measurement”

Execute the restart of measurement after this stop by referring to 3.1 “Procedure for Starting the Automatic Measurement”.

- [NOTE]
- It is recommended to insert a CF card to pull out the collected measurement data before stopping the measurement. >> 3.2(1) “Data collection by CF card”
 - It may be necessary to replace the lithium battery (for display unit) with a new one depending on the battery consumption due to measurement stop (in case of long-term stop or when the battery life is short even if the stop is short). >> 6.16 “Replacement of lithium battery”

(2) Long-term operation stop

To stop the operation for a few weeks or more, carry out the operation shown in 3.3 (1) “Measurement stop” and then turn off the power at supply source.

(3) Transportation and storage

In case of disconnection and storage, or transportation to other place, carry out the operation as shown below.

① **Stop the measurement.** >> 3.3(2) “Long-term operation stop”

② **Disconnect cables and tubes.** Disconnect cables and tubes connected to external of monitor.

③ **Ensure the installation location.** Prepare the indoor place under meeting the conditions shown below.

Ambient temperature: -10 to 50°C

Humidity: 0 to 80%RH or less

④ **For the restart of measurement, refer to the corresponding item.**

>> 2. “Connection and Preparation”, 3.1 “Procedure for Starting the Automatic Measurement”

4. Calibration

4.1 Functions Related to Calibration

(a)Necessity of calibration This monitor can continue the measurement with higher reliability by calibrating PM2.5 measurement system and flow rate sensor. The functions related to calibration are shown in Table “Purpose and Contents of Functions Related to Calibration”.

(b)Usual calibration Usually, carry out “Internal calibration” and “Flow rate calibration by standard flowmeter” periodically. >> 4.2 “Internal Calibration”, 4.3 “Calibration of Flow Rate by Standard Flowmeter”, 6.1 “Maintenance List”

(c)Value after external calibration If the external calibration is carried out, the measurement value changes as shown below.

Value after external calibration = “value before calibration” x “internal span factor” x “external span factor” + “external zero deviation”

Internal span factor: factor calculated by calibrating calibration film

External span factor: span factor that is generally obtained by parallel measurement with standard measurement method, etc.

External zero deviation: zero deviation that is generally obtained by “blank test” and “parallel measurement with standard measurement method”

(d)Calibration by standard flow rate The flow rate sensor of this monitor displays the value converted to the flow rate in the standard status [20°C, 1 atmosphere (101.3kPa)]. Consequently, use the standard flowmeter that displays the flow rate in the standard status for the calibration of flow rate. >> 4.3 “Calibration of Flow Rate by Standard Flowmeter”

Purpose and Contents of Functions Related to Calibration

Item	Purpose	Contents	Contact output	Reset during operation
Internal calibration	Calibration of measurement function by calibration film	Calibration for inserting an calibration film in the measurement unit following the instructions on the calibration film calibration screen to adjust the span factor to the value described on the label on calibration film >> 4.2 “Internal Calibration”	Under adjustment Under calibration	Disabled
External calibration	Calibration of monitor in standard measurement method, etc.	Calibration for setting internal span factor, external span factor, and external zero deviation to arbitrary values by inputting keys >> 5.4(3) “External calibration”	—	—
Flow rate calibration	[Calibration of flow rate by standard flowmeter] Basic flow rate calibration	Calibration of flow rate sensor in the monitor for adjusting the indication to the value measured by standard flowmeter >> 4.3 “Calibration of Flow Rate by Standard Flowmeter”, 5.4(4) “Screen operation for flow rate calibration”	Under adjustment	Enabled

(To be continued)

(Continued from previous page)

Item	Purpose	Contents	Contact output	Reset during operation
	[Manual flow rate control] Check the operation by setting an arbitrary flow rate	- The sample flow rate can be freely increased and decreased by stopping the automatic flow rate control. - The difference of indication from that of standard flowmeter can be checked and the operation status of pump unit can be checked. >> 5.4(4) "Screen operation for flow rate calibration"	—	—
	[Flow rate correction factor] Calibration for setting the correction factor to arbitrary value	The correction factor that can be obtained by calibrating the flow rate by standard flowmeter can be set to an arbitrary value by operating keys. >> 5.4(4) "Screen operation for flow rate calibration"	—	—
Calibration of temperature / humidity / pressure	[Temperature] Calibration of outdoor air temperature	Change the outdoor air temperature correction factor so that the displayed value of outdoor air temperature corresponds to the indicated value of standard thermometer. >> 5.4(5) "Calibration of temperature, humidity, and pressure"	—	—
	[Humidity] Calibration of outdoor air humidity	Change the outdoor air humidity correction factor so that the displayed value of outdoor air humidity corresponds to the indicated value of standard hygrometer. >> 5.4(5) "Calibration of temperature, humidity, and pressure"	—	—
	[Pressure] Calibration of peripheral pressure sensor	Change the peripheral pressure correction factor so that the displayed value of peripheral pressure corresponds to the indicated value of standard barometer. >> 5.4(5) "Calibration of temperature, humidity, and pressure"	—	—
Blank test	Check the stability of indication by introducing zero gas	Connect DUB/FPM zero filter (option) to the sample air suction port and check the stabilization of indication for 24 hours or more continuous measurement. >> 6.15 "Blank Test"	—	—
Confirmation of coincidence	Check the difference of indication from that of standard measurement method	Check the difference of indication from that of standard measurement method by the measurement for introducing the same sample air to the sample air suction port in parallel to PM2.5 low-volume sampler.	—	—

4.2 Internal Calibration

(a) Carry out the internal calibration in the following the procedure. Since the internal calibration screen exists in the calibration menu, the procedure is also described in 5.4(2) “Screen operation for internal calibration”.

(b) The calibration film to be used for internal calibration has inherent calibration film value separately. It is described on the label affixed to the calibration film. The value described on the label for each monitor is set to the “Calibration Film Setup” screen of calibration menu when the product is shipped from the factory.

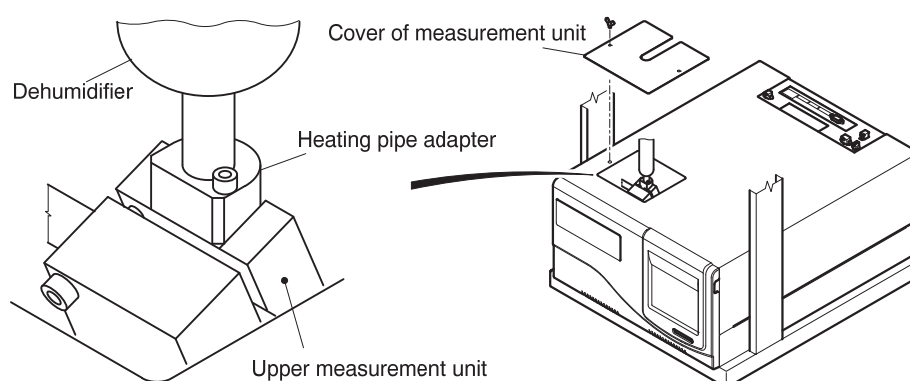
① **Check the continuous operation status.** Check the monitor is continuously operated for 3 hours or more (approx. 24 hours at start-up). >> 3.1 “Procedure for Starting the Automatic Measurement”

【IMPORTANT】 • If the monitor is not in stable continuous operation status, the internal calibration may not be carried out properly.

② **Prepare the calibration film.** Prepare the attached calibration film at hand.

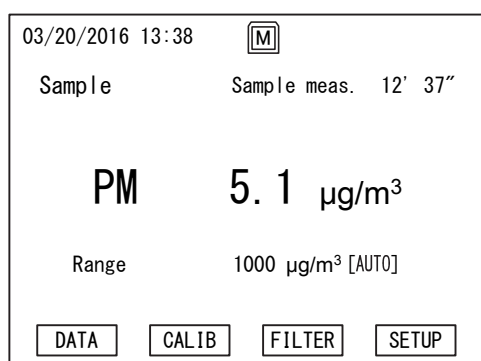
③ **Check the connection of dehumidifier.** Remove the cover of measurement unit and check that the dehumidifier is connected to the upper measurement unit with heating pipe adapter.

【IMPORTANT】 • If the heating adapter is not connected to the measurement unit, light enters the measurement unit and the measurement cannot be carried out normally.

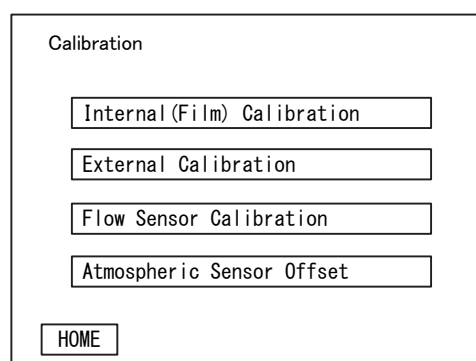


Dehumidifier Connection Check

④ **Display “Calibration Menu” screen.** Press **CALIB** on “Home” screen.



Home Screen



Calibration Menu Screen

⑤ **Display “Start Blank Measurement” screen.** Press the section within the frame of [Internal (Film) Calibration] on “Calibration Menu” screen.

⑥ **Check the calibration film value on the screen.** Check the calibration film value displayed on “Start Blank Measurement” screen corresponds to the “Calibration film value” shown in the calibration film at your hand.

Film Calibration

Confirm the calibration film value,
and touch [Start Blank Measurement] key.

Start Blank Measurement

Calib. Film Value 0.803mg/m³

BACK

Start Blank Measurement Screen

- If the calibration film value on “Start Blank Measurement” screen is different from “Calibration film value” shown in the label, change them so that they get correspondent, and then go to the step “⑦”. >> 5.4(1) “Check or change the calibration film value”

⑦ **Remove the front panel** >> ② Figure “Attachment and Removal of Front Panel” in 3.1 “Procedure for Starting the Automatic Measurement”

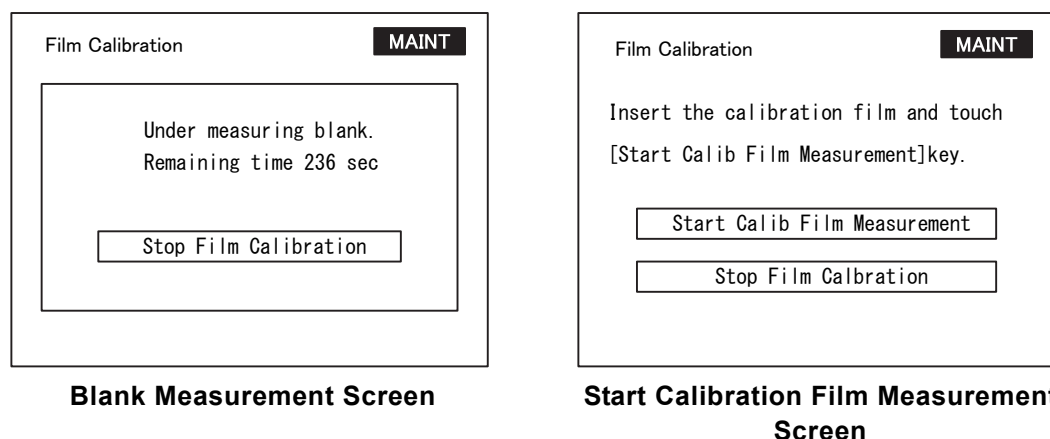
⚠ CAUTION

Injury

- While the power is distributed, do not touch the operating part of measurement unit, etc. The measurement unit is intermittently opened and closed and the reel rotates to feed filter. However, on “Start Calibration Film Measurement” screen for inserting the calibration film, the operation of measurement unit, etc. is stopped.
-

⑧ **Start the blank measurement.** Press the section within the frame of [Start Blank Measurement] on “Start Blank Measurement” screen.

- The operation of blank measurement starts and the operation progresses as shown below.
 - ① Lower measurement unit moves downward (for approx. 5 seconds) “Detector cell move downward” is displayed on the screen.
 - ② Feed filter (for approx. 7 seconds) “Filter tape is feeding” is displayed on the screen.
 - ③ Lower measurement unit moves upward (for 5 approx. seconds) “Detector cell move upward” is displayed on the screen.
 - ④ Blank measurement (for approx. 240 seconds = 4 minutes) “Blank measurement screen” is displayed and the remaining time (seconds) of blank measurement is displayed. The remaining time starts from “239”. If the time becomes “000”, the lower measurement unit moves downward and “Start Calibration Film Measurement” screen is displayed. On “Start Calibration Film Measurement” screen, the buzzer keeps sounding in 3-second intervals until the key is pressed.



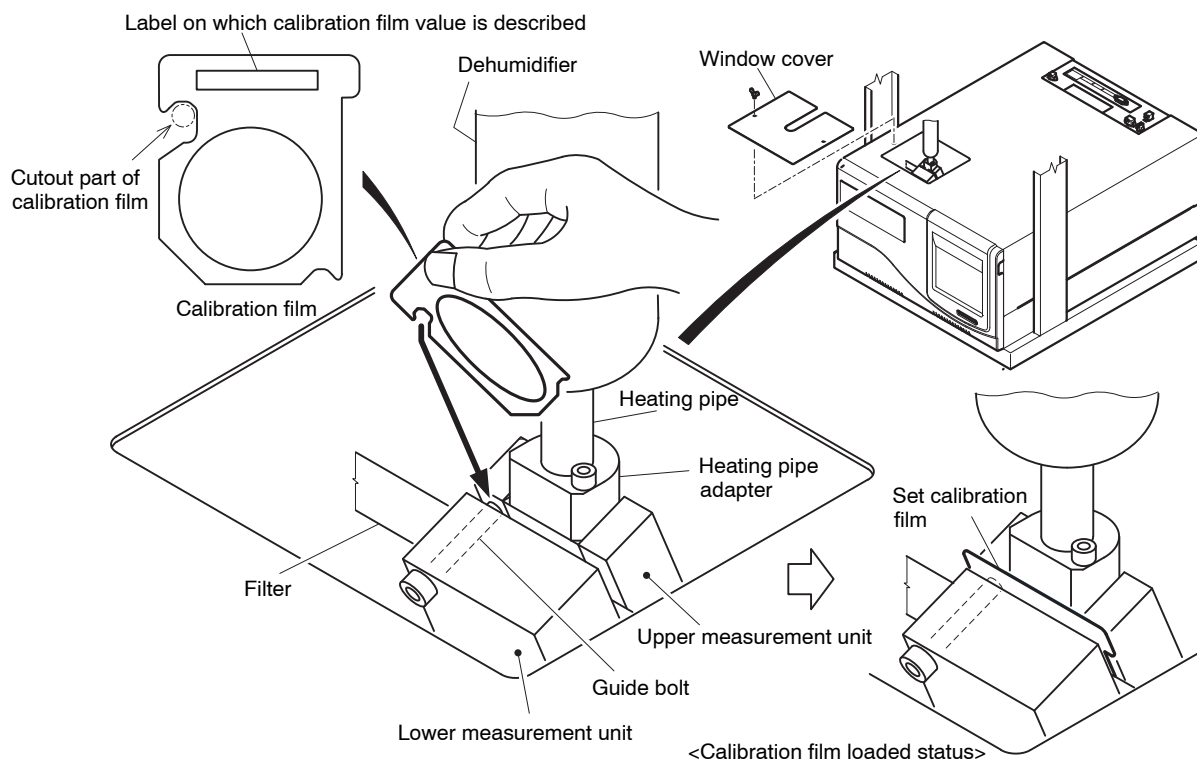
- If you press the section within the frame of [Stop Film Calibration] on “Blank Measurement” screen, the blank measurement can be stopped in the middle. In this case, carry out the operation following the instructions on the screen.

⑨ Load the calibration film

- Hitch the calibration film to the guide bolt Take an calibration film between your fingers as shown in the figure and insert it between opening part of measurement unit and radiation source cell. If the cutout part of calibration film is hitched to the guide bolt, release your fingers.

【IMPORTANT】 • Insert the calibration film carefully along the bottom surface of upper measurement unit not to damage the filter by calibration film.

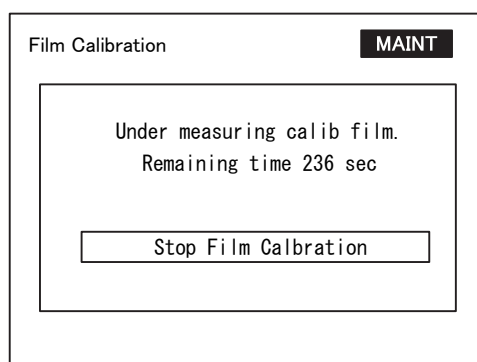
- Check the calibration film is properly set Check the calibration film and the surface of lower measurement unit become parallel.



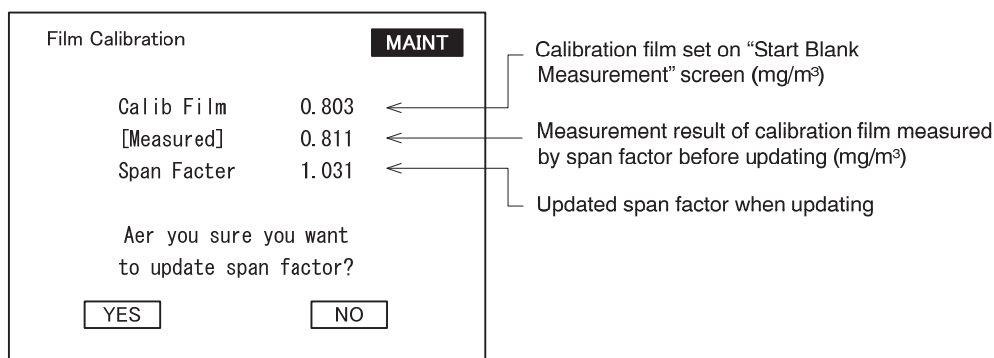
Loading of Calibration Film

⑩ **Start the measurement of calibration film.** Press the section within the frame of [Start Calib Film Measurement] on “Start Calibration Film Measurement” screen.

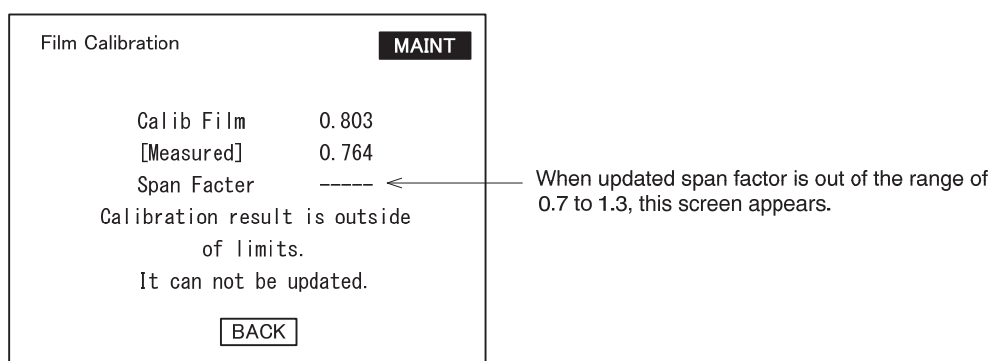
- The measurement of calibration film starts and the operation progresses as shown below.
 - (a) While the lower measurement unit moves upward (for approx. 5 seconds) “Detector cell move upward” is displayed.
 - (b) Measurement of calibration film (for approx. 240 seconds = 4 minutes) “Calibration Film Measured” screen is displayed and the remaining time (seconds) of calibration film measurement is displayed. If the remaining time starts from “239” and becomes “000”, the screen shifts to “Calibration Film within Calibration Range” screen (when the span factor is within 0.7 to 1.3) or “Calibration Film out of Calibration Range” screen via “Open Measurement Unit” screen. At this time, the buzzer keeps sounding in 3-second intervals until the key is pressed.
- If you press the section within the frame of [Stop Film Calibration] on “Calibration Film Measured” screen, the internal calibration can be stopped. In this case, carry out the operation following the instructions on the screen.



Calibration Film Measured Screen

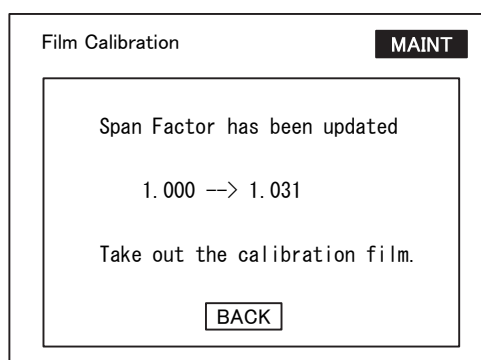


Within Internal Calibration Range Screen

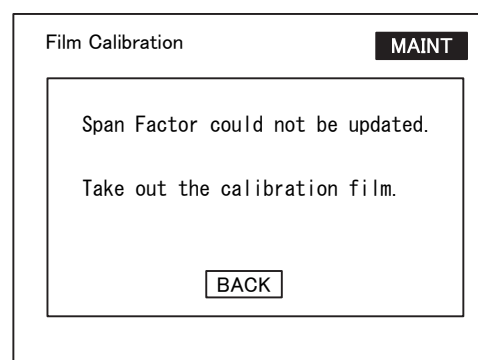


Out of Internal Calibration Range Screen

- ⑪ **Make a note of calibration film measurement value.** Make a note of [Measured] value on “Within Internal Calibration Range” screen or “Out of Internal Calibration Range” screen.
- ⑫ **Return to “Start Blank Measurement” screen.** Press on “Within Internal Calibration Range” screen or press on “Out of Internal Calibration Range”.
- ⑬ **Repeat blank measurement and calibration film measurement.** Repeat the steps “⑧ Start the blank measurement” to “⑫ Return to “Start Blank Measurement” screen” and make a note of total three [Measured] values.
- ⑭ **Check calibration film value, etc. and go to next step.** If all [Measured] values are within $\pm 2\%$ of calibration film value, the values are normal. It is not necessary to rewrite the span factor. If the values are considerably different, a trouble may be generated. After the steps up to “⑰ Attach the front panel” are finished, refer to 7. “Troubleshooting”. If a proper measure cannot be found, please request the engineering service.
- When the span factor is rewritten Press on “Within Internal Calibration Range” screen. The span factor is updated and the screen shifts to “Remove Calibrated Calibration Film” screen.
 - When the span factor is not rewritten Press on “Within Internal Calibration Range” screen. The screen shifts to “Remove Not-calibrated Calibration Film” screen.
 - When “Out of Internal Calibration Range” screen is displayed Press . The screen shifts to “Remove Not-calibrated Calibration Film” screen.



Remove Calibrated Calibration Film Screen



Remove Not-calibrated Calibration Film Screen

- ⑮ **Pull out the calibration film** Pull out the calibration film from measurement unit carefully.
- ⑯ **Display “Home” screen** Press on “Remove Calibrated Calibration Film” screen or “Remove Not-calibrated Calibration Film” screen and press on the displayed “Start Blank Measurement” screen. Then press on the continuously displayed “Calibration Menu” screen.
- ⑰ **Attach the front panel** >> ② Figure “Attachment and Removal of front Panel” in 3.1 “Procedure for Starting the Automatic Measurement”

4.3 Calibration of Flow Rate by Standard Flowmeter

- (a) Calibrate periodically using the standard flowmeter that is calibrated in the defined method so that the suction flow rate of sample air is within the standard.
- (b) The flow rate sensor of this monitor displays the flow rate in the standard status [20°C, 1 atmosphere (101.3kPa)]. The “Sample flow rate” on “Data Display Menu” screen (1/3) is also the standard flow rate.

	DATA	
	Real Conc.	197.2 ug/m ³
	Count	1049
	Noise	860 mV
	Cell Temp.	31.6 °C
	Cell Pressure	75.5 kPa
	STD Flow Rate	16.7 L/min
	HOME	CF Log Data

Standard flow rate →

[Flow rate at 20°C, 1 atmosphere (101.3kPa)]

“Sample flow rate (standard flow rate)” on “Data Display Menu” screen (1/3)

- 【IMPORTANT】** • For confirmation and calibration of flow rate, be sure to use the standard flowmeter that displays the standard flow rate. The flowmeter that displays the actual flow rate cannot execute the calibration properly.

- (c) Calculation of actual flow rate of sample PM2.5 is proportional to the actual flow rate of sample. For this reason, this monitor calculates and obtains the “actual flow rate of sample” from temperature and pressure at that time based on the sample flow rate (standard value) measured by flow rate sensor in the monitor by the following formula.

$$\text{Actual flow rate of sample (L/min)} = \text{Sample flow rate (L/min)} \times \frac{273 + \text{Temperature (°C)}}{273 + 20 \text{ (°C)}} \times \frac{101.3 \text{ kPa}}{\text{Atmosphere pressure (kPa)}}$$

- (d) The actual flow rate is calculated from standard flow rate (measurement value of flow rate sensor), outdoor air temperature, and atmosphere pressure. To carry out the measurement accurately, it is necessary to calibrate the peripheral pressure and outdoor air temperature periodically.
- >> 5.4(5) “Calibration of temperature, humidity, and pressure”

① Prepare the device.

Standard flowmeter It is calibrated in the defined method and can display the standard flow rate

Blade hose φ15 × φ 22, length to which standard flowmeter can be connected

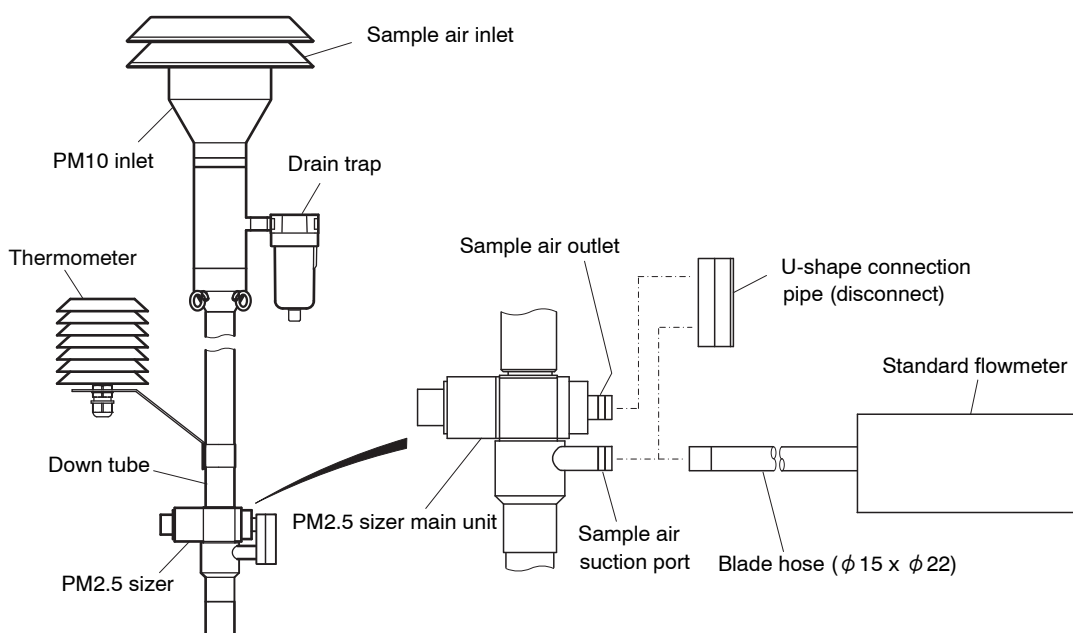
Name of product: Techno Blade Hose TB-15 15 × 22 Code No. 116B050

- ② **Warm up the standard flowmeter.** Carry out the warm-up operation sufficiently by referring to the instruction manual of standard flowmeter to be used.

- ③ **Disconnect U-shape connection pipe** Pull out the U-shape connection pipe from PM2.5 sizer unit.

- 【IMPORTANT】** • Carry out disconnection and connection carefully not to damage the O-ring.

- ④ **Connect the blade hose** Connect the blade hose with length to which standard flowmeter can be connected to the sample air suction port of PM2.5 sizer unit.
- ⑤ **Connect the standard flowmeter** Connect the standard flowmeter to the other end of blade hose.



Connection of Standard Flowmeter

- ⑥ **Check “Sample flow rate”**. Press DATA on “Home” screen and check that the value of “Sample flow rate” on the displayed “Data Display Menu” screen (1/3) is within $\pm 2\%$ of indication of standard flowmeter.

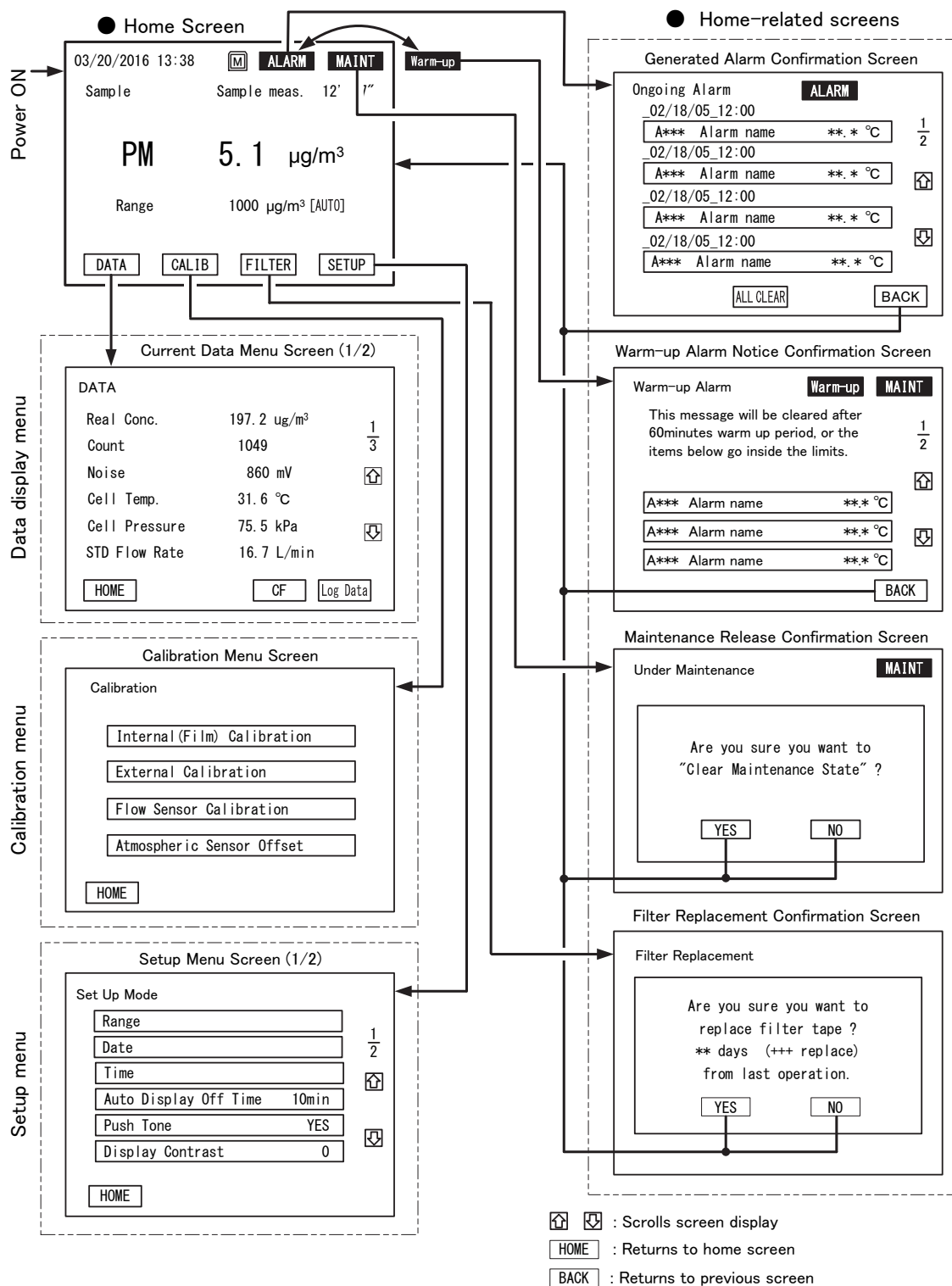
-
- 【IMPORTANT】**
- Carry out the confirmation after the value of “Sample flow rate” is sufficiently stabilized.
 - “Actual flow rate of sample” on “Data Display Menu” screen (2/3) is not the standard flow rate. Do not misunderstand it.
-

- ⑦ **Return to “Home screen”** Press HOME on “Data Display Menu” screen (1/3).
- If the value of “Sample flow rate” is within “⑥ Check “Sample flow rate””, it is not necessary to calibrate the flow rate. Go to the step “⑨ Disconnect the standard flowmeter”.
- ⑧ **Execute the flow rate calibration** On the “Flow Rate Calibration” screen, carry out the flow rate calibration. >> 5.4(4) “Screen operation for flow rate calibration”
- ⑨ **Disconnect the standard flowmeter** Disconnect the blade hose from the sample air suction port of PM2.5 sizer unit and connect U-shape connection pipe in the original state.

5. Modes of Operations

5.1 Screen Map

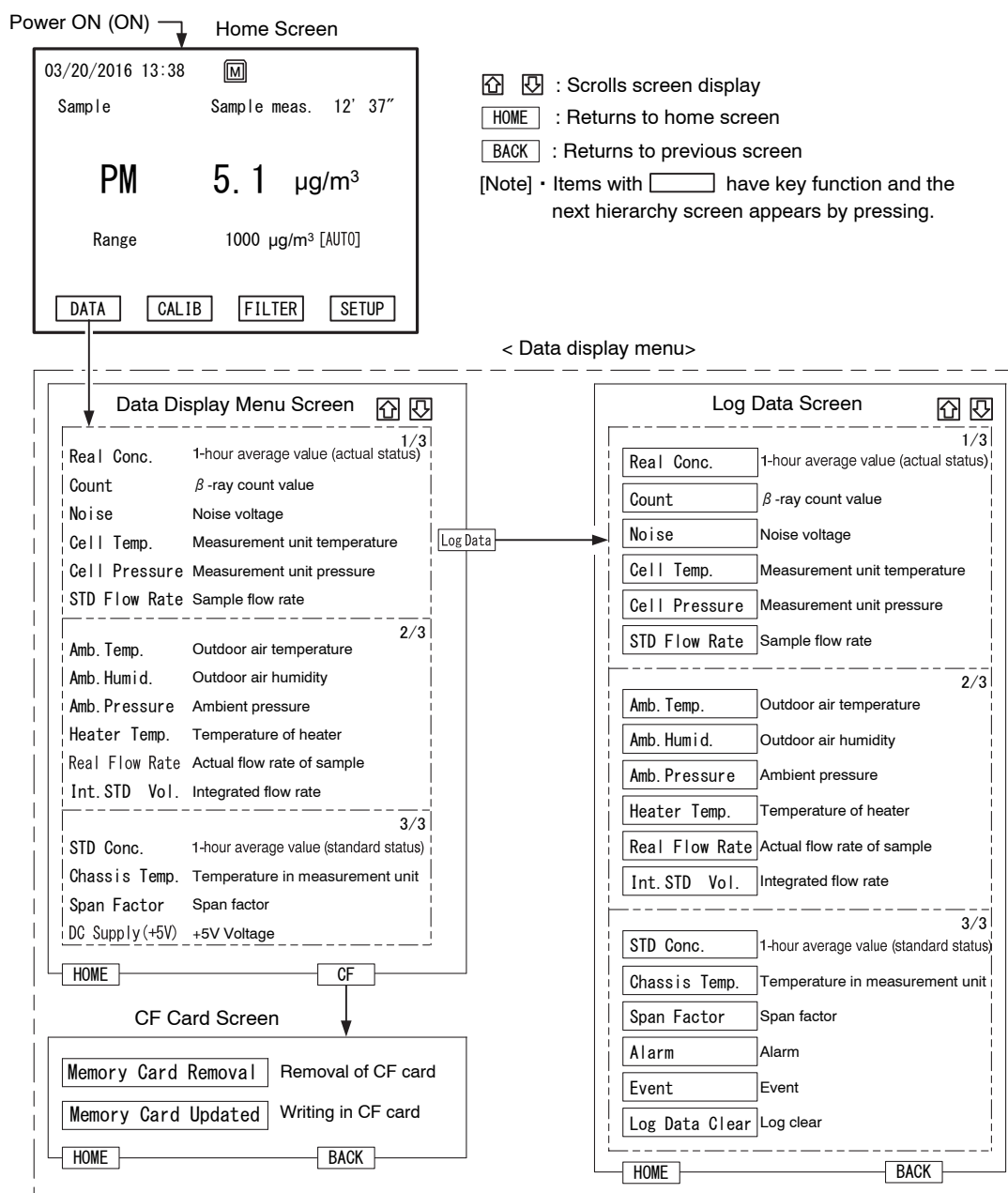
(1) Map of home screen and home-related screens



Map of Home Screen and Home-related Screens

- (a) If electric power is supplied to the monitor, the basic screen “Home” screen appears in a few seconds. If **DATA**, **CALIB**, or **SETUP** key is pressed on “Home” screen, you can enter each menu and its top screen appears.
- (b) For the procedure for operating “Home” screen and “Home-related” screens, refer to 5.2 “Operation of Home Screen and Home-related screens”.

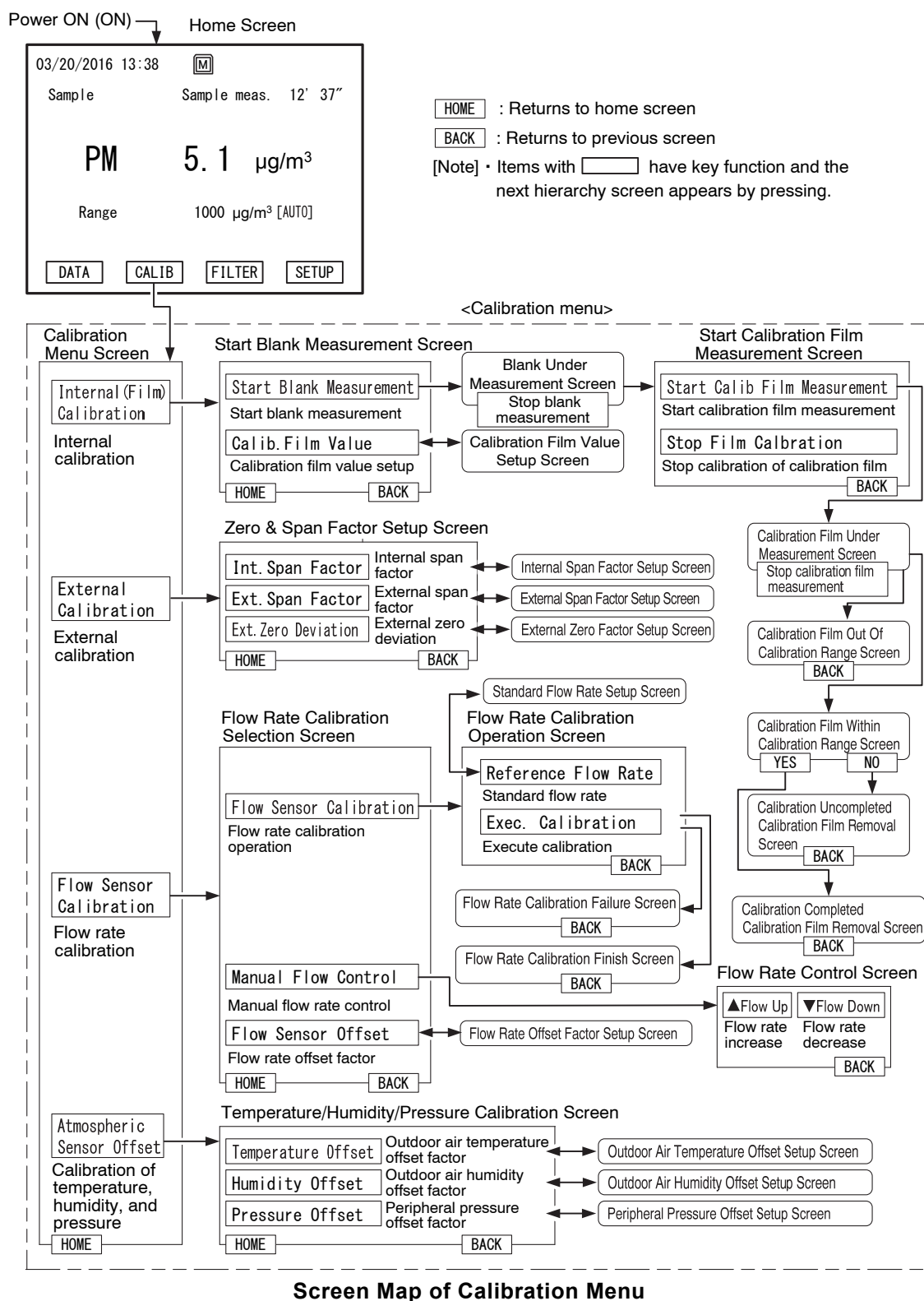
(2) Screen map of data display menu



Screen Map of Data Display Menu

- (a) If **DATA** is pressed on “Home” screen, “Data Display Menu” screen (1/3) appears.
- (b) For each procedure for operating “Data Display Menu” screen, refer to 5.3 “Operation of Data Display Menu”.

(3) Screen map of calibration menu

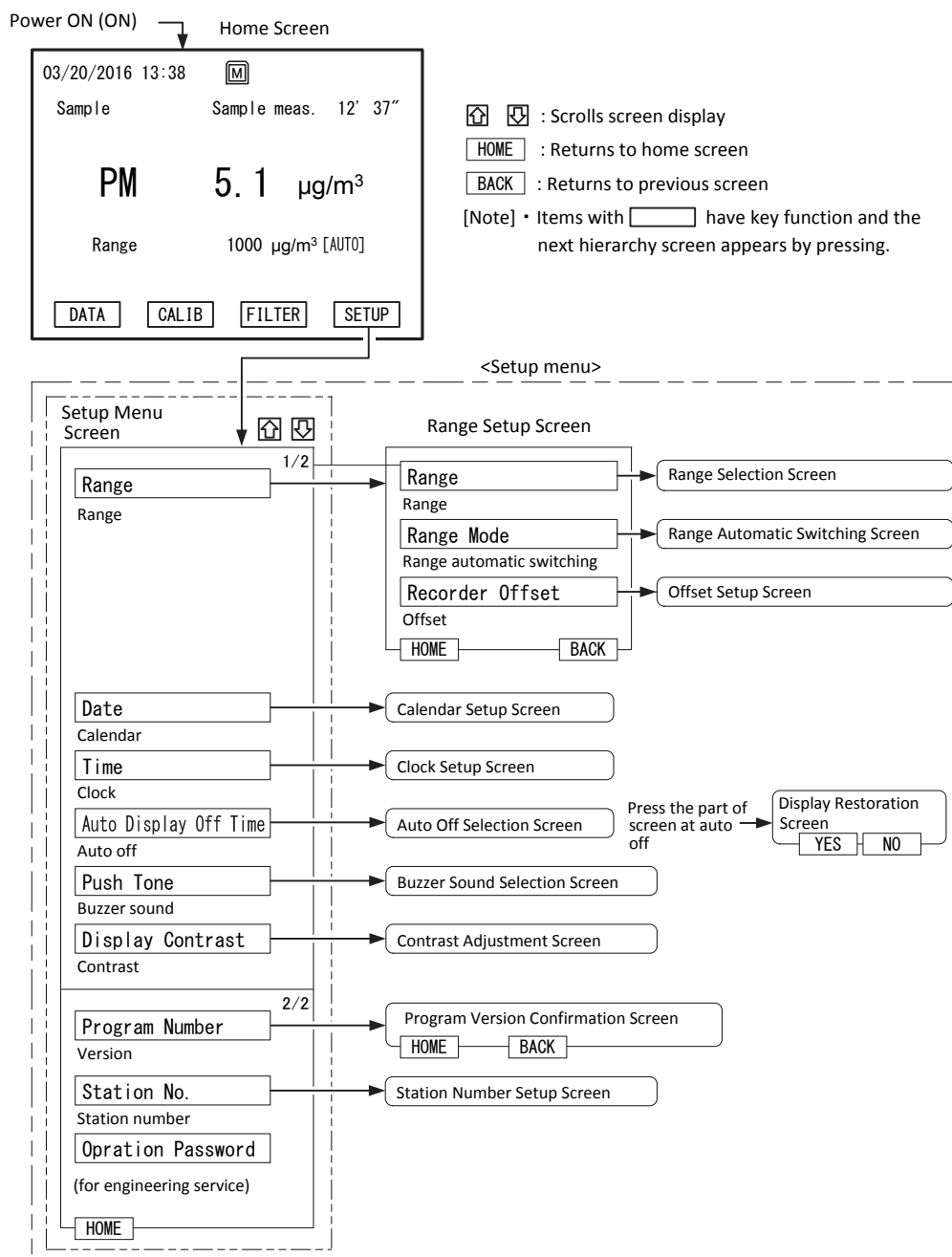


(a) If [CALIB] is pressed on "Home" screen, "Calibration Menu" screen appears.

(b) For each procedure of operating "Calibration Menu" screen, refer to 5.4 "Operation of Calibration Menu".

- (c) For overview of calibration, calibration of peripheral pressure, outdoor air temperature, and outdoor air humidity, refer to each item separately. >> 4.1 “Functions Related to Calibration”

(4) Screen map of setup menu



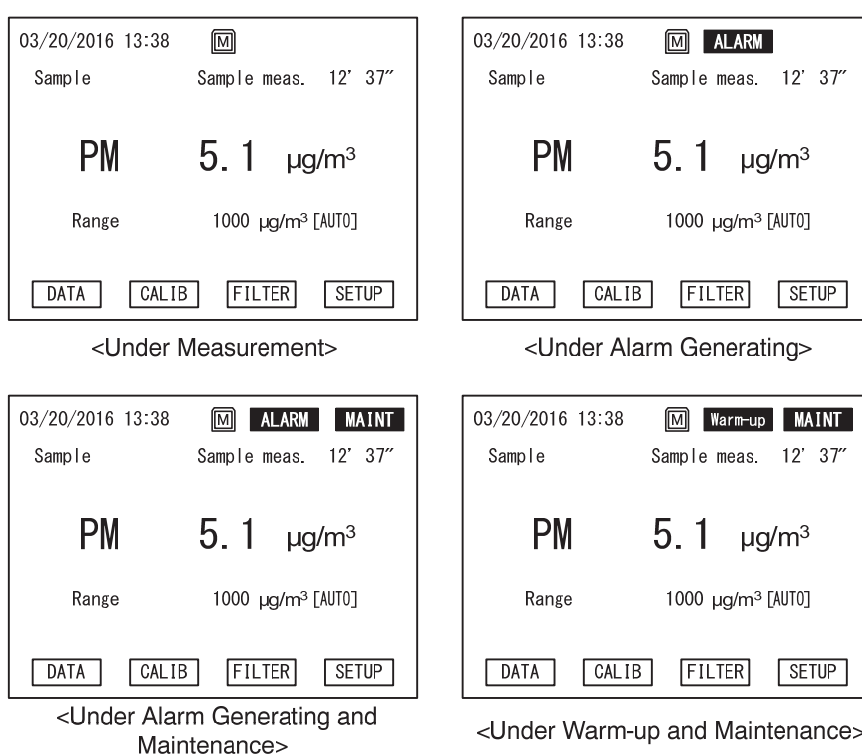
Screen Map of Setup Menu

- (a) If [SETUP] is pressed on “Home” screen, “Setup Menu” screen appears.
- (b) For each procedure for operating “Setup Menu” screen, refer to 5.5 “Operation of Setup Menu”.
- (c) The setting of each item can be changed on the popup screen that is displayed on the right side of the screen. If the popup screen is closed, the former screen can be operated.

5.2 Operation of Home Screen and Home-related screens

(1) Display and keys of home screen

- (a) On “Home” screen, you can check current date and time, measurement value of last 1-hour average value and average value range you set, etc.
- (b) If an alarm is generated, **ALARM** is displayed. If the monitor enters maintenance status, **MAINT** is displayed.
- (c) If the power is turned on, the monitor enters warm-up operation status until the temperature in the monitor, etc. enters normal range and **Warm-up** and **MAINT** are displayed at the same time (for max. 60 minutes).
- (d) If a CF card is inserted, **M** is displayed.
- (e) **ALARM**, **Warm-up**, and **M** are displayed on screens other than “Home” screen. The operation method is common.



Example of Display of Home Screen

- (f) Keys on “Home” screen If **DATA** key, etc. is pressed (touched) in the lower part of screen, the top screen of the corresponding menu screen or filter replacement confirmation screen is displayed. **ALARM** in the upper part of the screen is equipped with a function for opening the relevant screen in addition to display function.

DATA Displays “Data Display Menu” screen (1/2).

CALIB Displays “Calibration Menu” screen.

SETUP Displays “Setup Menu” screen (1/2).

FILTER Displays “Filter Paper Replacement Confirmation” screen related to home screen.

ALARM Displays “Generated Alarm Confirmation” screen related to home screen.

Warm-up Displays “Warm-up Alarm Notice Confirmation” screen related to home screen.

MAINT Displays “Maintenance Release Confirmation” screen related to home screen.

M Displays “CF Card Removal Execution” screen.

(2) Confirmation and reset of alarm information

(a) If an alarm is generated, **ALARM** is displayed.

(b) The alarms are classified into alarm 1 and alarm 2 depending on the importance.

- Alarm 1 (AL-1) This is a defect of monitor that does not affect the measurement. The measurement is continued without any changes. If the cause is eliminated and the operation returns to normal, the alarm is automatically reset. This status can be also reset by operating a key. However, if the cause is not eliminated, the alarm is generated again even if it is reset. If the alarm 1 and alarm 2 are generated at the same time, the alarm cannot be reset.

[IMPORTANT] • Three alarm symbols shown below are classified into alarm 2 (AL-2). If the alarm is generated, the pump stops and the measurement also stops, however, the alarm is automatically reset on the hour (00 minute).

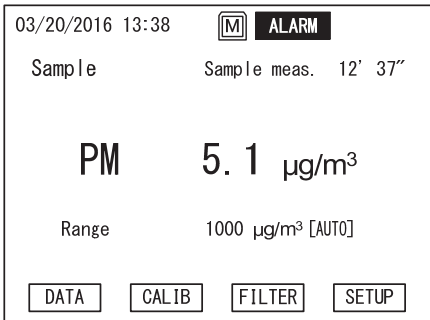
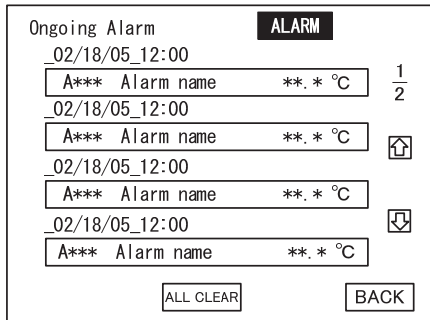
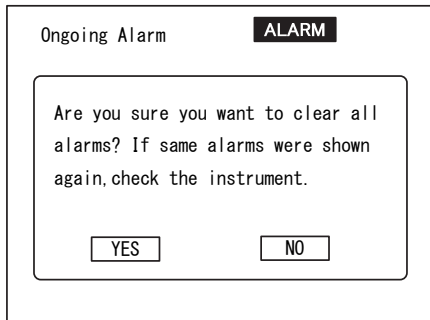
A734	pump startup	A740	filter paper shortage
A741	clogging of filter paper		

- Alarm 2 (AL-2) This is a defect of monitor with which generation of a problem that will affect the measurement is forecasted. The monitors such as heater and pump stop and the measurement is immediately stopped. The alarm can be reset by eliminating the cause and turning off and on the power switch, etc.

(c) This **ALARM** is equipped with key function in addition to display function. If this key is pressed, “Generated Alarm” screen is displayed. On “Generated Alarm” screen, the content of alarm can also be checked by opening “Help” screen for each item.

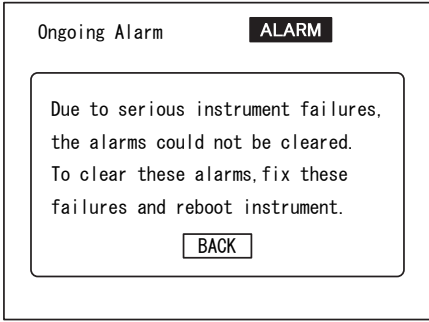
(d) For the measures against each alarm, refer to 7.1 “Alarm Symbols and Measures”.

Procedure for Checking and Resetting the Alarm Information

Procedure and operation	Example of screen
① Check the alarm display Check there is ALARM in the upper right of “Home” screen. ② Display “Generated Alarm” screen Press ALARM on “Home” screen.	
 Home Screen (under alarm generating)	 Generated Alarm Screen
③ Check the alarm information Check the alarm information (date and time of generation, alarm classification, and alarm name) on “Generated Alarm” screen. , : Switches pages of screen (when 5 items or more are displayed) ALL CLEAR: Displays the screen for resetting the alarm display. BACK: Returns to the previous screen. ④ Display “Alarm 1 Reset” screen, etc. To reset the alarm display, press ALL CLEAR on “Generated Alarm” screen. • When only alarm 1 is generated: “Alarm 1 Reset” screen is displayed. • When alarm 2 (or alarms 1 and 2) is generated: “Alarm 2” screen is displayed. ⑤ Check alarm 1 reset or alarm 2 generation Carry out any of following operations depending on the displayed screen.	
(1) If “Alarm 1Reset” screen is displayed ① Check the description on “Alarm 1 Reset” screen. ② Press YES. • Alarm 1 is completely reset and “Generated Alarm” screen is displayed. NO: Returns to the previous screen without resetting the alarm.	 Alarm 1 Reset Screen

(To be continued)

(Continued from previous page)

Procedure and operation	Example of screen
(2) If “Alarm 2” screen is displayed ① Check the description on “Alarm 2” screen. ② Press BACK. Since the alarm 2 is a serious alarm, it is not reset and the simultaneously generated alarm 1 is not reset and “Generated Alarm” screen is displayed.	 Alarm 2 Screen
⑥ Return to “Home” screen Press BACK on “Generated Alarm” screen. ⑦ Respond to the alarm Eliminate the cause of alarm. - When the inside of monitor is touched, turn off the power for safety, and then eliminate the cause of alarm. - In case of alarm 2, eliminate the cause of alarm and turn on the power (reboot) again to the monitor.	

(3) Confirmation of warm-up alarm

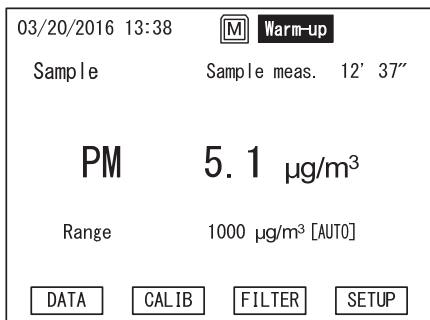
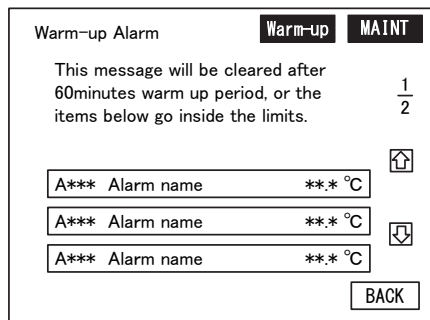
- (a) If the power is turned on, the monitor executes warm-up operation until the flow rate enters the normal range. At this time, the monitor automatically enters maintenance status and **Warm-up** and **MAINT** are displayed in the upper right of “Home” screen, etc.
- (b) If this **Warm-up** is pressed, the screen is switched to “Warm-up Alarm” screen and alarm classification that does not meet the warm-up conditions, alarm name, and current measured value are displayed.
- (c) The content of alarm can be checked by opening “Help” screen for each item from “Warm-up Alarm” screen.
- (d) The warm-up operation is continued for 60 minutes at a maximum. Usually, if the monitor meets the warm-up conditions, the warm-up operation is automatically terminated and the monitor enters the measurement status, and then **Warm-up** and **MAINT** are hidden. If the monitor does not meet the conditions in 60 minutes, the warm-up operation is forcibly terminated and **Warm-up** is hidden. Instead of it, an alarm is generated and **ALARM** is displayed.
- (e) The maintenance signal that is output during warm-up operation is automatically reset by termination of warm-up operation. It can be also reset forcibly. >> 5.2(4) “Reset of maintenance display”

Procedure for Checking the Warm-up Alarm

Procedure and operation	Example of screen
① Check the warm-up display Check there are Warm-up and MAINT in the upper right of “Home” screen.	

(To be continued)

(Continued from previous page)

Procedure and operation	Example of screen
② Display “Warm-up Alarm” screen Press Warm-up on “Home” screen.	
 03/20/2016 13:38 M Warm-up Sample Sample meas. 12' 37" PM 5.1 $\mu\text{g}/\text{m}^3$ Range 1000 $\mu\text{g}/\text{m}^3$ [AUTO] DATA CALIB FILTER SETUP	 Warm-up Alarm Warm-up MAINT This message will be cleared after 60minutes warm up period, or the items below go inside the limits. $\frac{1}{2}$ A*** Alarm name *** °C A*** Alarm name *** °C A*** Alarm name *** °C BACK
Home Screen (during warm-up operation)	Warm-up Alarm Screen
③ Check the alarm information during warm-up operation Check alarm classification and alarm name on “Warm-up Alarm” screen. , : Switches pages of screen (when 6 items or more are displayed) BACK: Returns to the previous screen.	
④ Return to “Home” screen Press BACK on “Warm-up Alarm” screen.	
⑤ Respond to the alarm Eliminate the cause of alarm.	

(4) Reset of maintenance display

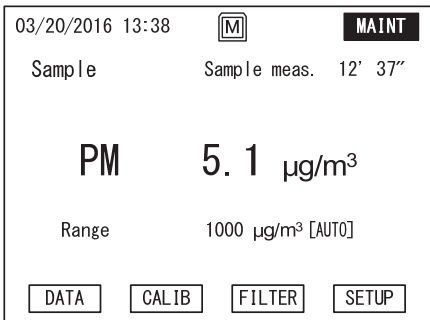
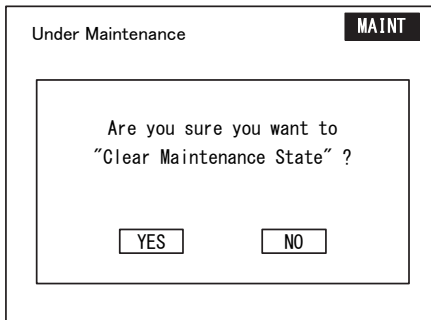
If this monitor is under maintenance, **MAINT** (maintenance display key) is displayed in the upper right of “Home” screen, etc. If **MAINT** is pressed, “Maintenance Release Confirmation” screen is displayed and the maintenance display can be reset. It is also interlocked with maintenance signal output of external output (option).

Procedure for Cancelling Maintenance Display

Procedure and operation	Example of screen
① Check the under maintenance display Check there is MAINT in the upper right of “Home” screen.	

(To be continued)

(Continued from previous page)

Procedure and operation	Example of screen
② Display “Maintenance Release Confirmation” screen Press MAINT on “Home” screen.	
 Home Screen (under maintenance)	 Maintenance Release Confirmation Screen

③ **Cancel under maintenance display.** Press **YES** on “Maintenance Release Confirmation” screen.

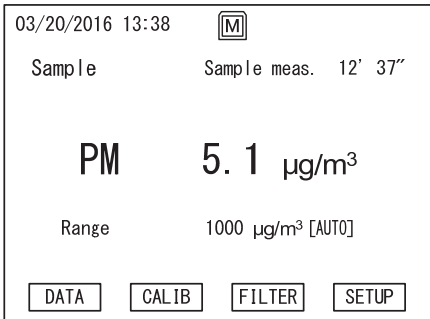
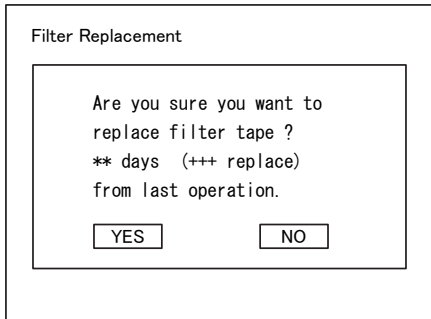
- The maintenance display is cancelled and **MAINT** is hidden, and then “Home” screen is displayed.

NO: The display is not cancelled and the screen returns to the previous screen.

(5) Screen operation for filter replacement

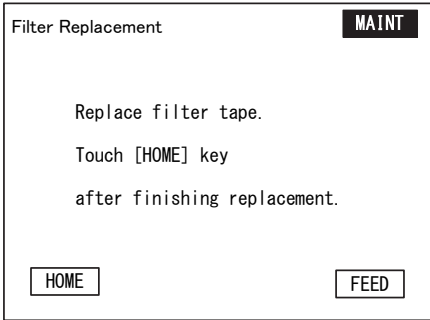
- (a) Here explains the screen operation for filter replacement. For the actual replacement work, refer to 6.7 “Replacement of Filter” because it is accompanied with removal of reel and other works.
- (b) If “Filter Replacement” screen is displayed, the replacement of filter is enabled. The pump stops and the lower measurement unit is at lower side.

Screen Function for Filter Replacement

Procedure and operation	Example of screen
① Display “Filter Replacement Confirmation” screen Press FILTER on “Home” screen.	
 Home Screen (under measurement)	 Filter Replacement Confirmation Screen

(To be continued)

(Continued from previous page)

Procedure and operation	Example of screen
② Display “Filter Replacement” screen Press YES on “Filter Replacement Confirmation” screen. - The pump stops and the lower measurement unit moves downward and “Filter Replacement” screen is displayed. NO : Filter replacement is cancelled and the screen returns to the previous screen. ③ Carry out the replacement Replace the filter with a new one on “Filter Replacement” screen. - If it is not necessary to feed one spot of filter after the filter is replaced with a new one, go to step “⑤”. ④ Feed one spot of filter Press FEED on “Filter Replacement” screen. ⑤ Return to “Home” screen Press HOME on “Filter Replacement” screen.	 Filter Replacement Screen

5.3 Operation of Data Display Menu

(1) Essential point of data display menu

- (a) With the data display menu, the current data and log data for each item listed in the table shown below can be checked (1-hour average value, accuracy management information, and event data).
- (b) Press **DATA** on “Home” screen. The current data is displayed on the displayed “Data Display Menu” screen (1/3 to 3/3). If **Log Data** is pressed on “Data Display Menu” screen (1/3 to 3/3), “Log Data” screen is displayed.
- (c) If **CF** is pressed on “Data Display Menu” screen (1/2 to 2/2), “CF Card” screen is displayed and the operation such as “Removal of CF card” is enabled. If a CF card is not inserted, the message “There is No CF card” is displayed.
- (d) The method for displaying the items to be displayed on each screen of data display menu and the log data count are shown in the table below.
- (e) The item “DC voltage (5V)” exists only in the current data. Items of alarm, event, and log clear exist only in the log data.

Items to be Displayed on “Data Display Menu” Screen

Data item			Content	Usual display value	Data display menu		
					Data display	Log data	Number of log data
Real Conc. 1-hour average value			1-hour average value of PM2.5 concentration (mass concentration in actual status)	—	○	○	75-day data
Accuracy management information	Count	beta ray count value	β-ray intensity counted by detector	8000 or more	○	○	*1
	Noise	Noise voltage	Noise voltage of preamplifier	800mV or less	○	○	75-day data
	Cell Temp	Temperature of measurement unit	Temperature at PM2.5 collection part (filter paper)	55°C or less	○	○	*1
	Cell Pressure	Pressure of measurement unit	Pressure at PM2.5 collection part (filter paper)	55 to 90kPa	○	○	*1
	STD Flow Rate	Sample flow rate	Converted flow rate at 20°C, 1 atmosphere (101.3kPa)	15 to 19L/min	○	○	*1
	Amb. Temp.	Outdoor air temperature	Temperature of sample air	-10 to 40 °C	○	○	*1
	Amb. Humid.	Outdoor air humidity	Humidity of sample air	0 to 100%	○	○	*1
	Amb. Pressure	Ambient pressure	Pressure of sample air	90 to 110kpa	○	○	*1
	Heater Temp	Temperature of heater	1-hour average temperature of dehumidifier (heater)	70 °C or less	○	○	*1
	Real Flow Rate	Actual flow rate of sample	Volume flow at temperature and atmosphere pressure at that time	16.7±0.3 L/min	○	○	*1
	Int. STD Vol.	Integrated flow rate	Integrated value of actual flow rate of sample at that time	0 to1000L	○	○	*1
	STD Conc.	1-hour average value	1-hour average value of PM2.5 concentration (mass concentration in standard status)	—	○	○	75-day data
	Chassis Temp.	Temperature in measurement unit	Temperature in chassis	45°C or less	○	○	*1

(To be continued)

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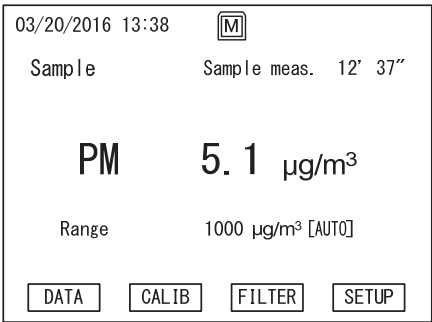
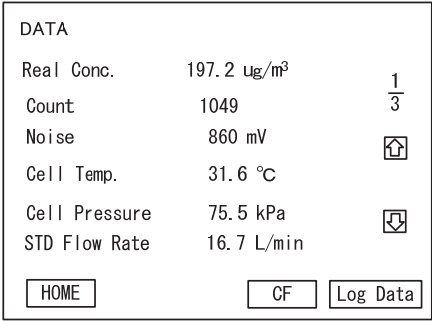
Data item		Content	Usual display value	Data display menu			
				Data display	Log data	Number of log data	
Event data	Span Factor	Span factor	Latest span factor	Normally 1.0 to 1.3	○	○	100 times data
	DC Supply (+5V)	DC voltage (+5V)	Status of supply to internal power supply (there is no log data)	OK	○	×	—
	Alarm	Alarm	Log data of alarm (only log data)	—	×	○	100 times data
	Event	Event	Log data of event (internal calibration, power ON/OFF) (only log data)	—	×	○	100 times data
	Log Data Clear	Log clear	All clearance of log data (only log data)	—	×	○	—

○: Displayed item ×: Hidden item *1: 1-hour value 10-day data, 1-day value 1095-day data

(2) Confirmation of current data

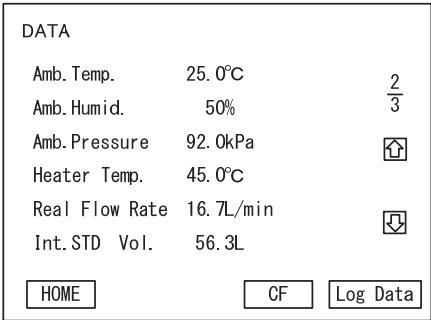
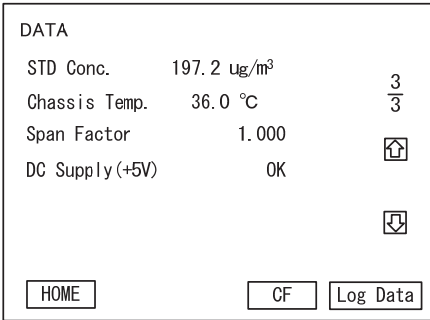
On “Data Display Menu” screen, various types of current data can be checked.

Procedure for Checking the Current Data

Procedure and operation	Example of screen
① Display “Data Display Menu” screen Press DATA on “Home” screen.	 Home Screen (under measurement)
② Check the current data Check various types of current data on “Data Display Menu” screen (1/3 to 3/3). ⏮, ⏭: Switches pages of screen. Log Data: “Log Data” screen is displayed. >> 5.3(3) “Confirmation and clearance of log data”	 Data Display Menu Screen (1/3)

(To be continued)

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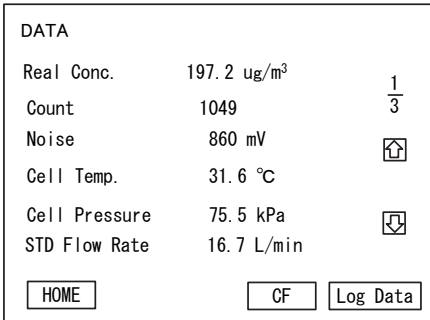
Procedure and operation	Example of screen
 Data Display Menu Screen (2/3)	 Data Display Menu Screen (3/3)

③ **Return to “Home” screen** Press HOME on “Data Display Menu” screen.

(3) Confirmation and clearance of log data

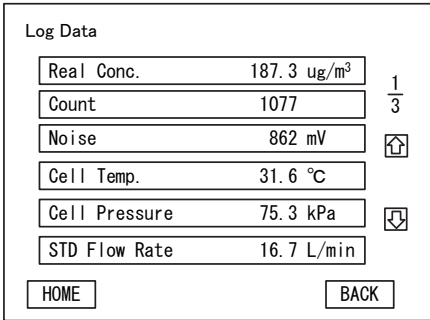
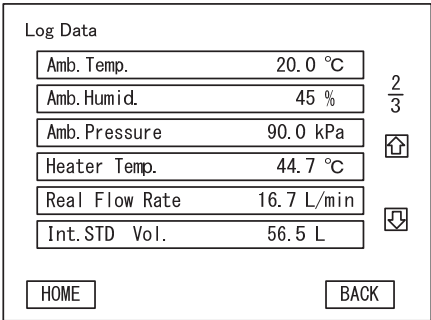
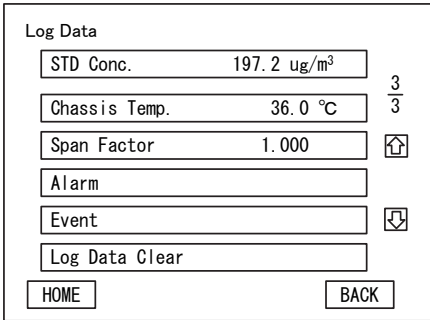
- (a) The log data of 1-hour average value, etc. can be checked on “Log Data” screen or other screen.
- (b) Items and latest values of log data can be checked on “Log Data” screen. If a key of each item is pressed on this screen, an individual log data screen is displayed. The item name in the table is described in [].
- (c) All log data can be cleared by [Log clear].

Procedure for Checking and Deleting the Log Data

Procedure and operation	Example of screen
① Display “Data Display Menu” screen Press DATA on “Home” screen. ② Display “Log Data” screen Press Log Data on “Data Display Menu” screen (1/3). ↑, ↓: Switches pages of screen.	 Data Display Menu Screen (1/3)

(To be continued)

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Procedure and operation	Example of screen
③ Check or delete Select necessary items and check or delete them. ⏮, ⏭: Switches pages of screen. ⏮: Returns to the previous screen.	 Log Data Screen (1/3)
 Log Data Screen (2/3)	 Log Data Screen (3/3)

(1) To check log data

- ① Press the section within frame of log data item to be checked on “Log Data” screen (1/3 to 3/3).
 - “Individual Log Data” screen is displayed.
- ② Check the log data on “Individual Log Data” screen.

⏮, ⏭: Switches pages of screen.

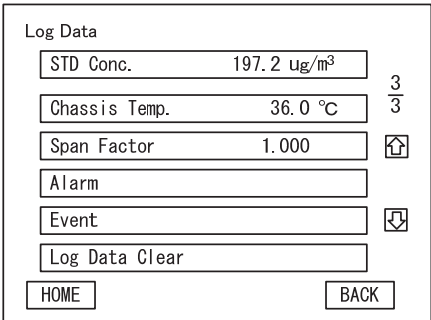
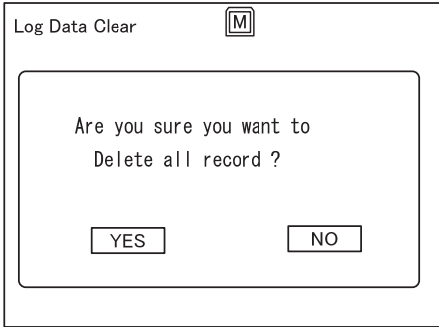
⏮: Returns to the previous screen.

16/03/20 12:00	0. 0254	μg/m³	ADJ	1
16/03/20 12:00	0. 0209	μg/m³	PON	2
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	
16/03/20 12:00	-----	μg/m³	NON	

HOME

BACK

(Continued from previous page)

Procedure and operation	Example of screen
 Log Data Screen (3/3)	 Log Clear Execution Screen

- ② Press on “Log Clear Execution” screen.
- All log data is cleared and “Log Data” screen (2/2) is displayed.
- : Returns to the previous screen without clearing the log data.

④ **Return to “Home” screen** Press on “Log Data” screen or “Individual Log Data” screen.

(4) Removal of CF card

- (a) To pull out a CF card, be sure to carry out the operation shown in this “Removal of CF Card”. The access to data is cancelled and the removal of card is enabled.

【IMPORTANT】 • If you pull out a CF card forcibly without carrying out “Removal of CF Card”, the data in the CF card may not be read properly.

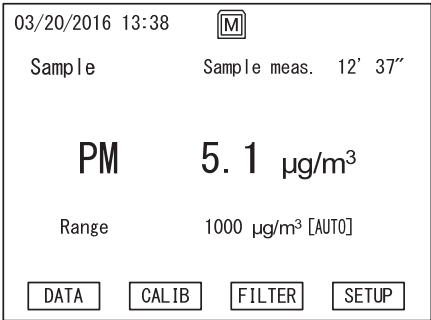
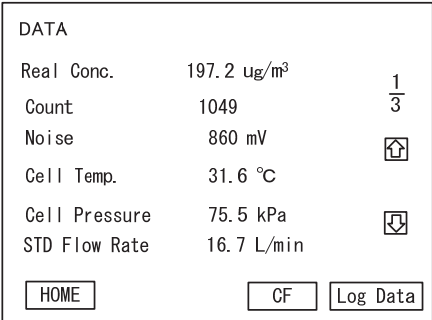
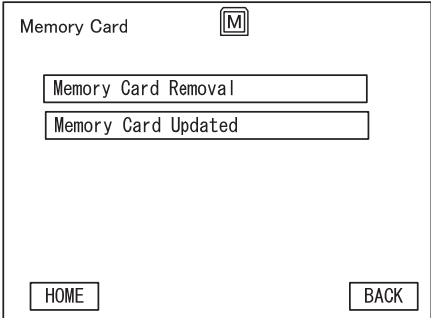
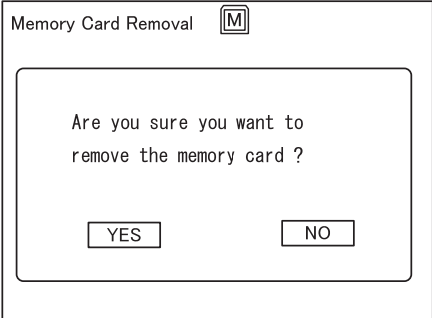
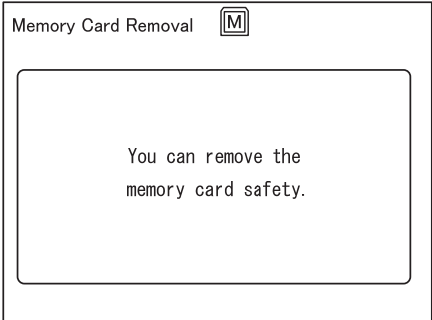
- (b) For the method for collecting data with CF card and the content of data, refer to 3.2 “Data in CF Card”.
- (c) If is pressed on “Home” screen or other screen, “③ CF Card Removal Execution” screen shown in Table “Procedure for Removing CF Card” is directly displayed.

Procedure for Removing CF Card

Procedure and operation	Example of screen
① Display “Data Display Menu” screen (1/3) Check M is displayed on “Home” screen and press DATA .	

(To be continued)

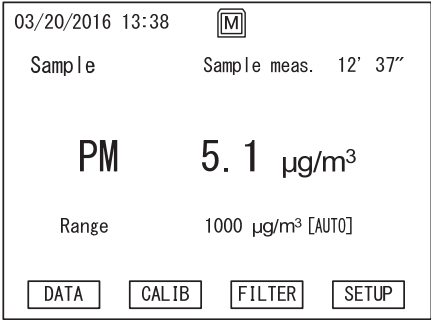
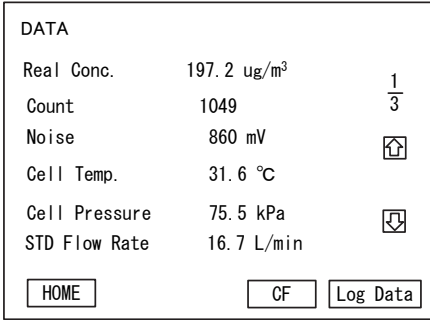
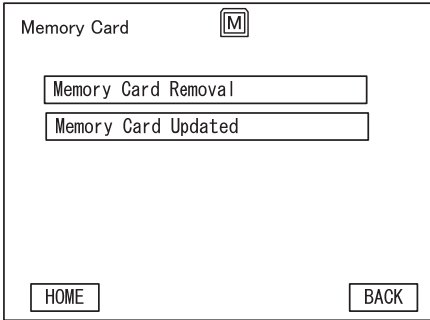
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Procedure and operation	Example of screen
 Home Screen	 Data Display Menu Screen (1/3)
② Display “CF Card” screen Press CF on “Data Display Menu” screen.	
③ Display “CF Card Removal Execution” screen Press [Memory Card Removal] on “CF Card” screen.	
 CF Card Screen	 CF Card Removal Execution Screen
④ Display “CF Card Removal Confirmations” screen, etc. Press YES on “CF Card Removal Execution” screen. NO : Returns to “CF Card” screen.	
⑤ Remove CF card Pull out a CF card from CF card insertion slot on “CF Card Removal Confirmation” screen. If a CF card is not pulled out for 10 seconds or more, the alarm buzzer sounds.	
	 CF Card Removal Confirmations Screen
⑥ Check M is hidden Check M is not displayed on “CF Card” screen.	
⑦ Return to “Home” screen Press HOME on “CF Card” screen.	

(5) Backup by writing in CF card

- (a) The data created by this operation is saved in the backup folder (back up) of CF card. On the other hand, the data that is automatically written by just inserting a CF card is saved in the data folder (data). >> 3.2(2) “CF card folder”
- (b) With this operation, all data saved by CPU are collectively written in the backup folder of CF card.
- (c) If this operation is executed, the data that was backed up previously is overwritten. If the previous data is necessary, save it in a PC, and then carry out this time’s backup.

Procedure for Backing up the Data by Writing in CF Card

Procedure and operation	Example of screen
① Insert a CF card Insert a CF card on “Home” screen or other screen. If  is blinking, wait until it switches to lighting. • If a CF card has been inserted, check  is not blinking. ② Display “Data Menu” screen (1/3) Check  is displayed on “Home” screen and press .	 
③ Display “CF Card” screen Press  on “Data Display Menu” screen. ④ Display “Backup Update Execution” screen Press [Memory Card Updated] on “CF Card” screen.	
⑤ Start update Press  on “Backup Update Execution” screen. • “Under CF Card Update” screen is displayed. : Returns to “CF Card” screen.	

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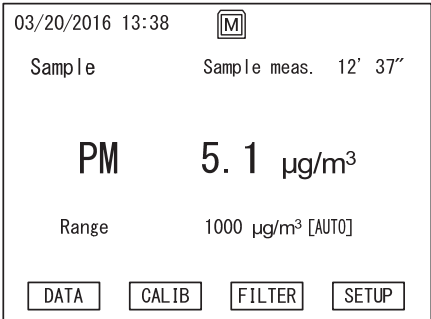
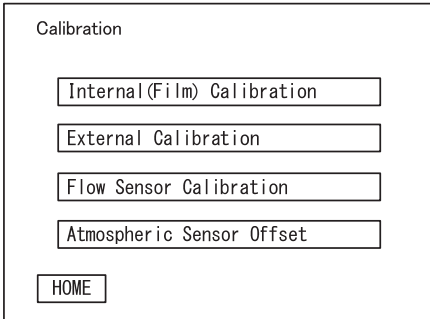
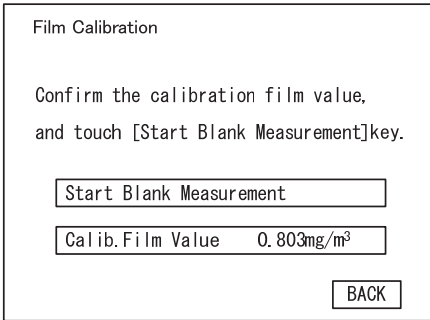
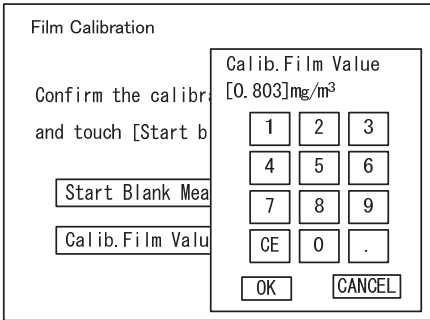
Procedure and operation	Example of screen
Backup Update Execution Screen	Under CF Card Update Screen
⑥ Respond to the displayed screen Carry out any of following operations depending on the displayed screen.	
(1) If “CF Card Updated Finish” screen is displayed ① Press BACK on “CF Card Updated Finish” screen. ② Check “CF Card” screen is displayed.	CF Card Updated Finish Screen.
(2) If “CF Card Update Error” screen is displayed ① Press BACK on “CF Card Update Error” screen. • “CF Card” screen is displayed. ② Carry out steps ④ to ⑥ on “CF Card” screen again. ③ If the error screen is displayed again even after the operation “②”, press BACK to return to “CF Card” screen and carry out 5.3 (4) “Removal of CF card” and initialize the CF card, and then carry out the operation again. ④ If the error screen is displayed again even after the operation “③”, the CF card may be damaged or the main indicator may be broken. Please contact the repair service (distributor).	CF Card Update Error Screen (*: 1 to 5)
⑦ Remove the CF card When you pull out the CF card, refer to “③ Display “CF Card Removal Execution” screen” or later in 5.3 (4) “Removal of CF card”. • When the CF card is not removed, press HOME on “CF Card” screen.	

5.4 Operation of Calibration Menu

(1) Check or change the calibration film value

- (a) You can check whether the calibration film value memorized in the monitor corresponds to the calibration film value described on the calibration film. For the overview of calibration, refer to 4.1 “Functions Related to Calibration”.
- (b) The calibration film value memorized by monitor can be checked on “Blank Measurement Start” screen. If it is different from the value described on the calibration film, change it so that they become correspondent.

Procedure for Checking or Changing the Calibration Film Value

Procedure and operation	Example of screen
① Display “Calibration Menu” screen Press [CALIB] on “Home” screen.	 Home Screen (under measurement)
② Display “Blank Measurement Start” screen Press [Internal (Film) Calibration] on “Calibration Menu” screen.	 Calibration Menu Screen
③ Check the calibration film value On “Blank Measurement Start” screen, check the value of [Calib.Film Value] corresponds to the value described on the calibration film. • If the values are correspondent, go to step “⑦”. [BACK]: Returns to the previous screen.	 Blank Measurement Start Screen
④ Display “Calibration Film Value Setup” screen Press [Calib.Film Value] on “Blank Measurement Start” screen.	 Calibration Film Value Setup Screen
⑤ Input a new calibration film value On “Calibration Film Value Setup” screen, input the described value using a numeric keypad. • Setting range: 0.001 to 20.000 mg/m³ [CE]: Blinking input value is reset to the original value. [CANCEL]: Returns to the previous screen without changing the value.	
⑥ Fix the calibration film value after change Press [OK] on “Calibration Film Value Setup” screen. • “Blank Measurement Start” screen is displayed	

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Procedure and operation	Example of screen
⑦ Return to “Home” screen Press BACK on “Blank Measurement Start” screen and press HOME on the displayed “Calibration Menu” screen.	

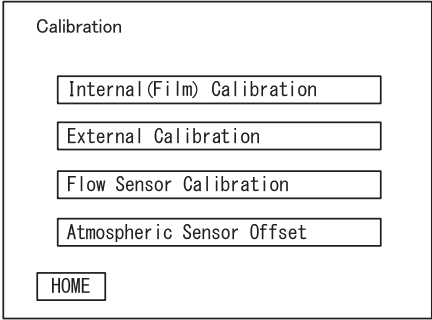
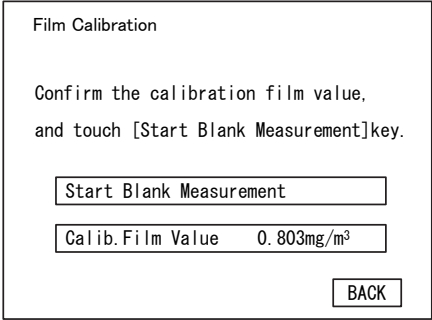
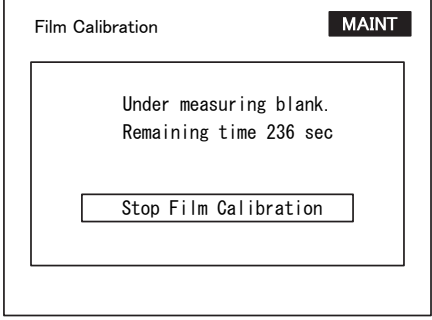
(2) Screen operation for internal calibration

- (a) You can promote the procedure for calibrating the calibration film on the screen related to calibration film calibration such as “Blank Measurement Start” screen. Here explains the functions of the relevant screens. For the operation including the insertion of calibration film, etc., refer to 4.2 “Internal Calibration”.
- (b) If the calibration of calibration film is started (blank measurement is started) during the measurement, the measurement is immediately stopped and the calibration of calibration film is started.
- (c) The calibration of calibration film is carried out following the sequence shown in the table below and the screens are switched depending on the process. For the process No. 6 “Calibration film measurement start”, it is necessary to insert a calibration film and press the key. For the process No. 10 “Wait for calibration / confirmation”, it is necessary to evaluate whether the span factor should be rewritten and operate the key to pull out the calibration film.
- (d) Usually, calibration film measurement is carried out three times and three span factors are checked, and then whether the span factor should be rewritten is evaluated depending on whether the calibration film measured value is within calibration film value $\pm 2\%$. To repeat the calibration film measurement, press for the process No. 10 “Wait for calibration / confirmation”. > >

Sequence of Calibration of Calibration Film

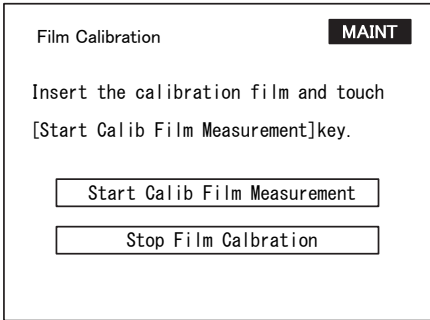
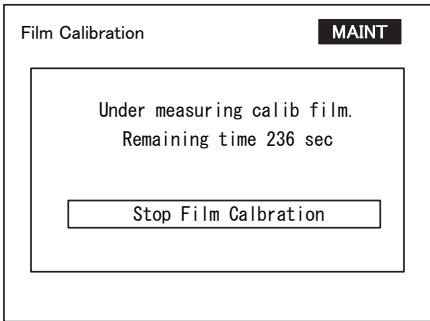
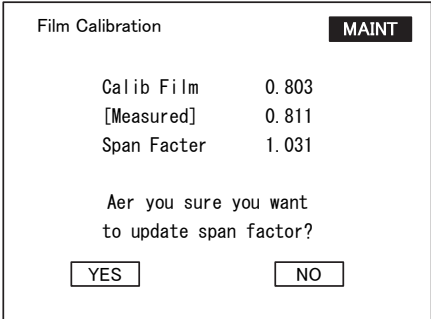
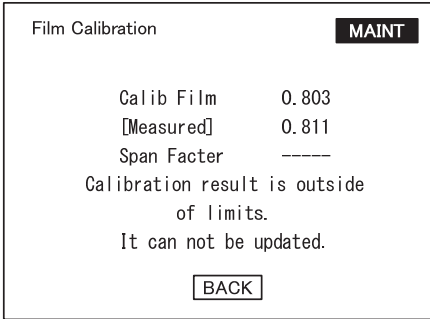
Process No.	Process name	Time (seconds)	Screen and display
	—	—	“Blank Measurement Start” screen
1	Measurement unit descent (open)	5	17 “Measurement Unit Moving Downward” screen “Filter Paper Feeding” screen “Measurement Unit Moving Upward” screen
2	Filter paper feeding	7	
3	Measurement unit ascent (close)	5	
4	Blank measurement	240	“Under Blank Measurement” screen
5	Measurement unit descent (open)	5	“Measurement Unit Moving Downward” screen
6	Calibration film measurement start	—	“Calibration Film Measurement Start” screen • Monitor waits until a calibration film is inserted and the key is pressed. During this operation, the buzzer sounds.
7	Measurement unit ascent (close)	5	“Measurement Unit Moving Upward” screen
8	Calibration film measurement	240	“Calibration Film Measurement” screen
9	Measurement unit descent (open)	5	“Measurement Unit Moving Downward” screen
10	Wait for calibration / confirmation	—	“Calibration Film within Calibration Range” screen or “Calibration Film out of Calibration Range” screen • Monitor wait until the buzzer sound and the key is pressed. “Calibration Completed Calibration Film Removal” screen or “Calibration Uncompleted Calibration Film Removal” screen • Monitor wait until the calibration film is pulled out and the key is pressed.
11	Measurement unit ascent (close)	5	“Measurement Unit Moving Upward” screen
	—	—	“Blank Measurement Start” screen

Procedure for Operating Screens for Internal Calibration

Procedure and operation	Example of screen
① Display “Calibration Menu” screen Press [CALIB] on “Home” screen. ② Display “Blank Measurement Start” screen Press [Internal (Film) Calibration] on “Calibration Menu” screen.	 Calibration Menu Screen
③ Check the calibration film value On “Blank Measurement Start” screen, check the value of [Calib.Film Value] corresponds to the value described on the label of calibration film. • If the value is not correspondent, change it ($> > 5.4(1)$ “Check or change the calibration film value”). [BACK] : Returns to the previous screen. ④ Start the blank measurement Press [Start Blank Measurement] on “Blank Measurement Start” screen. • “Under Blank Measurement” screen is displayed via a screen displaying the status such as “Measurement unit moving downward”.	 Blank Measurement Start Screen
⑤ Wait for the termination of blank measurement On “Under Blank Measurement” screen, wait for the displayed remaining time to become “0”. • If the remaining time becomes “0”, the buzzer sounds and “Calibration Film Measurement Start” screen is displayed. • To stop the blank measurement for some reasons, press [Stop Film Calibration] and carry out the operation following the display.	 Under Blank Measurement Screen

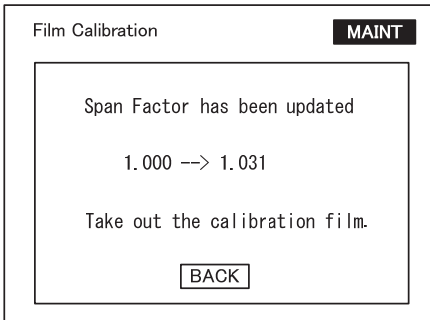
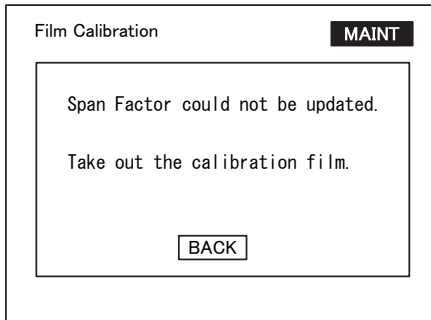
(To be continued)

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Procedure and operation	Example of screen
⑥ Set the calibration film in the measurement unit When “Calibration Film Measurement Start” screen is displayed, pull out the calibration film from calibration film storage part and insert it in the measurement unit. ⑦ Start the calibration film measurement Press [Start Calib Film Measurement] on “Calibration Film Measurement Start” screen. • “Calibration Film Measurement” screen is displayed via screen displaying the current status such as “Measurement Unit Moving Upward”. • To stop the calibration of calibration film for some reason, press [Stop Film Calibration] and carry out the operation following the display. [BACK]: Returns to “Blank Measurement Start” screen.	 Calibration Film Measurement Start Screen
⑧ Wait for the termination of calibration film measurement Wait for the remaining time displayed on “Calibration Film under Measurement” screen to become “0”. • If the remaining time becomes “0”, the buzzer sounds and “Calibration Film within Calibration Range” screen or “Calibration Film out of Calibration Range” screen is displayed. • To stop the calibration for some reasons, press [Stop Film Calibration] and carry out the operation following the display.	 Calibration Film under Measurement Screen
⑨ Check the calibration film measured value Check the calibration film measured value and other values on “Calibration Film within Calibration Range” screen or “Calibration Film out of Calibration Range” screen. • If “Calibration Film out of Calibration Range” screen is displayed, press [BACK] and carry out the operation from “③” again. • Repeat the calibration film measurement three times and check the error from calibration film value is within $\pm 2\%$. If it is exceeded, carry out the calibration (change the span factor and press [YES]).	 Calibration Film within Calibration Range Screen  Calibration Film out of Calibration Range Screen

(To be continued)

(Continued from previous page)

Procedure and operation	Example of screen
⑩ Update the span factor To update the span factor, press YES on “Calibration Film within Calibration Range” screen. • “Calibration Completed Calibration Film Removal” screen is displayed. NO : “Calibration Uncompleted Calibration Film Removal” screen is displayed without updating the span factor.	 Calibration Completed Calibration Film Removal Screen  Calibration Uncompleted Calibration Film Removal Screen ⑪ Remove the calibration film When “Calibration Completed Calibration Film Removal” screen or “Calibration Uncompleted Calibration Film Removal” screen is displayed, pull out the calibration film from the measurement unit and press BACK . • “Blank Measurement Start” screen is displayed. ⑫ Return to “Home” screen Press BACK on “Blank Measurement Start” screen and press HOME on the displayed “Calibration Menu” screen.

(3) External calibration

(a) This is the calibration for setting the external span factor etc. to an arbitrary value by input from keys.

(b) There are following items on “Zero & Span Factor Setup” screen of external calibration and they can be arbitrarily changed.

Internal span factor Factor calculated by calibration film calibration. It is changed when it is especially required. It changes by internal calibration in each case.

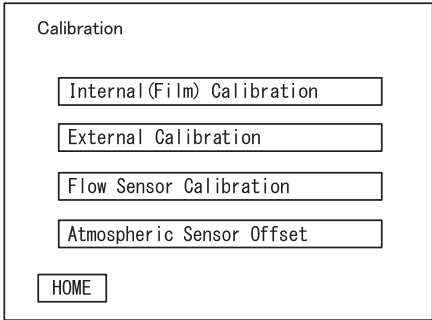
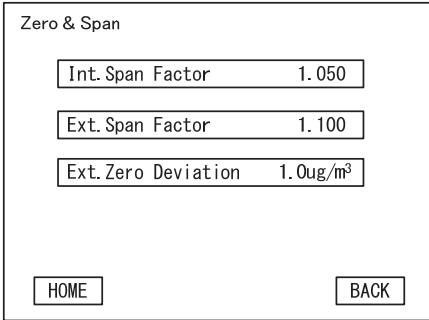
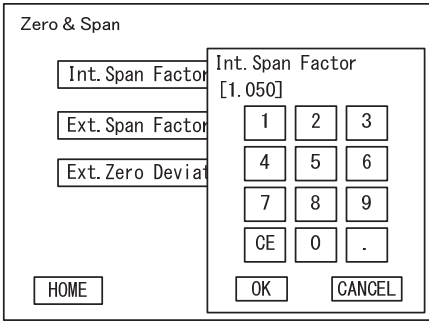
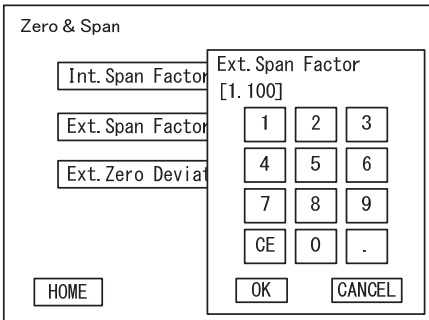
External span factor Factor calculated by “Parallel Measurement with Standard Measurement Method”

External zero deviation Factor calculated by “Blank Test” or “Parallel Measurement with Standard Measurement Method”

(c) The value after calibration is shown below.

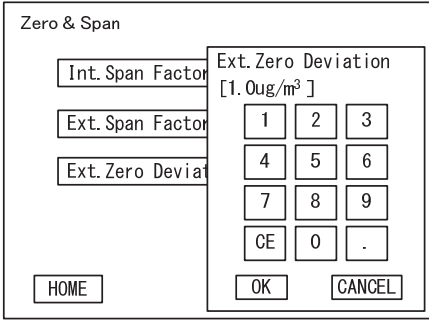
Value after calibration = Measured value x Internal span factor x External span factor + External zero deviation

Procedure of External Calibration

Procedure and operation	Example of screen
① Display “Calibration Menu” screen Press [CALIB] on “Home” screen. ② Display “Zero & Span Factor Setup” screen Press [External Calibration] on “Calibration Menu” screen.	 The Calibration Menu screen displays four options: Internal (Film) Calibration, External Calibration, Flow Sensor Calibration, and Atmospheric Sensor Offset. A HOME button is at the bottom. Calibration Menu Screen  The Zero & Span Factor Setup screen shows three fields: Int. Span Factor (1.050), Ext. Span Factor (1.100), and Ext. Zero Deviation (1.0ug/m³). HOME and BACK buttons are at the bottom. Zero & Span Factor Setup Screen
③ Display the setup screen for an item to be calibrated On “Zero & Span Factor Setup” screen, press the frame of an item to be calibrated. [BACK]: Returns to the previous screen	
(1) To change the internal span factor ① Press [Int.Span Factor]. ② On “Internal Span Factor Setup” screen, input the internal span factor after change and press [OK]. “Zero & Span Factor Setup” screen is displayed. Setting range: 1.00 to 10.00 [CANCEL]: Returns to the previous screen without changing the value.	 The Internal Span Factor Setup screen shows a numeric keypad for inputting the internal span factor. The current value is 1.050. The keypad includes digits 1-9, 0, a decimal point, and a CE button. OK and CANCEL buttons are at the bottom. Internal Span Factor Setup Screen
(2) To change the external span factor ① Press [Ext.Span Factor]. ② On “External Span Factor Setup” screen, change the external span factor after change and press [OK]. “Zero & Span Factor Setup” screen is displayed. Setting range: 0.5 to 1.5 [CANCEL]: Returns to the previous screen without changing the value.	 The External Span Factor Setup screen shows a numeric keypad for inputting the external span factor. The current value is 1.100. The keypad includes digits 1-9, 0, a decimal point, and a CE button. OK and CANCEL buttons are at the bottom. External Span Factor Setup Screen

(To be continued)

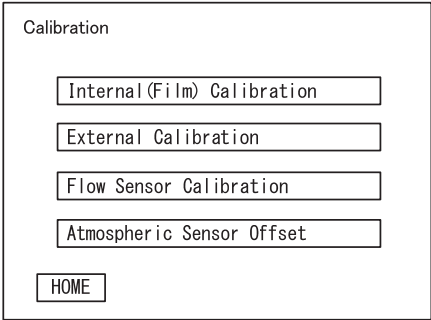
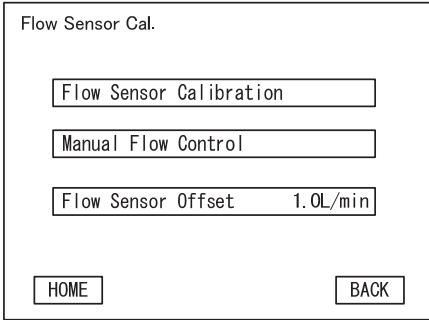
(Continued from previous page)

Procedure and operation	Example of screen
(3) To change the external zero deviation ① Press [Ext.Zero Deviation]. ② On “External Zero Deviation Setup” screen, input the external deviation after change and press [OK]. “Zero & Span Factor Setup” screen is displayed. Setting range: 1.00 to 2.00ug/m³ [CANCEL]: Returns to the previous screen without changing the value.	 External Zero Deviation Setup Screen
④ Return to “Home” screen Press [HOME] on “Zero & Span Factor Setup” screen.	

(4) Screen operation for flow rate calibration

- (a) Check the flow rate of sample air is within the standard with the standard flowmeter periodically. Carry out the calibration as occasion demands. For the calibration of flow rate, it is necessary to connect the standard flowmeter to the monitor before operating the screen. First, carry out the operation shown in 4.3 “Calibration of Flow Rate by Standard Flowmeter” and refer to this screen operation when it is required.
- (b) The flow rate sensor of this monitor indicates the value converted to the flow rate in the standard status [20°C, 1 atmosphere (101.3kPa)]. Use the standard flowmeter which can indicate the standard flow rate.
- (c) It is possible to set an arbitrary flow rate by using “Manual flow rate control” function. It is also possible to check the difference from standard flowmeter for different flow rates.

Procedure for Operating the Screens for Flow Rate Calibration

Procedure and operation	Example of screen
① Display “Calibration Menu” screen Press [CALIB] on “Home” screen. ② Display “Flow Rate Calibration” screen Press [Flow Sensor Calibration] on “Calibration Menu” screen.	 Calibration Menu Screen  Flow Rate Calibration Screen

(To be continued)

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Procedure and operation	Example of screen
③ Select the flow rate calibration Select a necessary item on “Flow Rate Calibration” screen. [BACK]: Returns to the previous screen (calibration menu screen).	
(1) To execute the flow rate calibration ① Press [Flow Sensor Calibration] on “Flow Rate Calibration Operation” screen. • “Flow Rate Calibration Operation” screen is displayed. ② Press [Reference Flow Rate] on “Flow Rate Calibration Operation” screen. • “Measured flow rate” displayed in the upper part of “Flow Rate Calibration Operation” screen is the current sample flow rate (standard flow rate). [BACK]: Returns to the previous screen. • After this operation, “Standard Flow Rate Setup” screen is displayed.	
③ Change the blinking value on “Standard Flow Rate Setup” screen using a numeric keypad and input the current standard flow rate indicated by the standard flowmeter and press [OK]. • After this operation, “Flow Rate Calibration Operation” screen is displayed.	
④ Press [Exec. Calibration] on “Flow Rate Calibration Operation” screen. • “Flow Rate Calibration Finish” screen or “Flow Rate Calibration Failure” Screen is displayed.	
⑤ Check the flow rate offset on “Flow Rate Calibration Finish” screen or “Flow Rate Calibration Failure” screen. • On “Flow Rate Calibration Finish” screen, flow rate offset before calibration (current) and flow rate offset after calibration (new) are displayed. • On “Flow Rate Calibration Failure” screen, the same offset as flow rate offset before calibration (current) is displayed as flow rate offset after calibration (new) (flow rate offset is not rewritten). The reason is that the flow rate offset is too large, for example.	

Flow Sensor Cal. **MAINT**

(1) Set reference value (**°C)

(2) Push execute key

FPM-377 Flow Rate 18.2 L/min

Reference Flow Rate 18.24L/min

Exec. Calibration

[BACK]

Flow Rate Calibration Operation Screen

Flow Sensor Cal. **MAINT**

(1) Set reference

(2) Push execute

FPM-377 Flow Rate

Reference Flow Rate [16.8]L/min

Exec. C

1 2 3

4 5 6

7 8 9

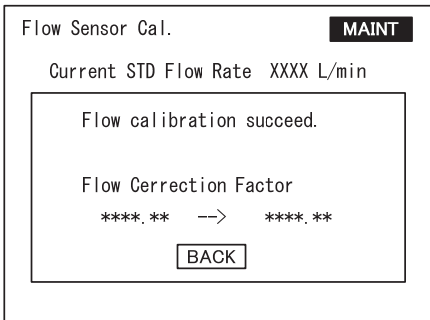
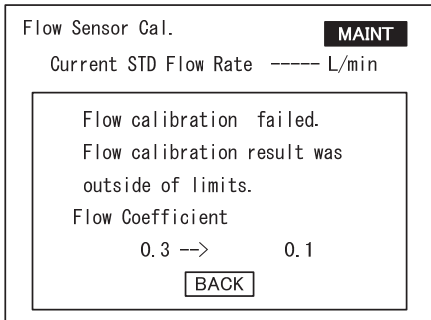
CE 0 .

[OK] **[CANCEL]**

Standard Flow Rate Setup Screen

(To be continued)

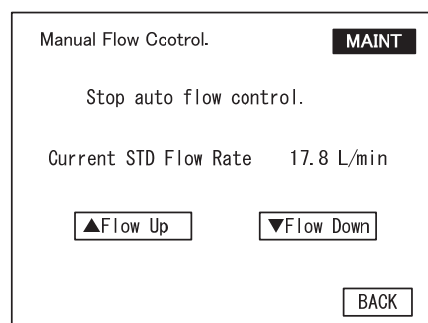
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Procedure and operation	Example of screen
 Flow Sensor Cal. MAINT Current STD Flow Rate XXXX L/min Flow calibration succeed. Flow Correction Factor ****. ** --> ****. ** [BACK]	 Flow Sensor Cal. MAINT Current STD Flow Rate ----- L/min Flow calibration failed. Flow calibration result was outside of limits. Flow Coefficient 0.3 --> 0.1 [BACK]
Flow Rate Calibration Finish Screen	Flow Rate Calibration Failure Screen

- ⑥ Press [BACK] on “Flow Rate Calibration Finish” screen or “Flow Rate Calibration Failure” screen.

(2) To change the flow rate manually (set an arbitrary flow rate)

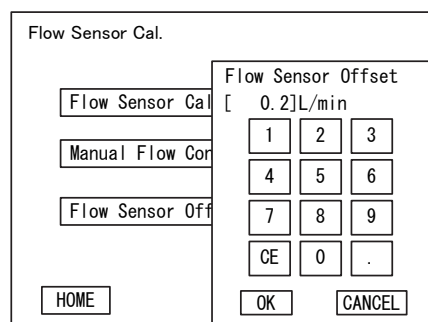
- ① Press [Manual Flow Control] on “Flow Rate Calibration Operation” screen.
- “Flow Rate Control” screen is displayed.
- ② Press the following keys on “Flow Rate Control” screen.
- Change the value of “Sample flow rate”. You can check the difference of indication from that of standard flowmeter and the operation status of pump unit, etc.
- [▲Flow Up]: Pressing this key once keeps change to flow rate increase direction and stops in 2 seconds.
- [▼Flow Down]: Pressing this key once keeps change to flow rate decrease direction and stops in 2 seconds.
- [BACK] Pressing this key returns to the previous screen.



Flow Rate Control Screen.

(3) To change the flow rate offset (input an arbitrary offset)

- ① Press [Flow Sensor Offset] on “Flow Rate Calibration Operation” screen.
- “Flow Rate Offset Setup” screen is displayed.
- ② On “Flow Rate Offset Setup” screen, input the flow rate offset after change and press [OK]. “Flow Rate Calibration” screen is displayed.
- Setting range: -5.0 to +5.0
- [CANCEL]: Returns to the previous screen without changing the value.



Flow Rate Offset Setup Screen

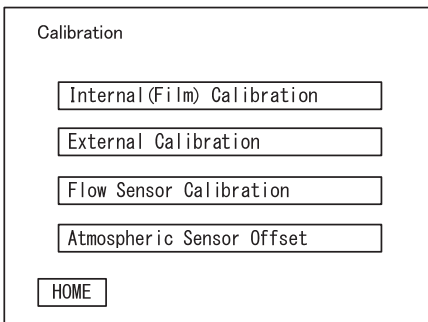
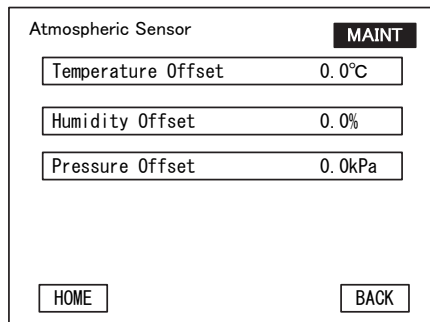
- ④ **Return to “Home” screen** Press [HOME] on “Flow Rate Calibration” screen.

(5) Calibration of temperature, humidity, and pressure

- (a) This is an operation for changing each offset to adjust the measured value of outdoor air temperature, outdoor air humidity, and peripheral pressure to the value of reference.
- (b) Prepare the references for measuring outdoor air temperature, outdoor air humidity, and peripheral pressure preliminarily so that the measurement can be carried out at the same time with this monitor.
- (c) Insert the sensor of reference for measuring outdoor air temperature and outdoor air humidity in the shade (projection hood) of temperature sensor of this monitor. In case of a sensor that cannot be inserted in the shade, prepare the same shape of shade separately and carry out the measurement by placing the sensor in it.

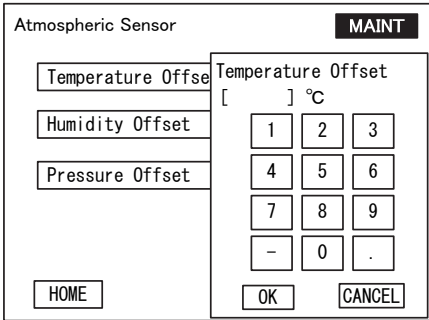
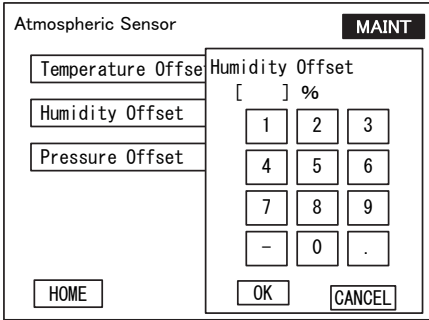
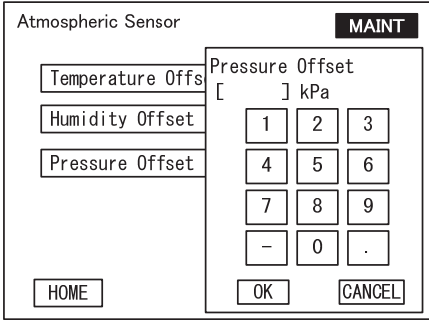
【IMPORTANT】 • If the measurement is carried out without inserting a sensor of reference for measuring outdoor air temperature and outdoor air humidity in the shade, the measurement cannot be carried out correctly.

Procedure for Calibrating Pressure, Temperature, and Humidity

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press [DATA] on “Home” screen, and then press [<] to display “Data Display Menu” screen (2/3). Wait until the values of outdoor air temperature, outdoor air humidity, and atmospheric pressure get stabilized and read the values and make a note of the values. ② Display “Calibration Menu” screen Press [HOME] on “Data Display Menu” screen. Press [CALIB] on the displayed “Home” screen. ③ Display “Temperature / Humidity / Pressure Offset” screen Press [Atmospheric Sensor Offset] on “Calibration Menu” screen.	 
④ Display the setup screen of item to be calibrated Press the frame of item to be calibrated on “Temperature / Humidity / Pressure Offset” screen. [BACK]: Returns to the previous screen	

(To be continued)

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Procedure and operation	Example of screen
(1) To calibrate the outdoor air temperature ① Press [Temperature Offset]. ② On “Outdoor Air Temperature Offset Setup” screen, input a offset value of following formula and press OK . “Temperature / Humidity / Pressure Offset” screen is displayed. New offset value = current offset value + measured value of reference – measured value of FPM-388 type • Setting range: -10.0 to 10.0°C CANCEL : Returns to the previous screen without changing the value.	 Outdoor Air Temperature Offset Setup Screen
(2) To calibrate the outdoor air humidity ① Press [Humidity Offset]. ② On “Outdoor Air Humidity Offset Setup” screen, input a offset value of following formula and press OK . “Temperature / Humidity / Pressure Offset” screen is displayed. New offset value = current offset value + measured value of reference – measured value of FPM-388 type • Setting range: -20.0 to 20.0% CANCEL : Returns to the previous screen without changing the value.	 Outdoor Air Humidity Offset Setup Screen
(3) To calibrate the peripheral pressure ① Press[Pressure Offset]. ② On “Peripheral Pressure Offset Setup” screen, input a offset value of following formula and press OK . “Temperature / Humidity / Pressure Offset” screen is displayed. New offset value = current offset value + measured value of reference – measured value of FPM-388 type • Setting range: -10.0 to 10.0kPa CANCEL : Returns to the previous screen without changing the value.	 Peripheral Pressure Offset Setup Screen,
⑤ Return to “Home” screen Press HOME on “Temperature / Humidity / Pressure Offset” screen.	

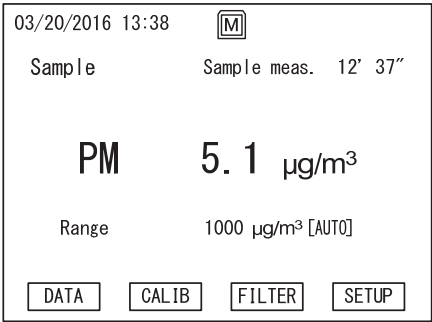
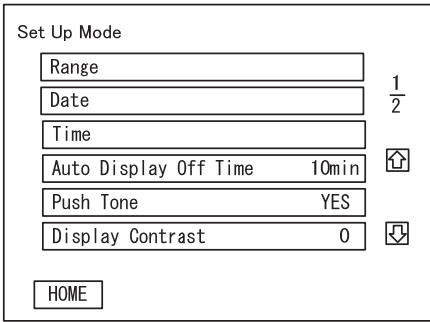
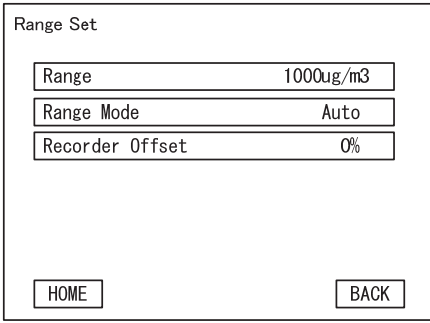
5.5 Operation of Setup Menu

(1) Change of range and offset

- (a) On “Range Setup” screen, you can select the average value range (range of 1-hour average value of PM2.5), switch the average value range, and offset average value (zero point shift).
- (b) For the average value range, any of following ranges can be selected.
- 0 to 1000 $\mu\text{g}/\text{m}^3$, 0 to 5000 $\mu\text{g}/\text{m}^3$
- (c) For the automatic switching of range, automatic range switching or fixed range can be selected.
- (d) The change on “Range Setup” screen is applied after the reset of next 1-hour average value. For the automatic switching of range, the range is increased if the measured value exceeds 100% of the current range (full scale value).
- (e) The setting of the offset of recorder is the shift of zero point of recorder, etc. Even if the measured value is negative, the record can be checked.

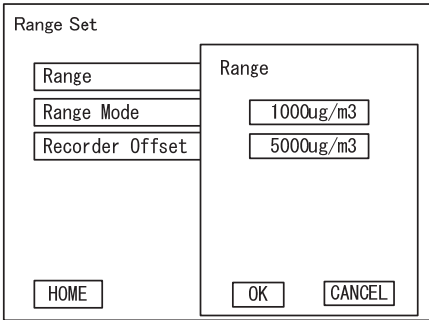
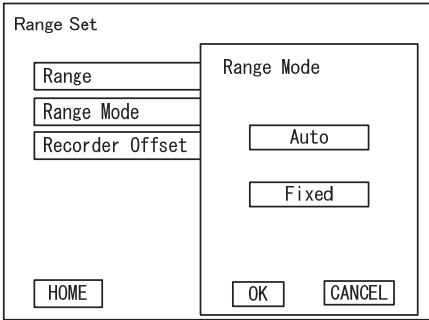
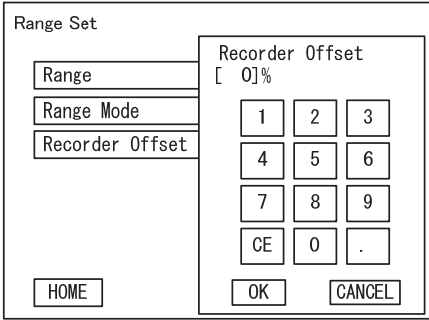
[NOTE] • The offset value to be set on “Range Setup” screen is the zero point of recorder, etc. It does not affect the transmission output.

Procedure for Changing Range and Offset

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press [SETUP] on “Home” screen.	 Home Screen (under measurement)
② Display “Range Setup” screen Press [Range] on “Setup Menu” screen (1/2).	 Setup Menu Screen (1/2)
③ Change the necessary item Check each setting value on “Range Setup” screen and change the necessary item. [BACK]: Returns to the previous screen	 Range Setup Screen

(To be continued)

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Procedure and operation	Example of screen
(1) To change the average value range ① Press [Range] on “Range Setup” screen. ② On the displayed “Average Value Range Selection” screen, press the target key. If it starts blinking, press [OK]. • After this operation, “Range Setup” screen is displayed. [1000μg/m³]: Range of average value becomes 0 to 1000μg/m³. [5000μg/m³]: Range of average value becomes 0 to 5000μg/m³. [CANCEL]: Returns to the previous screen without changing the value.	 Average Value Range Selection Screen,
(2) To change the average value range automatic switching ① Press [Range Mode] on “Range Setup” screen. ② On the displayed “Average Value Range Automatic Switching” screen, press the target key. If it starts blinking, press [OK]. • After this operation, “Range Setup” screen is displayed. [Auto]: Automatic switching of average value range is enabled (factory setting). [Fixed]: Automatic switching of average value range is disabled. [CANCEL]: Returns to the previous screen without changing the value.	 Average Value Range Automatic Switching Screen
(3) To change the offset value of average value of recorder ① Press [Recorder Offset] on “Range Setup” screen. ② On the displayed “Average Value Offset Setup” screen, change the blinking value using a numeric keypad and press [OK]. • After this operation, “Range Setup” screen is displayed. • Setting range: 0% (factory setting) to 100% [CE]: Blinking input value is reset. [CANCEL]: Returns to the previous screen without changing the value.	 Average Value Offset Setup Screen,
④ Return to “Home” screen Press [HOME] on “Range Setup” screen.	

(2) Change of date and time

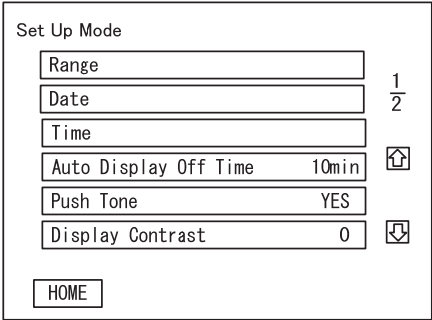
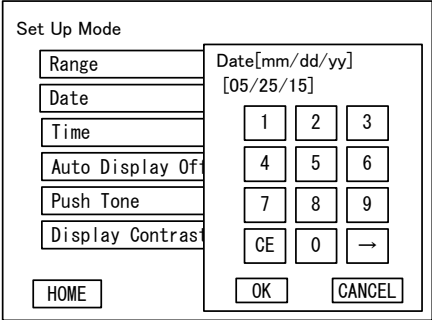
(a) On “Calendar Setup” screen, current year (dominical year), month, and day can be changed. On “Clock Setup” screen, the current time can be changed.

[IMPORTANT] • Avoid change from current time to past time. Otherwise, all measured value data during this time are cleared.

- (b) The clock embedded in the monitor is protected by backup battery. If lag is generated in date and time for some reasons, adjust them to current date and time.

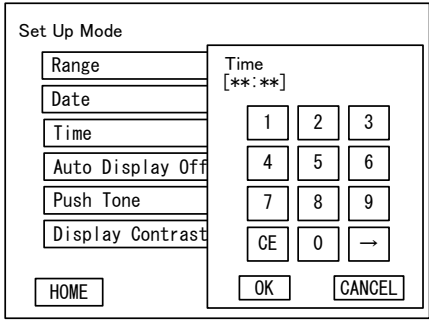
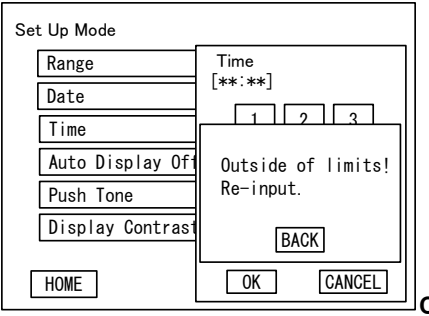
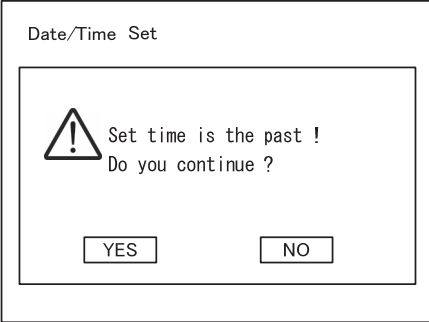
- [NOTE]
- The lunar inequality of clock embedded in the monitor is approximately 1 minute at a maximum.
 - If the measurement is carried out with one reset signal input per hour, the lag between external timer and built-in clock needs to be small.

Procedure for Changing Date and Time

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press [SETUP] on “Home” screen. ② Change the necessary item On “Setup Menu” screen (1/2), press [F4] to display “Setup Menu” screen (1/2) and change calendar or time.	 Setup Menu Screen (1/2)
(1) To change the calendar ① Press [Date] on “Setup Menu” screen (1/2). ② On “Calendar Setup” screen, change the blinking value using a numeric keypad and press [OK]. • After this operation, “Setup Menu” screen (1/2) is displayed. • If a value extending back the currently set date is input and [OK] is pressed, “Date / Time Change Confirmation” screen is displayed. Go to step “③”. • Setting range: 2000/01/01 to 2099/12/31 (year / month / day) [→]: Blinking digit moves. [CE]: Blinking input value is reset. [CANCEL]: Returns to the previous screen without changing the value.	 Calendar Setup Screen

(To be continued)

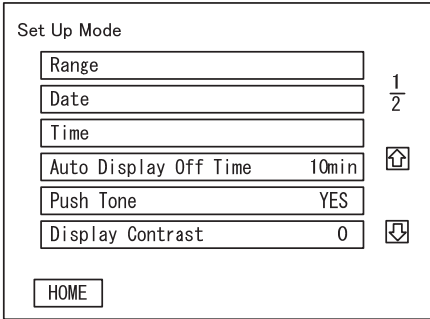
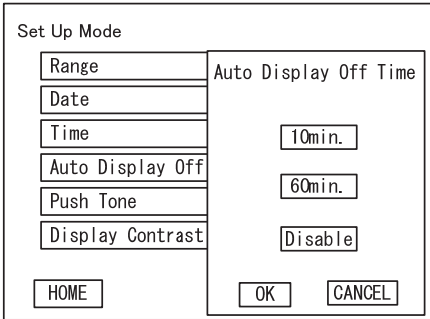
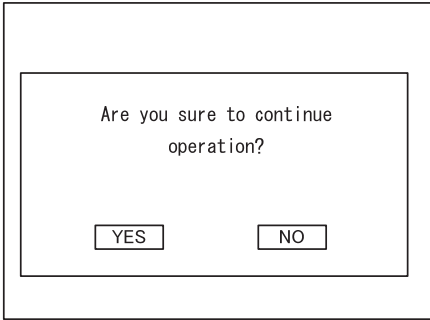
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Procedure and operation	Example of screen
(2) To change the time ① Press [Time] on “Setup Menu” screen (1/2). ② On “Clock Setup” screen, change the blinking value using a numeric keypad and press OK . - After this operation, “Setup Menu” screen (1/2) is displayed. - If a value extending back the currently set time (except minute) is input and OK is pressed, “Date / Time Change Confirmation” screen is displayed. Go to step “③”. - Setting range: 0:00 to 23:59 (time: minute) → : Blinking digit moves. CE : Blinking input value is reset. CANCEL : Returns to the previous screen without changing the value.	 Clock Setup Screen  Clock Setup out of Input Range Screen
③ Evaluate whether the log data can be cleared If “Date / Time Change Confirmation” screen is displayed, evaluate whether the date and time should be changed even if the log data (logging data) is cleared and press NO , etc. After this operation, “Setup Menu” screen (1/2) is displayed. YES : Past log data is deleted and new date and time are set. NO : Returns to “Setup Menu” screen (1/2) without changing date and time.	 Date / Time Change Confirmation Screen
④ Return to “Home” screen Press HOME on “Setup Menu” screen (1/2).	

(3) Change of auto display off / buzzer sound / contrast

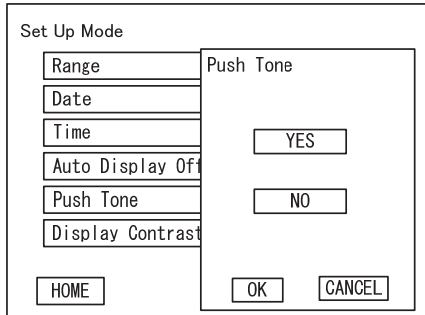
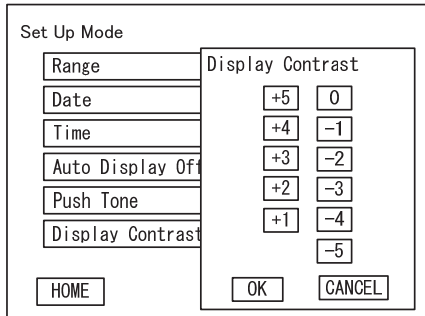
- (a) On the “Automatic Display off Selection” screen, the time to automatically turn off the backlight after termination of screen operation can be selected. When the backlight is turned off, it can be turned on by pressing a part of screen.
- (b) On “Buzzer Sound Selection” screen, you can select “Yes” or “No” of “Beep” sound to be sounded when the screen is pressed.
- (c) On “Contrast Adjustment” screen, the contrast of screen display can be adjusted.

Procedure for Changing Auto Display off / Buzzer Sound / Contrast

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press SETUP on “Home” screen. ② Change the necessary item Press ↓ on “Setup Menu” screen (1/2) to display “Setup Menu” screen (1/2), and then change necessary item in auto display off, buzzer sound, and contrast.	 Setup Menu Screen (1/2)
(1) To change the auto display off ① Press [Auto Display Off Time] on “Setup Menu” screen (1/2). ② On “Auto Display off Selection” screen, press the target key. If it starts blinking, press OK . • After this operation, “Setup Menu” screen (1/2) is displayed. 10 min : Enables the auto display off. The screen is automatically turned off in 10 minutes after the screen operation. 60 min : Enables the auto display off. The screen is automatically turned off in 60 minutes after the screen operation (factory setting). Disable : Disables the auto display off (always turned on). CANCEL : Returns to the previous screen without changing the value ③ To reset the display off, press a part of screen and press YES on the displayed “Display Restoration” screen. NO : Continues display off status without restoring the screen.	 Auto Display off Selection Screen  Display Restoration Screen

(To be continued)

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Procedure and operation	Example of screen
(2) To change the buzzer sound ① Press [Push Tone] on “Setup Menu” screen (1/2). ② Press the target key on “Buzzer Sound Selection” screen. If it starts blinking, press OK . After this operation, “Setup Menu” screen (1/2) is displayed. YES : Enables the buzzer sound. When the screen is pressed, “Beep” sound is output (factory setting). NO : Disables the buzzer sound. CANCEL : Returns to the previous screen without changing the value	 Buzzer Sound Selection Screen
(3) To change the contrast ① Press [Display Contrast] on “Setup Menu” screen (1/2). ② On “Contrast Adjustment” screen, press the target key. If it starts blinking, press OK . - After this operation, “Setup Menu” screen (1/2) is displayed. - Selection range: Indicates the contrasting density level of screen. (light) -5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5 (dark) CANCEL : Returns to the previous screen without changing the value.	 Contrast Adjustment Screen
③ Return to “Home” screen Press HOME on “Setup Menu” screen (1/2).	

(4) Confirmation of program version

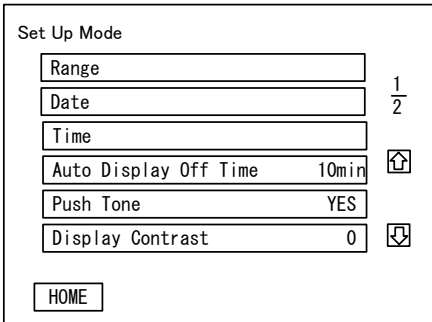
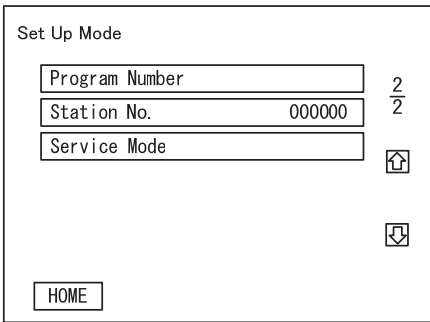
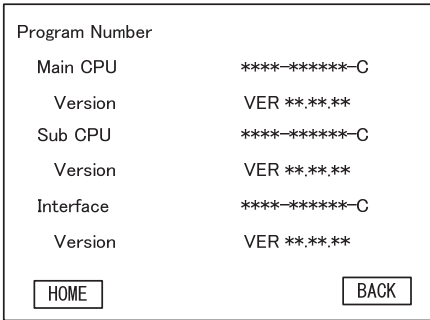
Management figure number and program version number of software of monitor can be checked on “Program Version Confirmation” screen.

Procedure for Checking the Program Version

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press SETUP on “Home” screen. ② Display “Program Version Confirmation” screen Press F1 on “Setup Menu” screen (1/2) to display “Setup Menu” screen (2/2), and then press [Program Number].	

(To be continued)

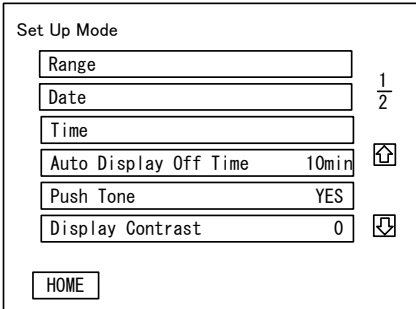
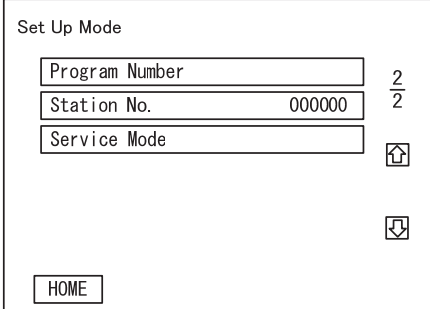
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Procedure and operation	Example of screen
 Setup Menu Screen (1/2)	 Setup Menu Screen (2/2)
③ Check figure number and program version Check figure number and program version of each item on “Program Version Confirmation” screen. Figure number: management figure number of software used by this monitor Program version: revision history number of software used by this monitor [BACK]: Returns to the previous screen. ④ Return to “Home” screen Press [HOME] on “Program Version Confirmation” screen.	 Program Version Confirmation Screen.

(5) Change of station number

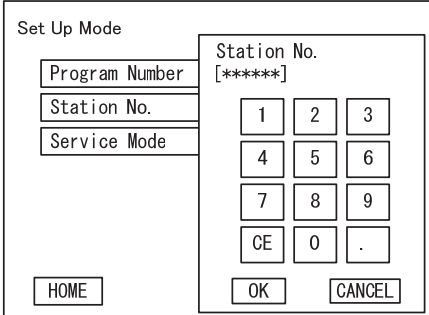
On “Station Number Setup” screen, the monitoring station number printed on the recorder can be changed.

Procedure for Changing the Station Number

Procedure and operation	Example of screen
① Display “Setup Menu” screen (1/2) Press [SETUP] on “Home” screen. ② Display “Station Number Setup” screen Press [↓] on “Setup Menu” screen (1/2) to display “Setup Menu” screen (2/2), and then press [Station No.].	 Setup Menu Screen (1/2)
	 Setup Menu Screen (2/2)

(To be continued)

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Procedure and operation	Example of screen
③ Change the station number On “Station Number Setup” screen, input a new station number from the numeric keypad and press OK. • After this operation “Setup Menu” screen (2/2) is displayed. • Setting range: 000000 to 999999 (factory setting: production number) CE: Blinking input value is reset to the original value. CANCEL: Returns to the previous screen without changing the value. ④ Return to “Home” screen Press HOME on “Setup Menu” screen.	 Station Number Setup Screen,

(6) Service menu

If [Service Mode] is pressed on “Setup Menu” screen, the service menu screen is displayed. The service menu should be used by engineers of manufacturer.

- 【IMPORTANT】** • Do not operate the service menu screen if it is not particularly essential. Otherwise, improper operation changes the setting value and the normal measurement may be disabled.

6. Maintenance

6.1 Maintenance List

- (a) To operate the product normally on a continuous basis and maintain its specified performance, understand its function, etc. thoroughly and perform maintenance work periodically.

[IMPORTANT] • Operating the product without performing maintenance periodically can result in a failure.

- (b) The “Maintenance cycle” described in the following table is based on the standard condition (conditional states to satisfy items such as the ones in 2.1 “Installation Location and Installation”). Since the maintenance cycle differs depending on the installation condition, modify the maintenance cycle according to the operating condition after operating the sensor more than several months.

- (c) For technical services equivalent such as repairs, please call our sales office or your sales representative. A specialist who is qualified for the technical certification system in our company or a person who has technical skills equivalent to that certification system must perform technical services.

Standard Maintenance List

No.	Subject	Item	Standard, etc.	Start		Maintenance cycle								Execution method, etc.
						1 week	1 month	3 months	4 months	6 months	1 year	2 years	3 years	
(Air introduction port / sizer)														
1	Inlet part	(1) Inspection in drain trap	Standard: no pool of water drops	○	○									Visual check
		(2) Inspection of impactor part	Standard: no condensation			○								Visual check
		(3) Cleaning of taint on impactor part	Standard: no remarkable taint	○		○								>> 6.4 “Cleaning of Impactor and Down Tube”
		(4) Cleaning in PM inlet	Disassemble and clean.				○							
		(5) Overhaul	Replace the O-rings (consumable parts)							○				
2	PM2.5 sizer	(1) Cleaning of dust cap	Standard: no remarkable accumulation of coarse grain		○									Visual check, cleaning
		(2) Inspection of taint on internal wall	Standard: no taint	○		○								Visual check, cleaning
		(3) Cleaning	Disassemble and clean the internal wall.				○							>> 6.5 “Cleaning of PM2.5 Sizer”
		(4) Overhaul	Replace the O-rings (consumable parts)							○				

(To be continued)

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No.	Subject	Item	Standard, etc.	Start	Maintenance cycle									Execution method, etc.
					1 week	1 month	3 months	4 months	6 months	1 year	2 years	3 years	When needed	
3	Down pipe, joint	(1) Inspection of taint on internal wall	Standard: no taint.	○		○								Visual check, cleaning
		(2) Cleaning	Clean the internal wall.				○						>> 6.4 “Cleaning of Impactor and Down Tube”	
		(3) Overhaul	Replace the O-rings (consumable parts)						○					
4	Dehumidifier	(1) Inspection of taint on internal wall	Standard: no taint.	○		○								Visual check, cleaning
		(2) Cleaning	Clean the internal wall.				○						>> 6.6 “Cleaning of Dehumidifier and Replacement of O-ring”	
		(3) Replacement	Replace the O-rings (consumable parts)						○					
		(4) Check of operation	Standard: target temperature is equal to actually measured temperature		○									Visual check
				Standard: temperature of dehumidifier is increasing			○							
(Particle collection part / detection part)														
5	Filter collection part	(1) Check of operation	Standard: there is no defect in opening / closing of collection pat, winding of filter	○	○									Visual check
		(2) Inspection of adhesion degree / taint	Standard: outline of spot is clear	○	○									Visual check
6	Filter	(1) Inspection of filter	Standard: intervals of spots are constant		○									Visual check
		(2) Inspection of remaining amount of filter	Standard: remaining amount of filter is sufficient		○									Visual check
		(3) Replacement	Replace the filter (consumable part)			○								6.7 “Replacement of Filter”

(To be continued)

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No.	Subject	Item	Standard, etc.	Start	Maintenance cycle									Execution method, etc.
					1 week	1 month	3 months	4 months	6 months	1 year	2 years	3 years	When needed	
7	Measurement unit	(1) Cleaning and grounding	Standard: There is no taint on upper and lower measurement units. They are surely grounded.						○					6.8 “Cleaning of Radiation Source Window and Detection Window” 6.9(1) “Check the grounding wire”
		(2) Check of insulation	Section between measurement unit and lead cable is insulated.						○					6.9(2) “Check the insulation of lower measurement unit”
(Suction pump)														
8	Sample air suction pump	(1) Check of operation	Standard: no abnormal sound or abnormal vibration		○									
		(2) Cleaning	Clean the flow control valve						○					6.10 “Cleaning of Flow Control Valve”
		(3) Replacement	Replace diaphragm and valve seat (consumable part)						○					6.11 “Replacement of Pump Unit Parts”
		(4) Replacement of main unit	Replace the pump main unit							○				•Contact sales dealer.
(Thermometer / hygrometer / barometer)														
9	Thermometer / hygrometer / barometer	Check / calibration	Standard: Outdoor air temperature, outdoor air humidity, and peripheral pressure on the screen correspond to the measured values of reference.	○		○								5.4(5) “Calibration of temperature, humidity, and pressure”
10	Humidity sensor	Replacement	Replace the humidity sensor assembly						○					6.12 “Replacement of Humidity Sensor”
(Other consumable parts)														
11	ZNR unit	Replacement	Replace ZNR unit							○			○	6.13 “ Replacement of ZNR Units” •When needed: at lightning strike

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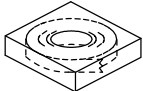
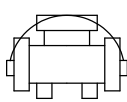
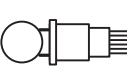
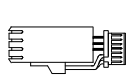
No.	Subject	Item	Standard, etc.	Start	Maintenance cycle									Execution method, etc.
					1 week	1 month	3 months	4 months	6 months	1 year	2 years	3 years	When needed	
12	Internal piping	Replacement	Replace the internal piping							○				6.14 “Replacement of Internal Piping”
(Display part / I/O part)														
13	Display part	Clock setting	Standard: display on screen corresponds to current time			○								5.5(2) “Change of date and time”
(General operation check / general adjustment)														
14	Gas flow	Leak test	Standard: no leakage	○					○					
15	Flow rate conversion	Calculation of actual flow rate of sample	Standard: actual flow rate is correctly calculated from sample flow rate, air temperature, and atmosphere pressure	○		○								4.3 “Calibration of Flow Rate by Standard Flowmeter”
16	Span adjustment	Calibration film calibration (check the static sensitivity)	Standard: span adjustment is available	○		○								4.2 “Internal Calibration”
17	Zero confirmation	(1) Blank test	Standard: if a filter is attached to sample air suction port, the concentration value gets close to 0	○					○					6.15 “Blank Test”
		(2) Replacement of radiation source	Replacement of β-ray source										○	Contact sales dealer.
18	Consistency check test	Check	Conduct parallel test with FRM							○				Based on the standard of installation location
(Other) (in case of outdoor type)														
19	Lithium battery	Replacement	Standard: lithium battery is not burnt								○			6.16 “Replacement of Lithium Battery”

6.2 Spare Parts and Replacement Parts

(1) Spare parts

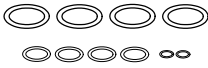
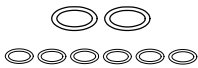
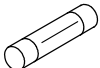
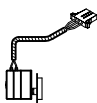
It is recommended that you always store consumable parts, periodical replacement parts, and spare parts shown in the following table. The following table shows standard example and the parts differ depending on the specification at ordering. The content may be changed without notice.

Spare Parts List

No.	Code No.	Part name (type name / standard)	Shape	Quantity			Replacement cycle	Remark
				Consumable parts	Periodical replacement parts	Spare parts		
1	7008500K	PTFE filter type AP-50		12			1 month	
2	125B851	Diaphragm 4M-901299 for MF-50D		2	2		6 months	For pump unit
3	125B852	Valve seat 4M-901097 for MF-50D		2	2		6 months	For pump unit
4	116B0481	Techno blade hose $\phi 6 \times \phi 11$ TB-6			6m (3m×2)		1 year	For between measurement unit - pump unit
5	116E022	High plain tube $\phi 7 \times \phi 11$			2m		1 year	
6	125B854	Tube TC-109 for MF-50D			2		1 year	For pump unit
7	125B853	Sleeve for MF-50D			4		1 year	For pump unit
8	125B385	Vacuum pump MF-50D			1		2 years	
9	6179800U	ZNR unit (single product)			3		1 year	
10	7205580K	Humidity sensor assembly		1	1		6 months	

(To be continued)

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No.	Code No.	Part name (type name / standard)	Shape	Quantity			Replacement cycle	Remark
				Consumable parts	Periodical replacement parts	Spare parts		
11	7219380K	PM2.5 cyclone O-ring set			1		1 year	
12	7226620K	PM10 inlet O-ring set			1		1 year	
13	7472240K	O-ring set for FPM-388			1		1 year	
14	106A067	Lithium battery				1	–	Display unit for backup
15	104A273	Fast-blow fuse 217 3.15 (3.15A / 250V)				4	–	For power supply of monitor For power supply of pump
16	7486840K	Open / close motor assembly				1	–	Replacement every 5 to 7 years recommended
17	7486830K	Filter feeding motor assembly				1	–	Replacement every 5 to 7 years recommended

(2) Replacement parts

The parts in the following table are the main parts that can be replaced with new ones if a part of product is damaged due to random failure or external factor. These parts are sold by our company. Please contact the sales dealer.

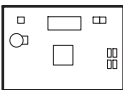
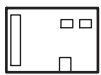
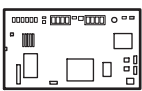
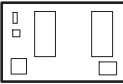
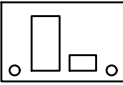
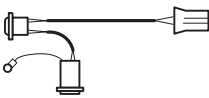
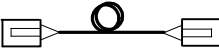
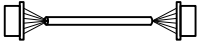
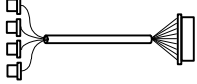
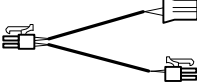
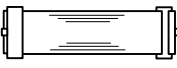
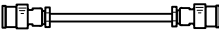
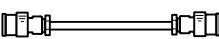
Replacement and adjustment of these parts require dispatch of engineering service staff.

Replacement Parts List

No.	Code No.	Part name	Sketch view	Quantity	Remark
1	134C283	Switching power supply LWD15-1515 (15V (2ch) / 15W)		1	

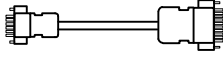
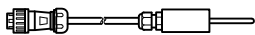
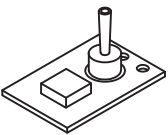
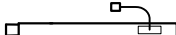
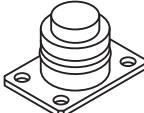
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No.	Code No.	Part name	Sketch view	Quantity	Remark
2	7382997U	Written ACPU unit		1	
3	7382877U	Written SCPU unit		1	
4	6961670F	I/O board unit		1	
5	7358831F	Power supply board unit		1	
6	7058090F	Amplifier board unit		1	
7	6963301F	External terminal strip unit		1	
8	7343800K	Power cable assembly GLN4-734380-*K		1	AC inlet – main power switch / power supply unit
9	7199630K	Cable assembly with connector FPM4-719963-*K		1	Display unit - power supply board (DC power supply)
10	7302000K	Cable assembly with connector GHC4-730200-*K		1	I/O board - power supply board (DC power supply)
11	7045800K	RS232C line cable GLN4-704580-*K		1	Display unit - I/O board (RS232C)
12	6919940K	Cable assembly with modular jack GLN4-691994-*K		1	Display unit - I/O board (modular jack)
13	7194430K	Flat cable with connector FPM4-719443-*K		1	I/O board – relay board
14	7382340K	Cable assembly with connector GLN4-738234-*K		1	I/O – power supply unit / relay board (AC power supply)
15	118G883	Flat cable with connector DUB4-692017-*D		1	I/O board – amplifier board unit
16	7486850K	Coaxial cable assembly FPM3-748685-*D		1	Amplifier board unit - I/O board
17	118G880	Coaxial cable assembly DUB4-692019-*D		1	Measurement part – amplifier board unit

(To be continued)

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No.	Code No.	Part name	Sketch view	Quantity	Remark
18	118N201	Grounding cable of measurement unit DUB4-703105-*D		1	Measurement unit - chassis
19	7030970K	Shielded cable with connector DUB4-703097-*K		1	Main unit – pump unit
20	7485110U	Thermometer FPM3-730791-U		1	In protection hood
21	113E014	Measurement unit pressure & pneumatic sensor		2	Directly connected to I/O board
22	7485120K	Flow rate sensor assembly for FPM-388 DUB4-730521-*K (name in this document: flow rate sensor)		1	I/O board – flow rate sensor assembly
23	7419720K	Detector assembly FPM4-741972-*K		1	Detector

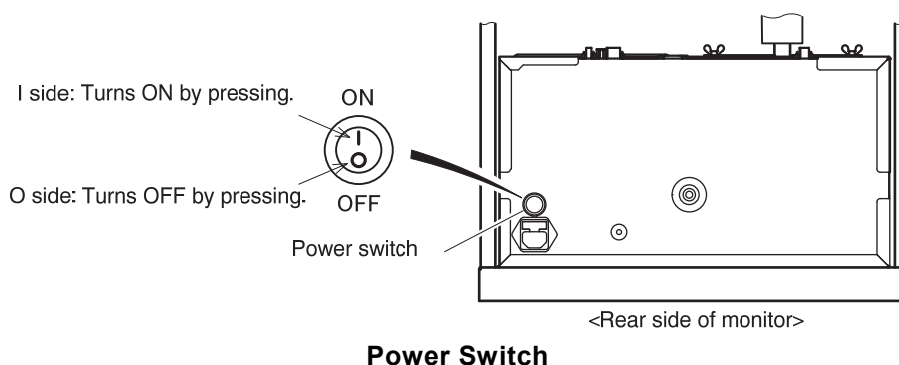
6.3 Operation before and after Maintenance

(1) Operation before maintenance

6.3 “Operation before and after Maintenance” is the operation before and after maintenance common to maintenance items. This operation does not fall under the category of some maintenance items. Refer to the procedure of each maintenance item first.

【IMPORTANT】 • If the reference of 6.3(1) “Operation before maintenance” or 6.3(2) “Operation after maintenance” is not specified for each maintenance item, the operation is not necessary.

- ① **Turn off the power.** Turn off the power (press O side). To enable safer operation, disconnect the power cable of the monitor from an outlet.



WARNING

Electric Shock

- Turn off the power (press O side) before removing the upper cover and operating the inside. To enable safer operation, disconnect the power cable from an outlet.
-

CAUTION

Tucking

- When the power is distributed, the measurement unit moves upward and downward intermittently and the reel rotates. If you may touch the measurement part, turn off the power preliminarily. If “Filter Replacement” screen is displayed, it is not necessary to turn off the power.
-

- ② **Remove the window cover.** Remove the butterfly bolts (2 pieces) and remove the window cover on the upper cover.

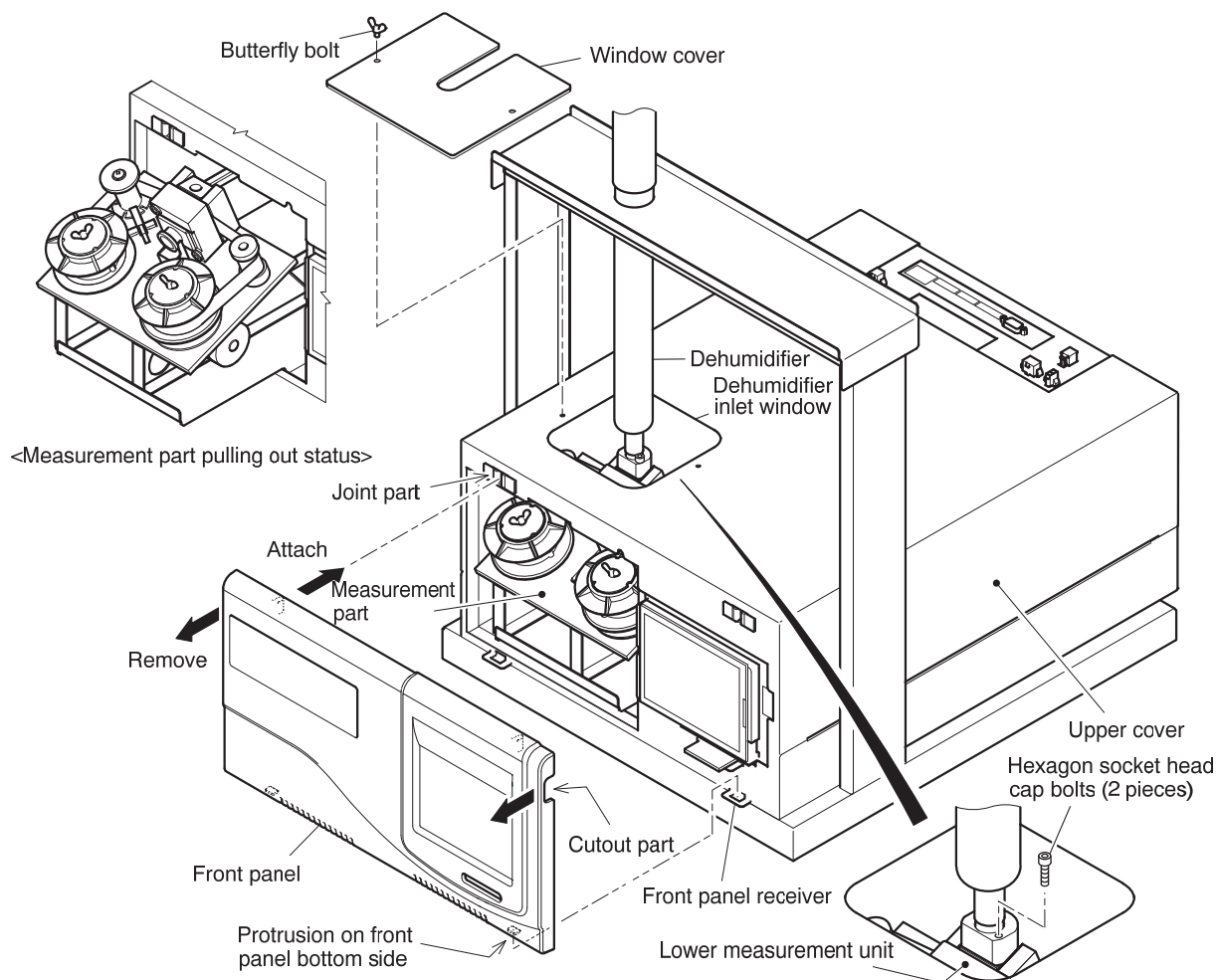
- ③ **Remove the retainer of dehumidifier.** Remove the hexagon socket head cap bolts (2 pieces) and separate measurement unit and dehumidifier, and then bring up the dehumidifier slightly.

④ **Remove the front panel.** Put your fingers on the cutout parts on both side surfaces of front panel and pull the front panel to remove it.

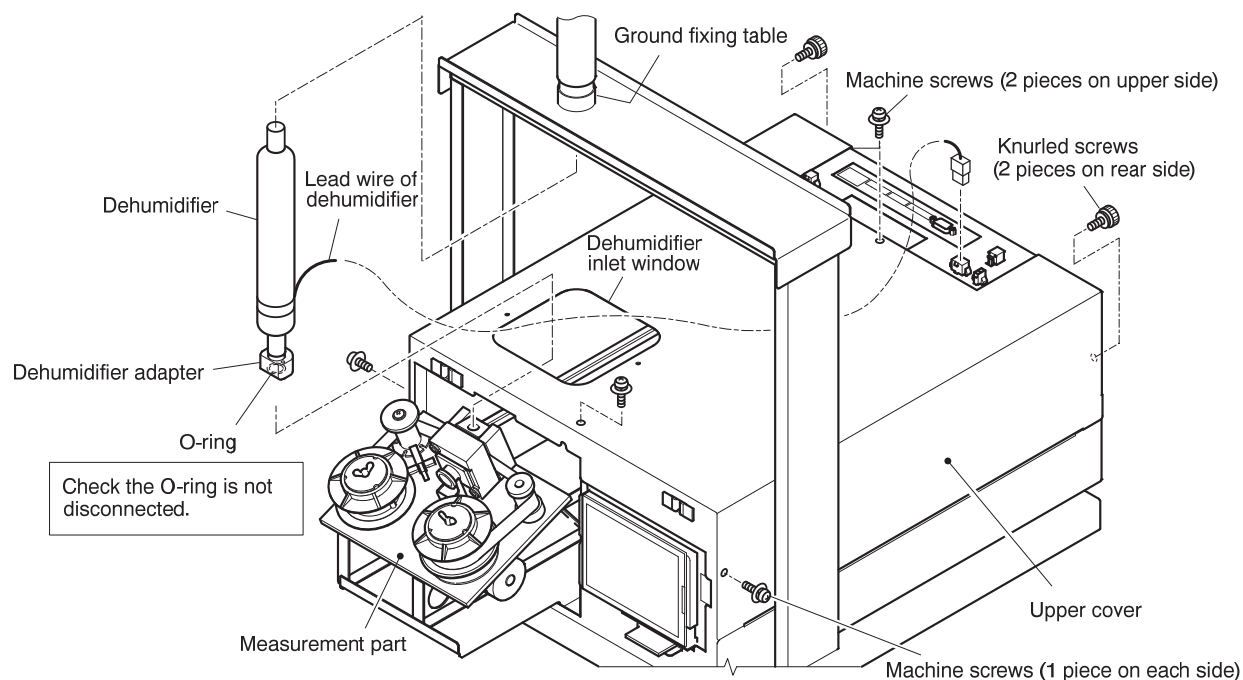
- If it is not necessary to pull out the measurement part, go to step ⑨ “Carry out the maintenance operation”.

⑤ **Check the O-ring.** Pull out the measurement part forward slowly by approximately 20mm and check the O-ring is fitted in the O-ring groove on the edge surface of dehumidifier adapter.

【IMPORTANT】 • Pull out the measurement part carefully not to damage or disconnect O-ring. The O-ring may be disconnected by small impact.



Pull out Measurement Part



Removal of Dehumidifier and Upper Cover

⑥ Pull out the measurement part. Pull out the measurement part slowly until it stops.

- If it is not necessary to remove the upper cover, go to step ⑨ “Carry out the maintenance operation”.

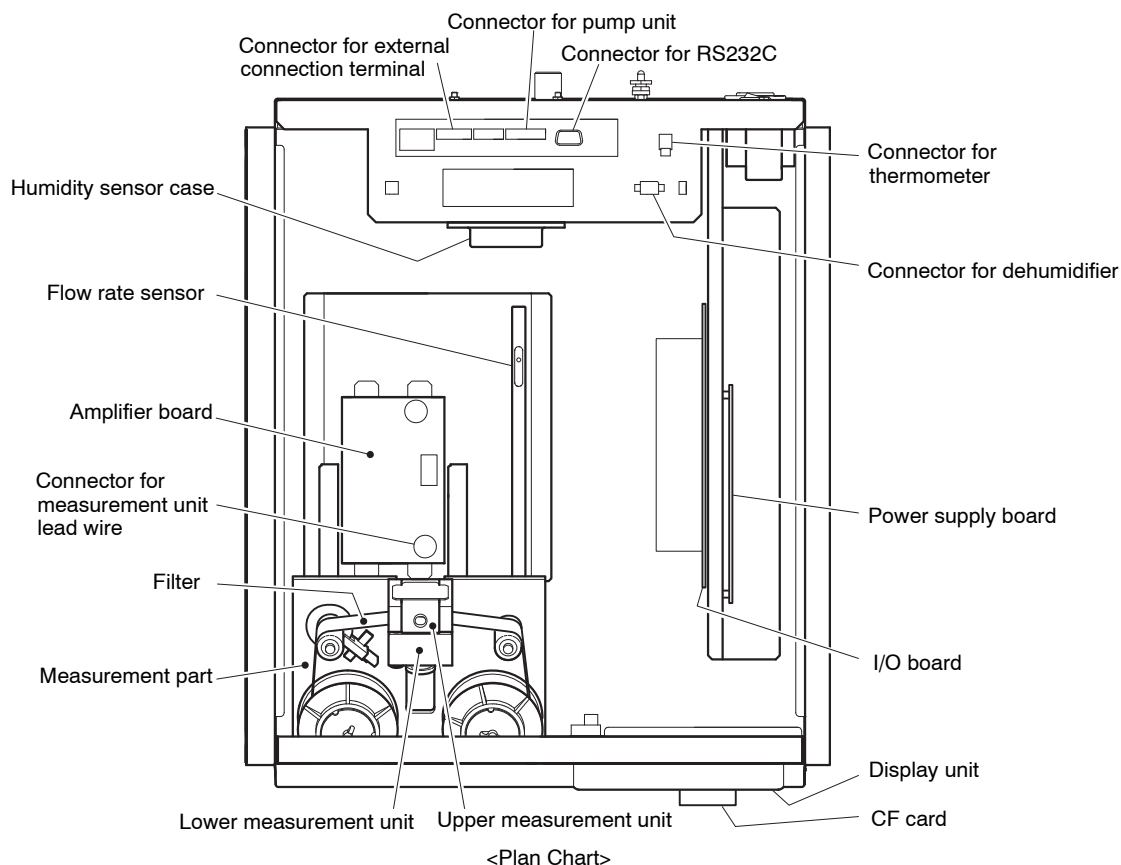
⑦ Pull out the dehumidifier.

- Disconnect the lead wire of dehumidifier Disconnect the connector of lead wire of dehumidifier.
- Disconnect the dehumidifier from ground fixing table Pull out the dehumidifier from lower part of ground fixing table downward slowly.
- Pull out the from dehumidifier inlet window Pull out the dehumidifier from dehumidifier inlet window carefully not to contact other parts.

⑧ Remove the upper cover. Remove the knurled screws fixing the upper cover (2 pieces on rear surface) and machine screws (2 pieces on upper surface, 1 piece on each side surface), total six screws and bring the upper cover upward to remove it.

⑨ Carry out the maintenance operation. Carry out the necessary maintenance operation referencing each item.

- The main parts in the monitor are shown in the figure “Main Parts in Monitor”.



Main Parts in Monitor

(2) Operation after maintenance

After the maintenance operation is carried out after 6.3(1) “Operation before maintenance”, carry out the following operation.

① **Attach the upper cover.** Put the upper cover on the monitor slowly not to tuck down wiring or piping in the monitor and fix it with knurled screws (2 pieces on rear surface) and machine screws (2 pieces on upper surface, 1 piece on each side surface), total six screws.

② **Attach the dehumidifier.**

① Insert the dehumidifier in the ground fixing table Insert the upper part of dehumidifier in the ground fixing table vertically. Insert it slowly until it stops.

【IMPORTANT】 • When you insert the dehumidifier in the lower part of ground fixing table, insert it in the central shaft vertically. If it is inserted at a tilt, O-ring or dehumidifier may be deformed or damaged and air may leak in it.

② Connect the lead wire of dehumidifier Connect the connector of lead wire of dehumidifier to the connector for dehumidifier (rear side of upper surface of main unit) the way it was.

③ Check the O-ring Check the O-ring is closely attached to the side surface on measurement part side of dehumidifier.

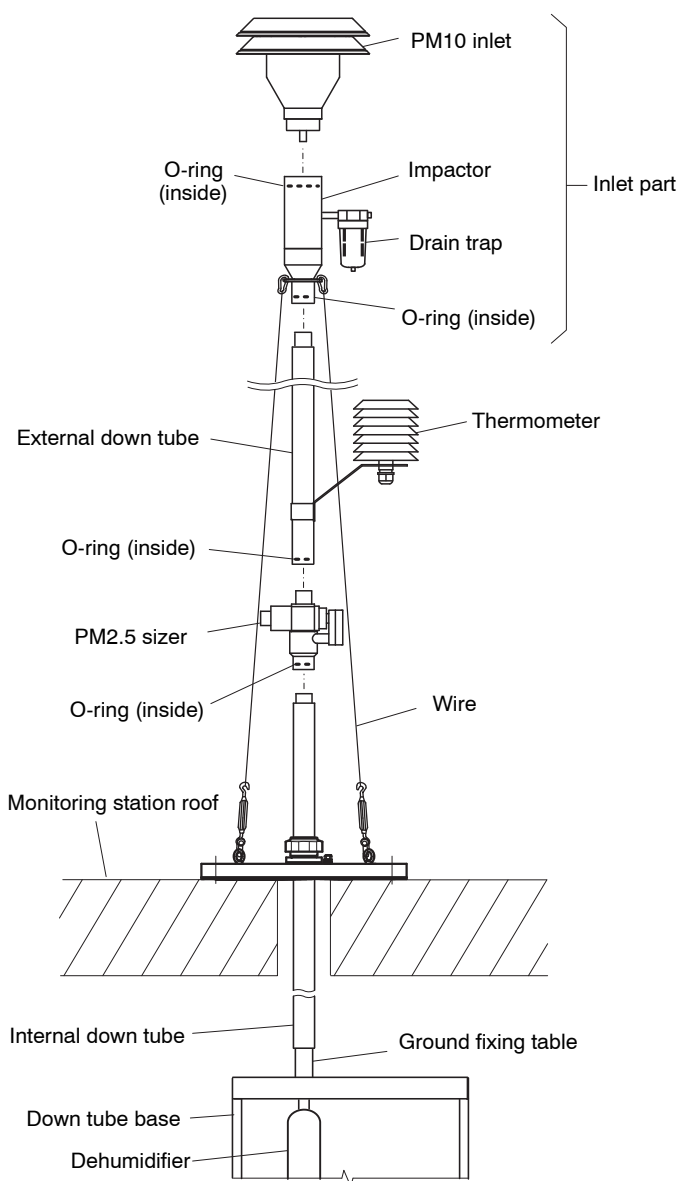
③ **Restore the measurement part.** Push back the measurement part slowly not to contact the lower part of dehumidifier until it stops.

【IMPORTANT】 • If the O-ring on the edge surface of lower part of dehumidifier is deformed or damaged, leakage may occur.

- ④ **Fix the dehumidifier.** Fix the lower part of dehumidifier (dehumidifier adapter) to the upper measurement unit with hexagon socket head cap bolts (2 pieces).
- ⑤ **Attach the front panel.** Attach the front panel in the original state. >> Figure “Pull out Measurement Part” in “⑤” in 6.3(1) “Operation before maintenance”
- ⑥ **Attach the window cover.** Attach the window cover to the upper cover with butterfly bolts (2 pieces).
- ⑦ **Turn on the power.** To restart the operation, connect the power cable to an outlet and turn on the power (press I side).
 - **Warm-up** and **MAINT** appear and if the monitor enters measurement status (in 60 minutes at a maximum), they disappear. >> “⑧ Wait for the measurement to be stabilized” in 3.1 “Procedure for Starting the Automatic Measurement”

6.4 Cleaning of Impactor and Down Tube

(1) Cleaning of impactor



Example of Inlet Part and Down Tube

WARNING

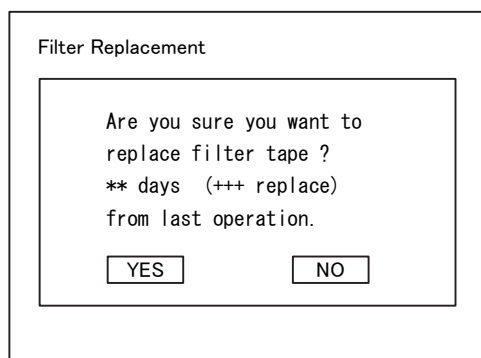
Fall Injury

- If there is a possibility of fall at high place such as roof of monitoring station, wear safety belt, etc. to prevent falling. In addition, to prevent injury, wear a helmet, life jacket, safety shoes, etc.
-

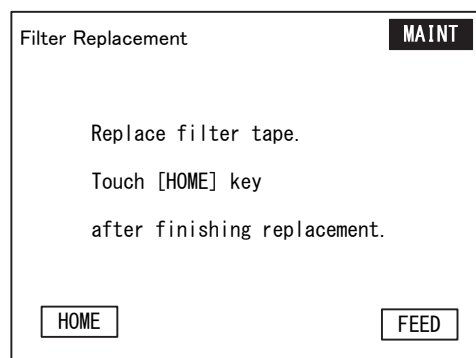
Here explains the cleaning of impactor. To carry out the cleaning of down pipe continuously, go to the operation shown in 6.4(2) “Cleaning of down tube” as explained in “⑦ Insert the Inlet in the Impactor”.

① **Display “Filter Replacement Confirmation” screen.** Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



Filter Confirmation Screen



Filter Replacement Screen

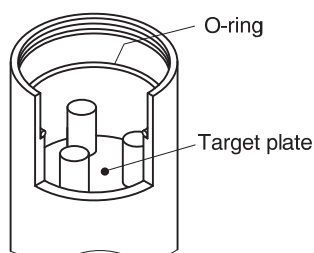
② **Remove the retainer of inlet part.** Loosen the wire, etc. fixing the inlet part.

③ **Remove the inlet part.** Pull out the inlet part upward from upper part of external down tube.

-
- 【IMPORTANT】**
- PM10 inlet is made of aluminum with alumite treatment surface. If it is damaged, it may be corroded. Do not rub it with a hard brush or other tool.
 - The upper and lower O-rings in the impactor are thin and fragile. Carry out the removal of inlet part carefully.
-

④ **Remove the impactor.** Pull out the impactor from the inlet part.

⑤ **Clean up the internal surface of impactor.** Clean up the target plate and its periphery in the impactor with cloth, etc. impregnated with alcohol.



Internal Surface of Impactor

⑥ **Apply grease to O-ring.** Check the grease applied to upper and lower O-rings in the impactor. Apply high-vacuum grease if necessary

-
- 【IMPORTANT】**
- The upper and lower O-rings in the impactor are thin and fragile. Apply grease carefully not to damage them.
 - When you insert PM inlet in the impactor, take care not to tuck down the O-ring by difference in grade at screw part or periphery.
-

⑦ **Insert the inlet in the impactor.** Insert PM10 inlet in the impactor.

- To clean up the down tube here, go to next step. >> Go to “④ Disconnect the external down tube” in 6.4(2) “Cleaning of down tube”.

⑧ **Insert the inlet part in the external down tube.** Insert the inlet part in the external down tube in the original state.

[IMPORTANT] • Insert the inlet part carefully not to damage the O-ring.

⑨ **Fix the inlet part.** Hook the wire, etc. in the original state to fix the inlet part.

[IMPORTANT] • Fix the inlet part securely. If it is loosened or stretched unevenly, it may cause a fall due to strong wind, etc.

⑩ **Display “Home” screen.** Press on “Filter Replacement” screen.

- The monitor goes into the normal measurement state.

(2) Cleaning of down tube

In the figure “Example of Impactor and Down Tube” in 6.4(1) “Cleaning of impactor and down tube”, there are external down tube and internal down tube. Here explains the example of external down tube, however, the internal down tube should be removed and carry out the cleaning complying with the cleaning method for external down tube.

① **Prepare the cleaning tool.** Cut the conducting wire (vinyl wire etc.) in the double length of down pipe + approx. 30cm and twist the gauze around the center and tie it.

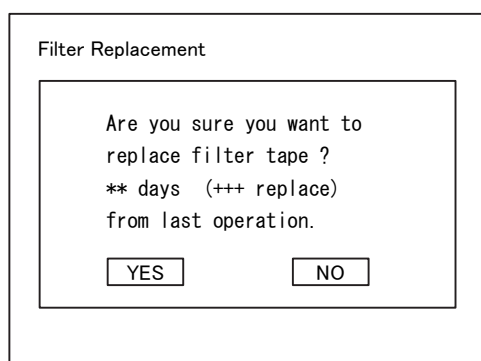
- Be creative with amount and twisting method so that the gauze is applied to the internal surface of down pipe properly.



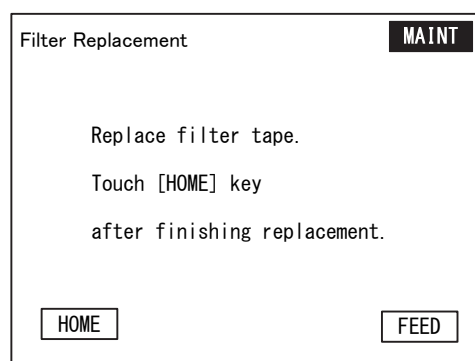
Cleaning Tool for Down Tube

② **Display “Filter Replacement Confirmation” screen.** Press on “Home” screen and press on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



Filter Confirmation Screen



Filter Replacement Screen

- ③ **Remove the retainer of inlet part.** Loosen the wire, etc. fixing the inlet part.
- ④ **Disconnect the external down tube.** Disconnect the inlet part from the external down tube and pull out the external down tube from PM2.5 sizer.
- ⑤ **Clean up the internal surface of down tube.** Pass the cleaning tool through the down tube and pull it out from the opposite side repeatedly to wipe off the taint on the internal surface of down tube.

- If it is heavily tainted, instill alcohol in the gauze.

【IMPORTANT】 • Do not damage the internal surface of down tube by edge of conducting wire, etc.

- ⑥ **Apply grease to O-ring.** Check the grease applied to O-ring in the lower part of down pipe. Apply high-vacuum grease if necessary.
- ⑦ **Restore the down tube.** Insert the down pipe in PM2.5 sizer and insert the inlet part in the down tube.

【IMPORTANT】 • Insert the down pipe carefully not to damage the O-ring.

- ⑧ **Fix the inlet part.** Fix the inlet part with wire, etc. in the original state.

【IMPORTANT】 • Fix the inlet part securely. If it is loosened or stretched unevenly, it may cause a fall due to strong wind etc.

- ⑨ **Display “Home” screen.** Press HOME on “Filter Replacement” screen.

- The monitor goes into the normal measurement state.

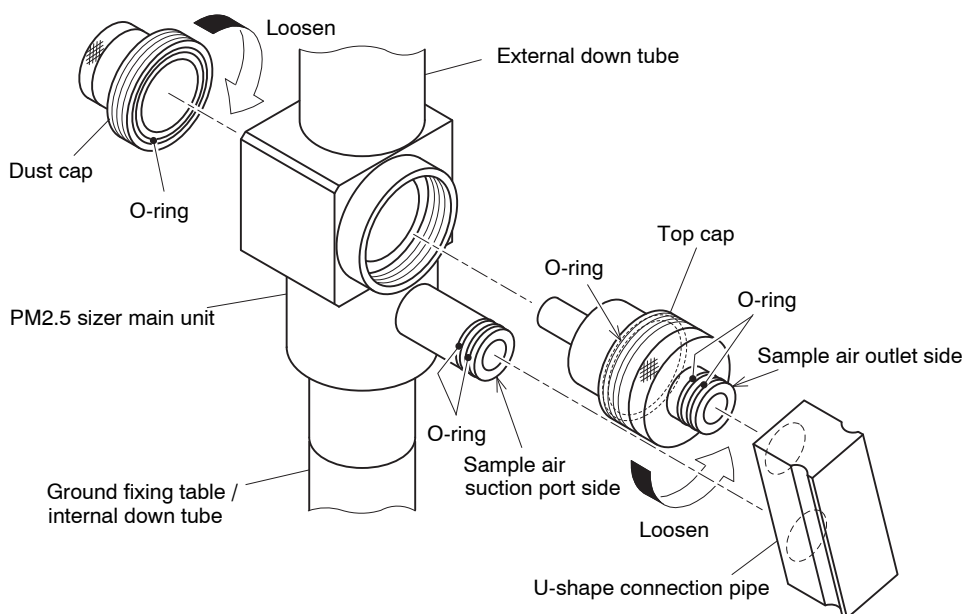
6.5 Cleaning of PM2.5 Sizer

PM2.5 sizer can be disassembled and cleaned up while it is attached to the monitor.

WARNING

Fall Injury

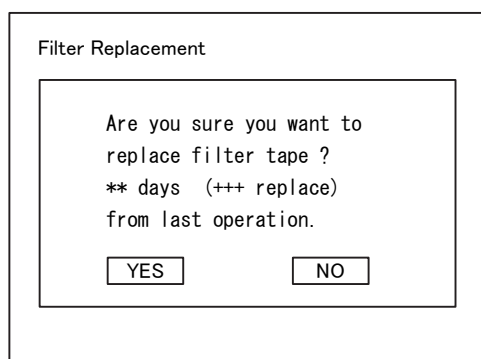
- If there is a possibility of fall at high place such as roof of monitoring station, wear safety belt, etc. to prevent falling. In addition, to prevent injury, wear a helmet, life jacket, safety shoes, etc.



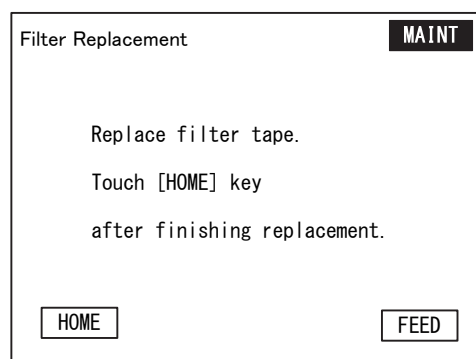
PM2.5 Sizer

① **Display “Filter Replacement Confirmation” screen.** Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



Filter Confirmation Screen



Filter Replacement Screen

- ② **Disconnect U-shape connection pipe.** Pull out U-shape connection pipe slowly and strongly to disconnect it from sample air discharge port side and sample air suction port side.
- ③ **Remove both caps.** Loosen top cap and dust cap and remove them from PM2.5 sizer.
- ④ **Carry out cleaning.** Clean up the internal surface of PM2.5 sizer with clean cloth, etc.
 - If it is heavily tainted, wipe it off with a paper wiper impregnated with alcohol.
- ⑤ **Assemble the parts in the original state.** Embed top cap, dust cap, and U-shape connection pipe in PM2.5 sizer main unit in the original state.

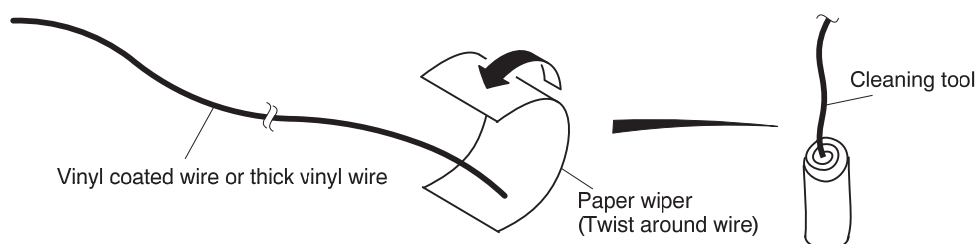
【IMPORTANT】 • Insert the parts carefully not to damage the O-ring.

- ⑥ **Display “Home” screen.** Press HOME on “Filter Replacement” screen.
 - The monitor goes into the normal measurement state.

6.6 Cleaning of Dehumidifier and Replacement of O-ring

- ① **Prepare the cleaning tool.** Twist a paper wiper around soft vinyl coated wire or thick vinyl wire securely.

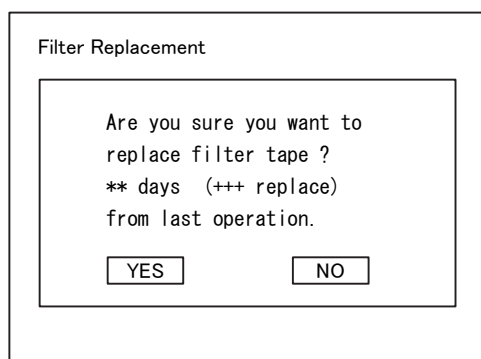
【IMPORTANT】 • Use a paper wiper (KIMWIPE <® product name>, etc.) which does not leave fine paper particles. Do not use tissue paper that frequently leaves fine paper particles in the dehumidifier.



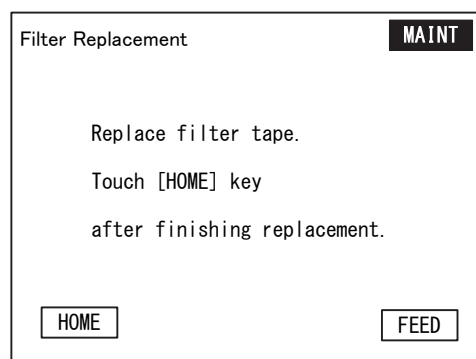
Cleaning Tool for Dehumidifier

- ② **Display “Filter Replacement Confirmation” screen.** Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



Filter Confirmation Screen



Filter Replacement Screen

- ③ **Carry out the operation before maintenance.** >> 6.3(1) “Operation before maintenance”

(The operation of 6.3(1) “① Turn off the power” is not necessary.)

6.3(1) “② Remove the window cover”

6.3(1) “③ Remove the retainer of dehumidifier”

6.3(1) “④ Remove the front panel”

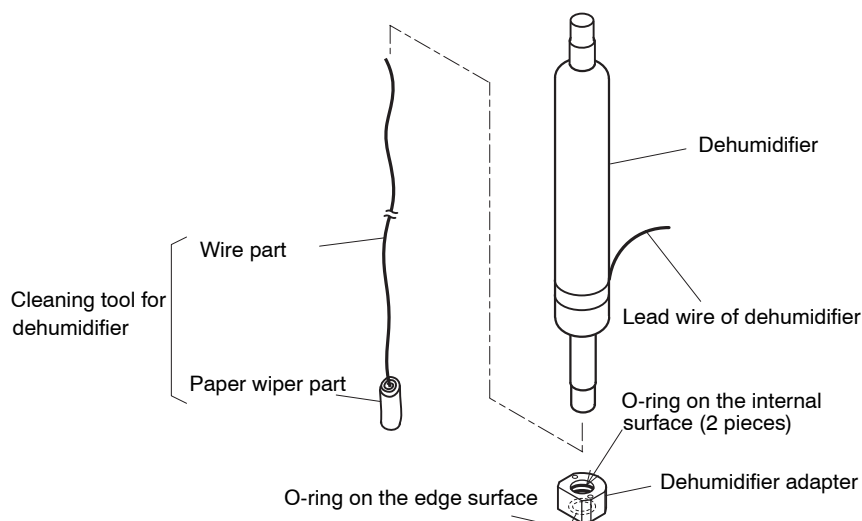
6.3(1) “⑤ Check the O-ring”

6.3(1) “⑥ Pull out the measurement part”

6.3(1) “⑦ Pull out the dehumidifier”

(The operation of 6.3(1) “⑧ Remove the upper cover” is not necessary.)

- ④ **Pull out the dehumidifier adapter.** Pull out the dehumidifier adapter from lower part of dehumidifier.



Cleaning of Dehumidifier

- ⑤ **Clean up the inside of dehumidifier.** Pass the cleaning tool, wire through the inside of dehumidifier and pull out the paper wiper. Repeat this step to clean up the inside of dehumidifier.

- If the paper wiper gets dirty, replace it with a new paper wiper and repeat the cleaning until the taint is completely removed.

[IMPORTANT]

- Carry out the cleaning carefully not to damage the inside of dehumidifier.
- If the inside is heavily tainted, clean it up with a paper wiper impregnated with small amount of alcohol and remove the alcohol with a dry paper wiper after the cleaning.

- ⑥ **Check O-ring.** Check there is no defect on an O-ring on the internal surface of ground fixing table to which upper part of dehumidifier is attached, an O-ring on internal surface of dehumidifier adapter, and an O-ring on edge surface of dehumidifier adapter.

- If an O-ring is damaged or changed, replace it with a new one.

[Replacement parts]

PM2.5 cyclone O-ring set No.7219380K

PM10 inlet O-ring set No.7226620K

O-ring set for FPM-388 No.7472240K

- ⑦ **Attach the dehumidifier adapter.** Insert the lower part of dehumidifier in the dehumidifier adapter until it stops.

- ⑧ **Carry out the operation after maintenance.** >> 6.3(2) “Operation after maintenance”

(The operation of 6.3(2) “① Attach the upper cover” is not necessary.)

6.3(2) “② Attach the dehumidifier”

6.3(2) “③ Restore the measurement part”

6.3(2) “④ Fix the dehumidifier”

6.3(2) “⑤ Attach the front panel”

6.3(2) “⑥ Attach the window cover”

(The operation of 6.3(2) “⑦ Turn on the power ” is not necessary.)

⑨**Display “Home” screen.** Press HOME on “Filter Replacement” screen.

- The monitor goes into the normal measurement state.

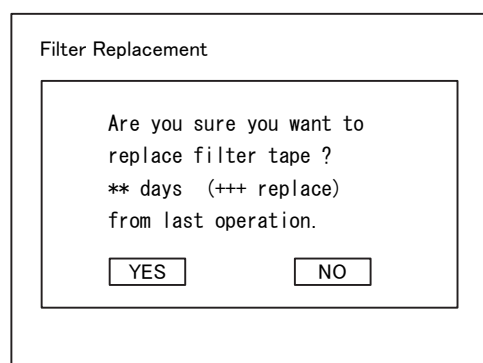
6.7 Replacement of Filter

① Prepare a new filter.

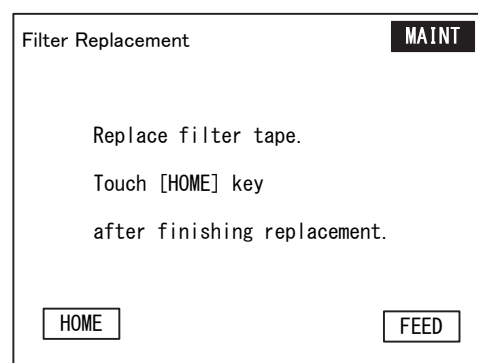
[Replacement part] Filter Name of product: PTFE filter type AP-50, No.7008500K

② Display “Filter Replacement Confirmation” screen. Press **FILTER** on “Home” screen and press **YES** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of measurement part stops, “Filter Paper Replacement” screen is displayed.



Filter Confirmation Screen



Filter Replacement Screen



Tucking

- Until “Filter Replacement” screen is displayed, do not touch the action part of measurement part. The lower measurement unit moves upward and downward intermittently and the reel rotates and your hand or other body part may get caught in the device.
-

③ Carry out the operation before maintenance. >> 6.3(1) “Operation before maintenance”

(The operation of 6.3(1) “① Turn off the power ” is not necessary.)

6.3(1) “② Remove the window cover”

6.3(1) “③ Remove the retainer of dehumidifier”

6.3(1) “④ Remove the front panel”

6.3(1) “⑤ Check the O-ring”

6.3(1) “⑥ Pull out the measurement part”

(The operation of 6.3(1) “⑦ Pull out the dehumidifier” is not necessary.)

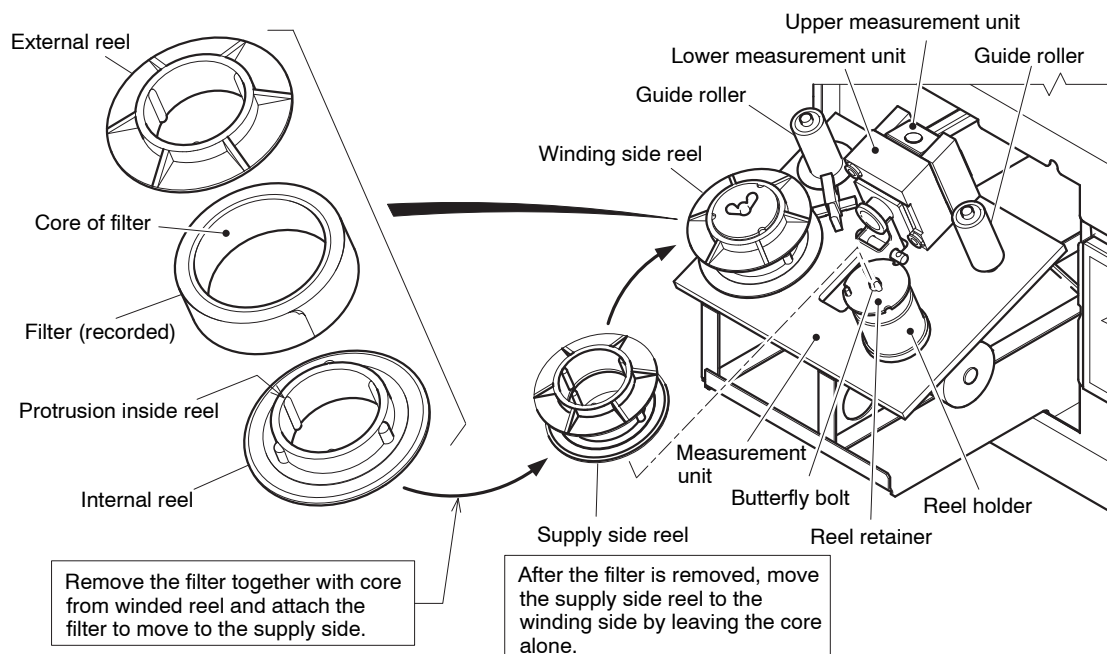
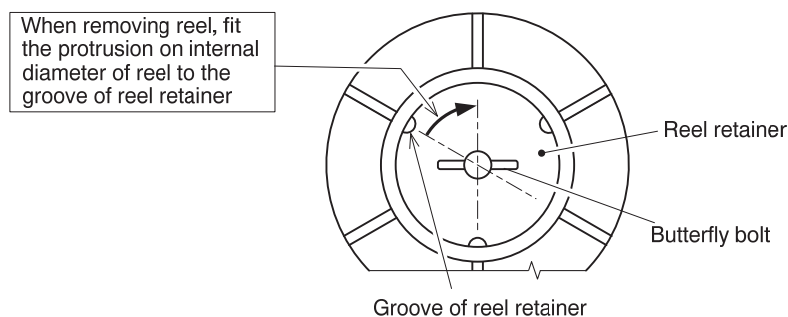
(The operation of 6.3(1) “⑧ Remove the upper cover” is not necessary.)

④ Rewind the used filter. Press **FEED** repeatedly to rewind the filter remaining on supply side and peel the end of filter glued on the reel on supply side.

- You can also cut and rewind the filter and remove the reel on supply side, and then peel the end of filter.

⑤ Remove the reel on supply side.

- ① Loosen the butterfly bolt. Loosen the butterfly bolt slightly.
- ② Fit the groove to the protrusion. Rotate the reel retainer and fit the groove of reel retainer to the protrusion inside of reel on supply side.
- ③ Remove the reel. Pull out the reel on supply side from the reel holder.

**Replacement of Filter****Fitting of Groove When Connecting or Disconnecting Reel****⑥ Replace the reel on winding side.**

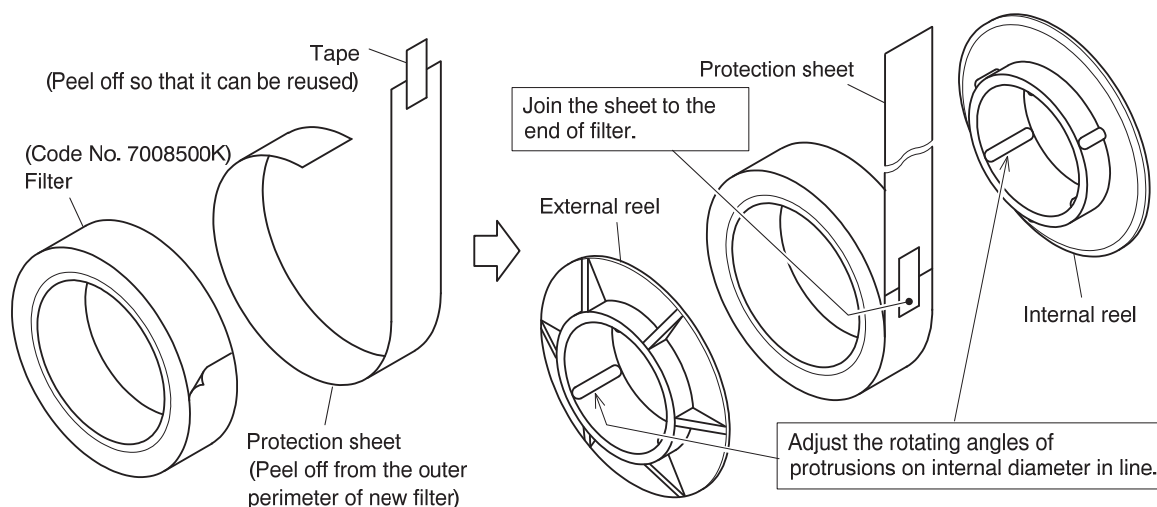
- ① Disconnect the reel. Disconnect the reel on winding side from reel holder in a same way as the step “⑤ Remove the reel on supply side”.
- ② Insert an empty reel. Put the reel removed from supply side (core of filter is left) in the reel holder on winding side. At this time, the protrusion on internal diameter of reel should fit to the groove of reel retainer.
- ③ Prevent the disconnection of reel. Rotate the reel retainer slightly to displace the groove position from the protrusion on internal diameter of reel and tighten the butterfly bolt.

⑦ Replace the filter with a new one.

- ① Remove the used filter. Remove external reel and internal reel of disconnected reel from

winding side and remove the used filter with the core together.

- ⑥ Joint the protection sheet with a filter. Peel off the tape fixing the protection sheet of a new filter so that it can be reused and joint it with the end of filter.



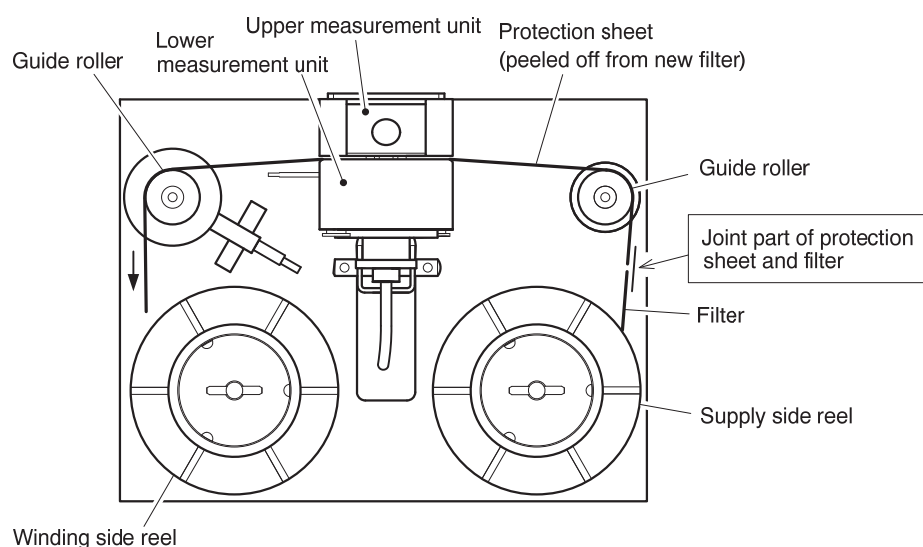
Joint the Protection Sheet with Filter

- ③ Embedding in reel. ----- Set the filter joining the protection sheet properly so that it is rolled out from right side to upward direction and embed in external reel and internal reel.
 - The reel on winding side becomes the reel on supply side by embedding a new filter.
- ④ Set the protrusions in line. ----- Adjust the rotating angles of protrusions on internal diameter of external reel and internal reel in line.

⑧ Embed the reel on supply side.

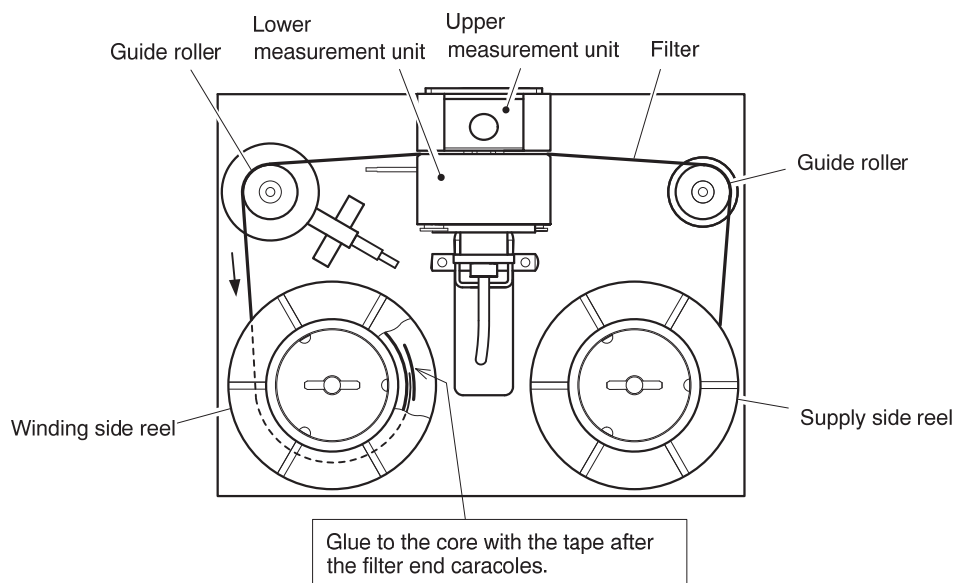
- ① Put the reel in reel holder. Embed the reel in which a new filter is set in the reel holder on supply side. At this time, the protrusion on internal diameter of reel should fit to the groove of reel retainer.
- ② Prevent the disconnection of reel. Rotate the groove position of reel retainer slightly and tighten the butterfly bolt.

⑨ Pass a new filter through the route.



Route of Filter

- ③ Pass the protection sheet through the arm. Pass the protection sheet jointed to the end of filter of reel on supply side through between upper measurement unit and lower measurement unit by rotating the guide roller from right side.
- ④ Remove the protection sheet. Pull the protection sheet by hand slowly to winding side. If it is pulled out until the length of filter reaches the winding reel, remove the protection sheet from the end of filter.
- ⑤ Glue the end of filter. After the filter end passes through the left guide roller and caracoles around the reel on winding side, glue it to the core with the tape peeled from the protection sheet.



Gluing of Filter End

- ⑩ **Rewind the filter.** Press FEED repeatedly and rotate the reel twice slowly to stretch the filter not to generate looseness.
- ⑪ **Carry out the operation after maintenance.** >> 6.3(2) “Operation after maintenance”
 (The operation of 6.3(2) “① Attach the upper cover” is not necessary.)
 (The operation of 6.3(2) “② Attach the dehumidifier” is not necessary.)
 6.3(2) “③ Restore the measurement part”
 6.3(2) “④ Fix the dehumidifier”
 6.3(2) “⑤ Attach the front panel”
 6.3(2) “⑥ Attach the window cover”
 (The operation of 6.3(2) “⑦ Turn on the power” is not necessary.)
- ⑫ **Display “Home” screen.** Press HOME on “Filter Replacement” screen.
 - The monitor goes into the normal measurement state.

6.8 Cleaning of Radiation Source Window and Detection Window

It is recommended to carry out the cleaning of radiation source window and detection window at the replacement of filters because the filter may be damaged by cleaning if there is a filter in the measurement unit.

WARNING

Radiant ray

- β -ray source in the measurement unit is the radioactive material of 10MBq or less sealed radiation source (^{14}C). For the handling, observe laws and regulations in the country where the product is used.
Since β -ray source is a radioactive material, never disassemble the upper measurement unit. Damage of radiation source window or removal of upper measurement unit may cause radiation exposure.
 - Do not move bare skin of hand or face etc. close within 250mm range from the radiation source window at the position where β -ray source in the radiation source window is seen. At the cleaning of upper measurement unit, be sure to wear latex gloves (for kitchen use) and do not use a sharp item (screwdriver, tweezers, etc.) that may damage the film surface of radiation source window.
 - When the measurement unit including β -ray source is replaced or the product is discarded, ask the sales company to pick it up.
 - If the radiation source window is damaged, carry out the emergency treatment shown below, and then contact the sales company immediately.
 - (a) Put the paper whose thickness is three name cards worth to the radiation source window and fix it with tape.
 - (b) If a part of β -ray source may be attached to the tip of tools or periphery, wipe it off with a piece of cloth (waste cloth) sufficiently. Put the used piece of cloth in a reclosable poly bag and put it in a cardboard box to store it strictly.
-

① Prepare tools.

Blow brush accessory

Latex gloves for kitchen use

② Display “Filter Replacement” screen. Press FILTER on “Home” screen and press YES on the displayed “Filter Replacement Confirmation” screen.

- The pump stops and the lower measurement unit moves downward.

【IMPORTANT】 • When cleaning the measurement unit, be sure to display “Filter Replacement” screen and check the lower measurement unit moves downward. If the upper measurement unit is opened when the lower measurement unit does not move downward, the filter may be damaged.

③ Carry out the operation before maintenance >> 6.3(1) “Operation before maintenance”

(The operation of 6.3(1) “① Turn off the power” is not necessary.)

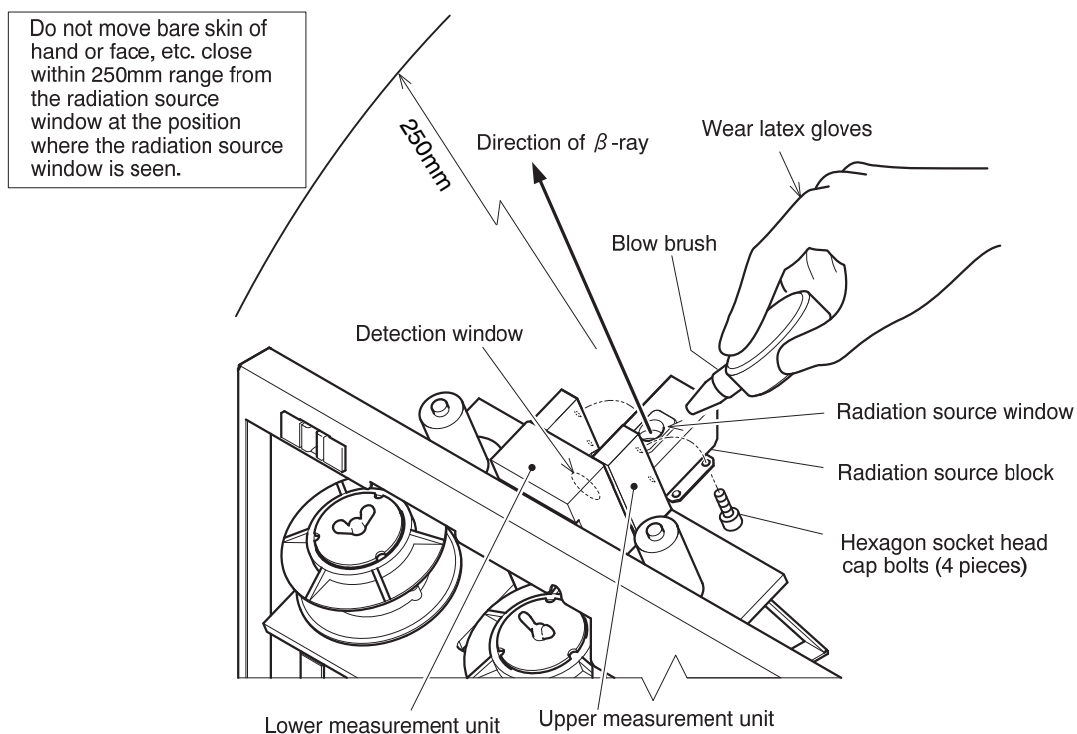
- 6.3(1) “② Remove the window cover”
- 6.3(1) “③ Remove the retainer of dehumidifier”
- 6.3(1) “④ Remove the front panel”
- 6.3(1) “⑤ Check the O-ring”
- 6.3(1) “⑥ Pull out the measurement part”
- 6.3(1) “⑦ Pull out the dehumidifier”
- 6.3(1) “⑧ Remove the upper cover”

⚠ WARNING

Electric shock

- After the upper cover is removed, do not touch the terminal of internal wiring, etc. by bare hand.
-

- ④ Open the radiation source block.** Wear latex gloves (for kitchen use) and remove the hexagon socket head cap bolts (4 pieces) fixing the radiation source block and rotate the radiation source block (approx. 90°) to open it.



Cleaning of Measurement Unit

- ⑤ Clean up radiation source window and detection window.** Clean up radiation source window and detection window using a blow brush.
- If a filter is attached, cut the filter once, and then pull it out from the measurement unit. After the cleaning, pass the filter through the measurement unit and joint the cut part with an adhesive tape, etc.

⑥ **Restore the radiation source block.** Close the radiation source block and fix it with hexagon socket head cap bolts (4 pieces) in the original state.

⑦ **Carry out the operation after maintenance.** >> 6.3(2) “Operation after maintenance”

6.3(2) “① Attach the upper cover”

6.3(2) “② Attach the dehumidifier”

6.3(2) “③ Restore the measurement part”

6.3(2) “④ Fix the dehumidifier”

6.3(2) “⑤ Attach the front panel”

6.3(2) “⑥ Attach the window cover”

6.3(2) “⑦ Turn on the power ”

⑧ **Display “Home” screen.** Press HOME on “Filter Replacement” screen.

- The monitor goes into the normal measurement state.

⑨ **Calibrate the calibration film.** >> 4.2 “Internal Calibration”

6.9 Check Grounding Wire and Insulation of Measurement Unit

(1) Check the grounding wire.

① **Carry out the operation before maintenance.** >> 6.3(1) “Operation before maintenance”

6.3(1) “① Turn off the power ”

6.3(1) “② Remove the window cover”

6.3(1) “③ Remove the retainer of dehumidifier”

6.3(1) “④ Remove the front panel”

6.3(1) “⑤ Check the O-ring”

6.3(1) “⑥ Pull out the measurement part”

6.3(1) “⑦ Pull out the dehumidifier”

6.3(1) “⑧ Remove the upper cover”

② **Check the connection of grounding wire of measurement unit.** Check each terminal of grounding wire of measurement unit is not loosened and the toothed lock washer is the closest to the screw hole side.

- If it is loosened, fix it securely. If the position of toothed lock washer is different, correct it.

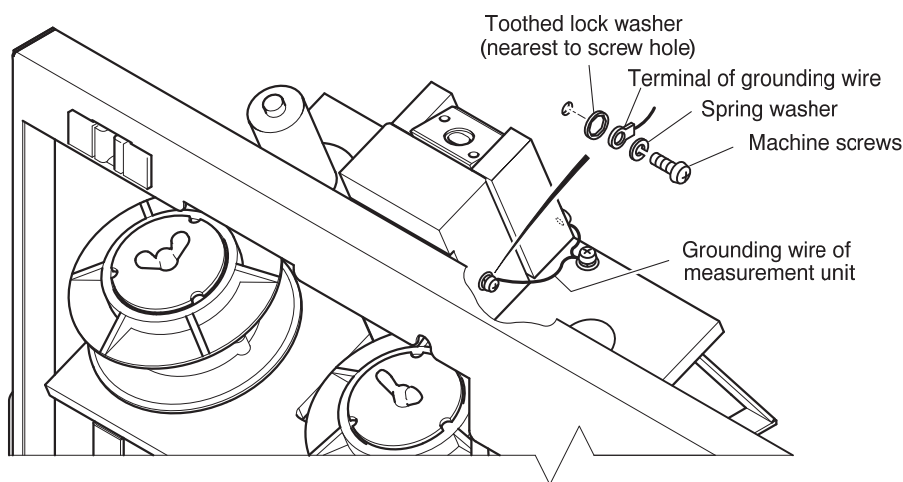
[IMPORTANT]

- Do not disconnect the grounding wire of measurement unit out of necessity.
- If the toothed lock washer of grounding wire of measurement unit is not closest to the screw hole side or other washer is used, the conduction may not be ensured.

③ **Carry out the following operation.** >> 6.3(2) “Operation after maintenance”

6.3 (2) “① Attach the upper cover”

6.3 (2) “② Attach the dehumidifier”

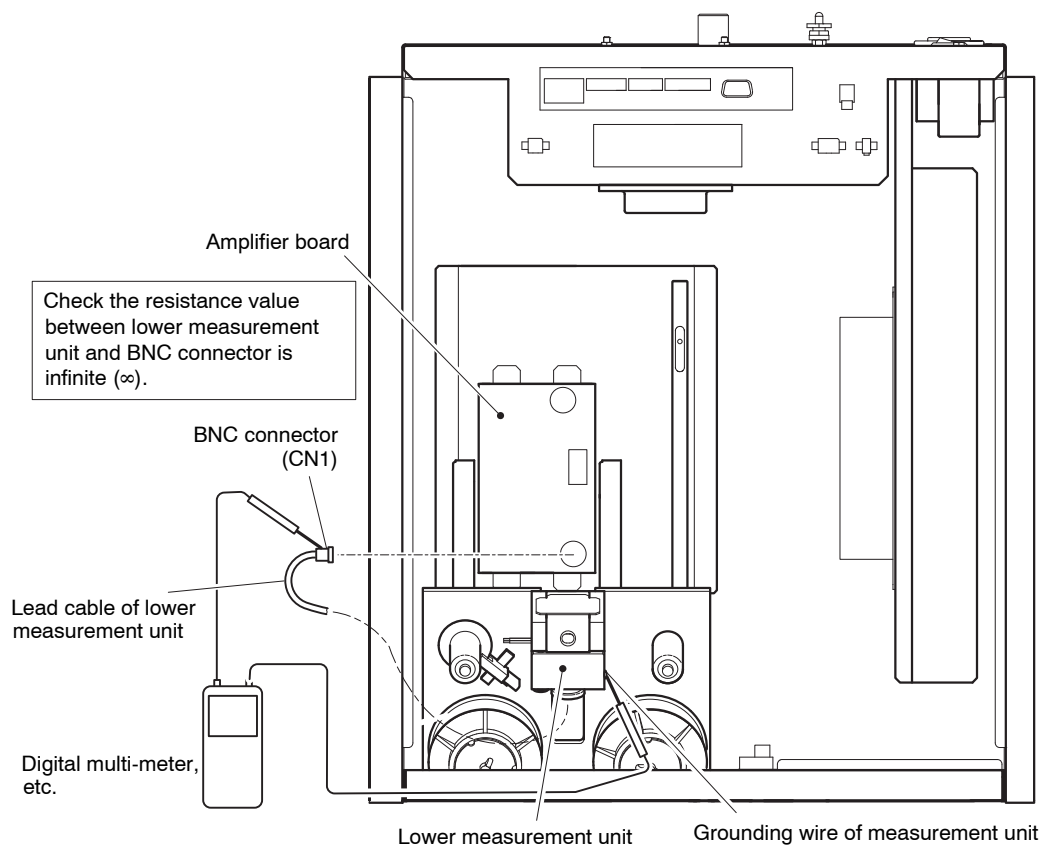


Grounding Wire of Measurement Unit

6.3(2) “③ Restore the measurement part”

- 6.3(2) “④ Fix the dehumidifier”
- 6.3(2) “⑤ Attach the front panel”
- 6.3(2) “⑥ Attach the window cover”
- 6.3(2) “⑦ Turn on the power ”

(2) Check the insulation of lower measurement unit



Check of Insulation Property

① Prepare the devices.

Digital multi-meter, etc. Measure the infinite of resistance value (Ω).

② Carry out the operation before maintenance. >> 6.3(1) “Operation before maintenance”

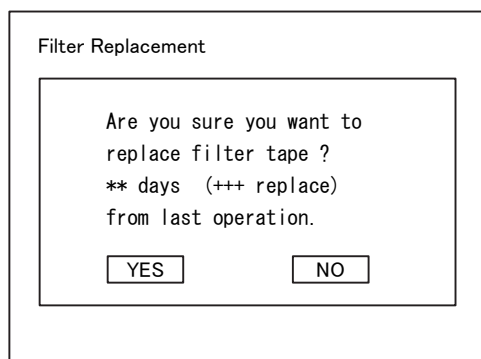
- 6.3(1) “① Turn off the power”
- 6.3(1) “② Remove the window cover”
- 6.3(1) “③ Remove the retainer of dehumidifier”
- 6.3(1) “④ Remove the front panel”
- 6.3(1) “⑤ Check the O-ring”
- 6.3(1) “⑥ Pull out the measurement part”
- 6.3(1) “⑦ Pull out the dehumidifier”
- 6.3(1) “⑧ Remove the upper cover”

- ③ **Disconnect BNC connector.** Disconnect BNC connector (CN1) of lead cable of lower measurement unit from the amplifier board
- ④ **Connect the meter.** Connect a digital multi-meter, etc. to between the disconnected BNC connector and a grounding wire of measurement unit at the lower measurement unit.
- ⑤ **Check the resistance value.** Set the measurement item of meter to resistance value (Ω) and check it indicates “ ∞ ” (infinite).
- If the resistance value is infinite, the insulation property of semiconductor detector is ensured. If this is not the case, refer to 7. “Troubleshooting”.
- ⑥ **Connect BNC connector.** Connect the BNC connector of lead cable of lower measurement unit, which was disconnected once, to the amplifier board again in original state.
- ⑦ **Carry out the operation after maintenance.** > > 6.3(2) “Operation after maintenance”
- 6.3(2) “① Attach the upper cover”
- 6.3(2) “② Attach the dehumidifier”
- 6.3(2) “③ Restore the measurement part”
- 6.3(2) “④ Fix the dehumidifier”
- 6.3(2) “⑤ Attach the front panel”
- 6.3(2) “⑥ Attach the window cover”
- 6.3(2) “⑦ Turn on the power”

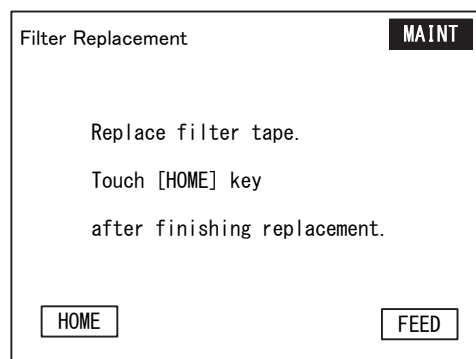
6.10 Cleaning of Flow Control Valve

- ① **Display “Filter Replacement Confirmation” screen.** Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.

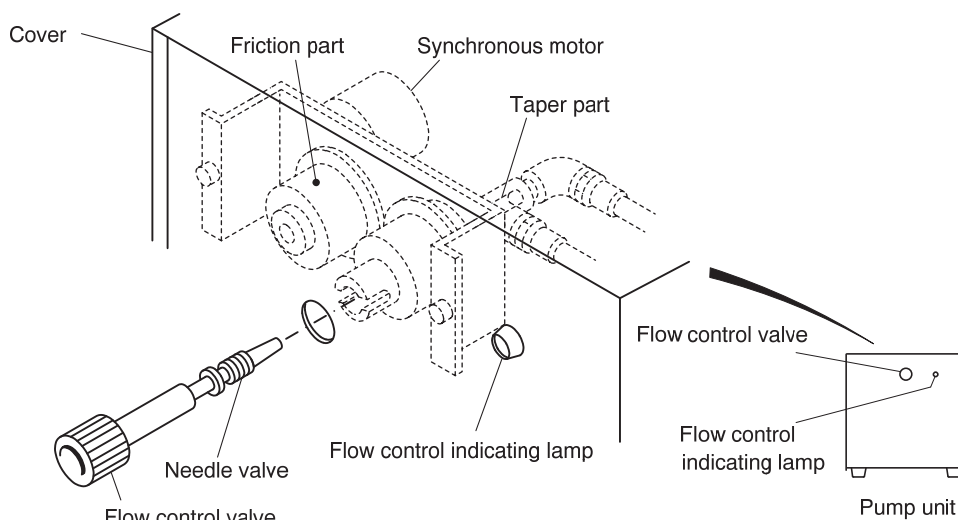


Filter Confirmation Screen



Filter Replacement Screen

- ② **Turn off the power of pump.** Disconnect the power cable of pump from an outlet for safety.
- ③ **Pull out the needle valve.** Rotate the flow control valve of pump unit in anticlockwise direction and pull out the needle valve.



Flow Control Valve

- ④ **Carry out cleaning.** Clean up the taper part at the tip of needle valve with a paper wiper instilled in alcohol, etc. Clean up the taper part of opposite side (concave side) in which taper part of needle valve enters with a paper wiper twisted around a bamboo stick, etc. instilled in alcohol.
- ⑤ **Restore the needle valve.** Embed the needle valve in the original state.
- ⑥ **Turn on the power of pump.** Connect the power cable of pump to an outlet.
- ⑦ **Display “Home” screen.** Press **[HOME]** on “Filter Replacement” screen.
- The monitor goes into the normal measurement state.

⑧ Check the flow rate.

- ① Display “Data Display Menu” screen (2/3). Press **DATA** on “Home” screen and press **↓** once on the displayed “Data Display Menu” screen (1/3).

	DATA	
	Amb. Temp. 25.0°C	$\frac{2}{3}$
	Amb. Humid. 50%	
	Amb. Pressure 92.0kPa	
	Heater Temp. 45.0°C	
Sample actual flow rate →	Real Flow Rate 16.7L/min	
	Int. STD Vol. 56.3L	
	HOME	CF Log Data

Actual Flow Rate of Sample on Data Display Menu Screen (2/3)

- ② Check the flow control indicating lamp. Check the flow control indicating lamp of pump is turned off.
- ③ Check the actual flow rate of sample. Check the actual flow rate of sample on “Data Display Menu” screen (2/3) is within the following range.

Range of actual flow rate of sample 16.7 ± 0.3 L/min

- If the flow rate enters this range within 5 seconds after the step “⑥ Turn on the power of pump”, the step “⑨ Embed the needle valve again” is not necessary. Go to “⑩ Display “Home” screen”.

- ⑨ **Embed the needle valve again.** If the actual flow rate of sample does not enter the range, carry out steps “② Turn off the power of pump” to “⑧ Check the flow rate” again.

- ⑩ **Display “Home” screen.** Press **HOME** on “Data Display Menu” screen (2/3).

- The monitor goes into the normal measurement state.

6.11 Replacement of Pump Unit Parts

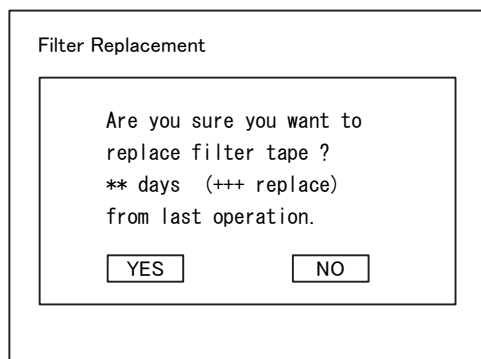
(1) Replacement of diaphragm

① Prepare the replacement parts.

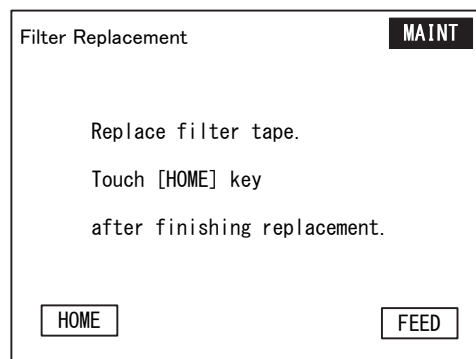
[Replacement part] Diaphragm Name of product: Diaphragm for MF-50D, No.125B851

② Display “Filter Replacement Confirmation” screen. Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the downward movement of lower measurement unit stops, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



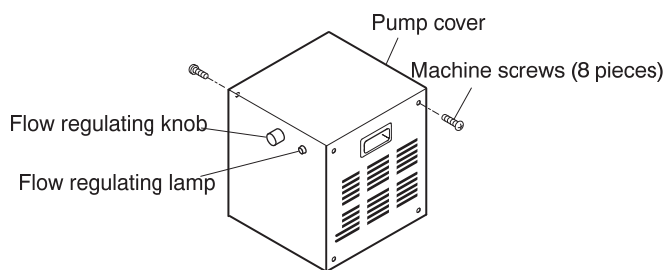
Filter Confirmation Screen



Filter Replacement Screen

③ Turn off the power of pump. Disconnect the power cable of pump from an outlet for safety.

④ Remove the pump cover. Loosen machine screws (right / left, 4 pieces each) and remove the pump cover.



Pump Unit

⑤ Remove the diaphragm cover. Loosen the hexagon bolts (4 pieces) and remove one diaphragm cover from the casing.

- There are diaphragms on both sides of pump. After the replacement of one diaphragm is finished, carry out the replacement of other diaphragm.

⑥ Pull out the diaphragm. Loosen a cross-recessed head countersunk bolt and remove the used diaphragm.

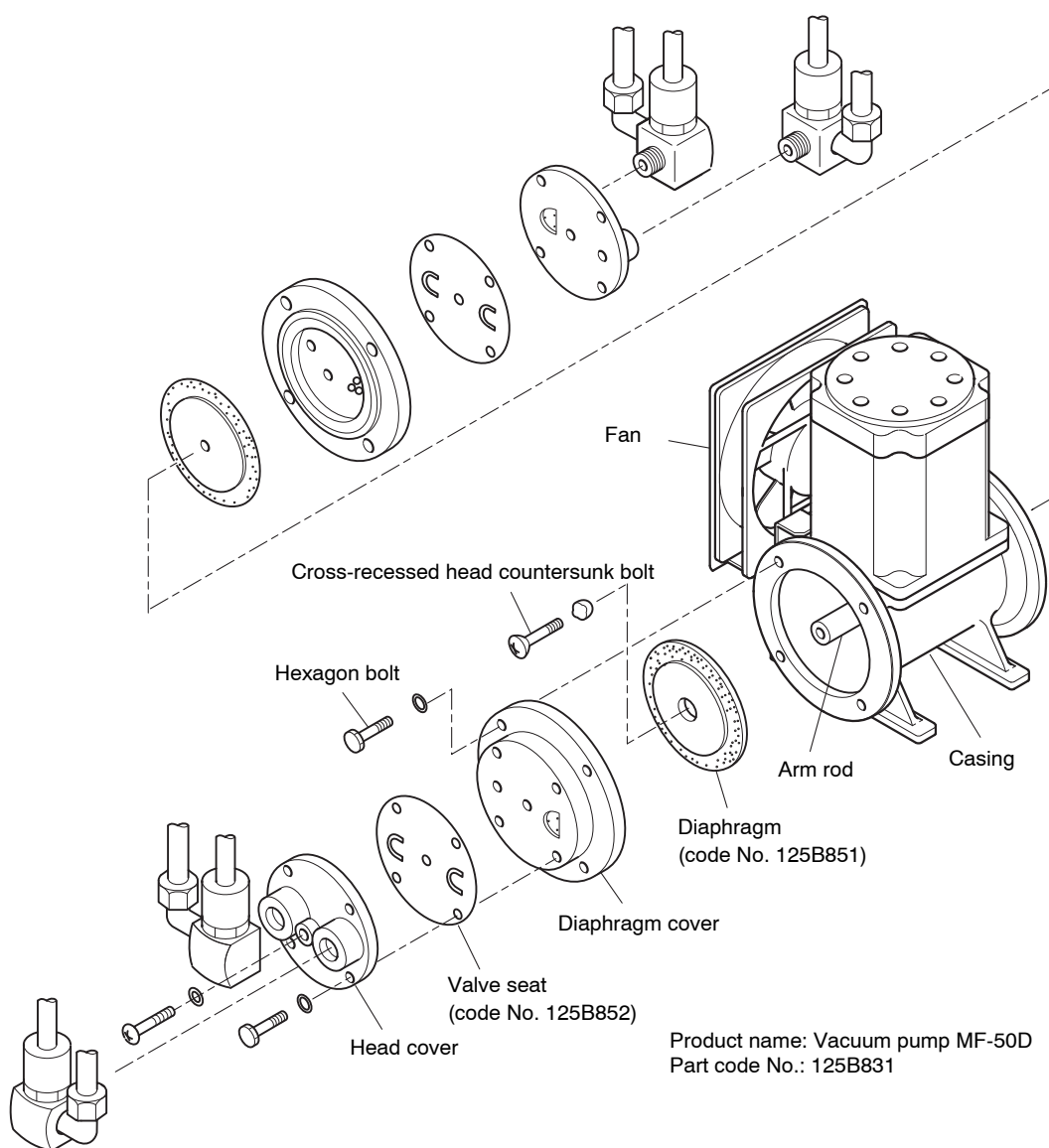
[NOTE] • If the diaphragm cover and the inside of casing is dirty, wipe them off carefully with a gauze dipped in alcohol, etc.

⑦ Pull out the arm rod. While rotating the fan with a screwdriver or other tool, pull out the arm rod forward to the end.

⑧ Embed the diaphragm. Fix the new diaphragm to the arm rod with a cross-recessed head countersunk bolt.

【IMPORTANT】 • Attach the diaphragm cover to the rotating position shown in the figure. If the rotation position is misaligned, it does not play the role.

- ⑨ **Restore the diaphragm cover.** Attach the diaphragm cover in the original state.
- ⑩ **Replace the other diaphragm.** Replace the diaphragm on opposite side according to the steps “⑥ to ⑨”.
- ⑪ **Attach the pump cover.** Attach the pump cover in the original state.
- ⑫ **Turn on the power of pump.** Connect the power cable of pump unit to the outlet again in the original state.
- ⑬ **Display “Home” screen.** Press HOME on “Date Display Menu” screen (2/3).
 - The monitor goes into the normal measurement state.
- ⑭ **Calibrate the flow rate.** >> 4.3 “Calibration of Flow Rate by Standard Flowmeter”



Disassembly of Pump Unit

(2) Replacement of valve seat

Refer to the figures “Pump Unit” and “Disassembly of Pump Unit” in 6.11 (1) “Replacement of diaphragm”.

① Prepare the replacement parts.

[Replacement part] Valve seat Name of product: Valve seat for MF-50D, No.125B852

② Carry out the operation before replacement. >> 6.11(1) “Replacement of diaphragm”

6.11(1) “② Display “Filter Replacement Confirmation” screen”

6.11(1) “③ Turn off the power of pump”

6.11(1) “④ Remove the pump cover”

③ Remove the head cover. Loosen the hexagon bolts (4 pieces) and remove the one head cover.

- The valve seats exist on both sides of pump. After the replacement of one side is finished, carry out the other replacement.

④ Remove the valve seat. Open the section between diaphragm cover and head cover using a flat-blade screwdriver or other tool and pull out the used valve seat.

⑤ Wipe off the taint on the surface. If the valve seat attachment surfaces of diaphragm cover and head cover are dirty, wipe them carefully with a paper wiper, etc. dipped in alcohol.

⑥ Assemble the valve seat. Set the rotating position of a new valve seat properly and put it on the diaphragm cover and place the head cover on it, and then assemble them with hexagon bolts.

【IMPORTANT】 • Assemble diaphragm cover, valve seat, and head cover keeping the proper rotating position separately.

⑦ Replace the other valve seat. Replace the other (opposite side) valve seat according to the steps “③ to ⑥”.

⑧ Carry out the operation after replacement. >> 6.11(1) “Replacement of diaphragm”

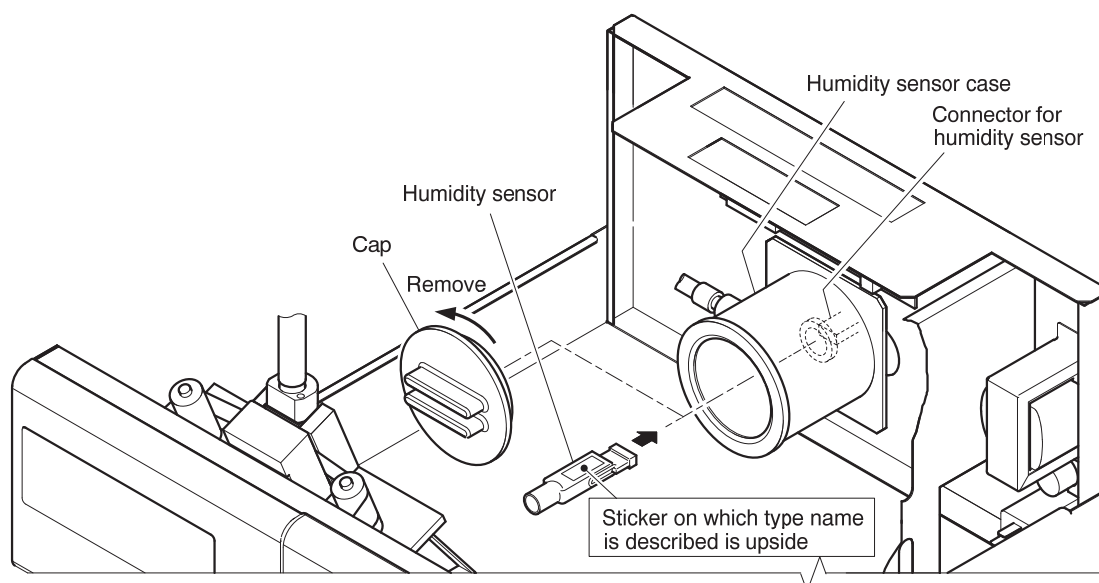
6.11(1) “⑪ Attach the pump cover”

6.11(1) “⑫ Turn on the power of pump”

6.11(1) “⑬ Display “Home” screen”

6.11(1) “⑭ Calibrate the flow rate”

6.12 Replacement of Humidity Sensor



Replacement of Humidity Sensor

① Prepare the replacement parts.

[Replacement part] Humidity sensor Name of product: Humidity sensor assembly,
No.7205580K

② Carry out the operation before maintenance. >> 6.3(1) "Operation before maintenance"

- 6.3(1) "① Turn off the power "
- 6.3(1) "② Remove the window cover"
- 6.3(1) "③ Remove the retainer of dehumidifier"
- 6.3(1) "④ Remove the front panel"
- 6.3(1) "⑤ Check the O-ring"
- 6.3(1) "⑥ Pull out the measurement part"
- 6.3(1) "⑦ Pull out the dehumidifier"
- 6.3(1) "⑧ Remove the upper cover"

③ Remove the cap. Remove the cap from the humidity sensor case.

④ Remove the used temperature sensor. Pull out the used temperature sensor slowly.

⑤ Connect a new sensor. Connect a new humidity sensor to the connector on back side of humidity sensor case.

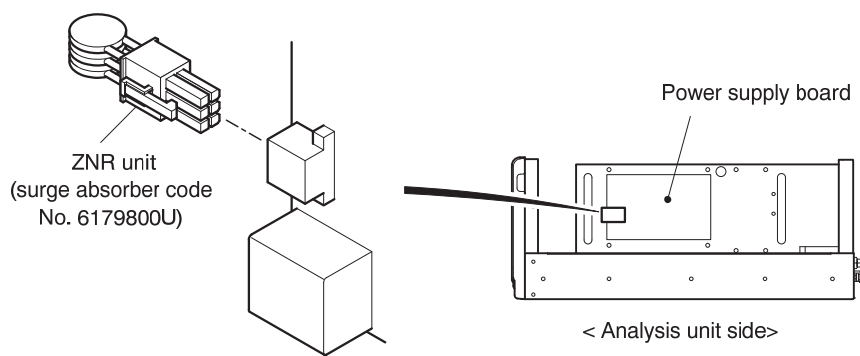
⑥ Attach the cap. Twist the cap in the back side of humidity sensor case slowly.

⑦ Carry out the operation after maintenance. >> 6.3(2) "Operation after maintenance"

- 6.3(2) "① Attach the upper cover"
- 6.3(2) "② Attach the dehumidifier"
- 6.3(2) "③ Restore the measurement part"
- 6.3(2) "④ Fix the dehumidifier"
- 6.3(2) "⑤ Attach the front panel"
- 6.3(2) "⑥ Attach the window cover"
- 6.3(2) "⑦ Turn on the power "

6.13 Replacement of ZNR Units

- (a) Replace the ZNR unit with a new one periodically. In addition, carry out the replacement when the lightning might affect the monitor.
- (b) In a region where is frequently struck by lightning, attach a commercially available product for taking measures against lightning to the system for supplying power to the monitor.



Replacement of ZNR Unit

① Prepare the replacement parts.

[Replacement part] ZNR unit Name of product: ZNR unit (single), No.7317890U, 3 pieces

② Carry out the operation before maintenance. >> 6.3(1) "Operation before maintenance"

- 6.3(1) "① Turn off the power"
- 6.3(1) "② Remove the window cover"
- 6.3(1) "③ Remove the retainer of dehumidifier"
- 6.3(1) "④ Remove the front panel"
- 6.3(1) "⑤ Check the O-ring"
- 6.3(1) "⑥ Pull out the measurement part"
- 6.3(1) "⑦ Pull out the dehumidifier"
- 6.3(1) "⑧ Remove the upper cover"

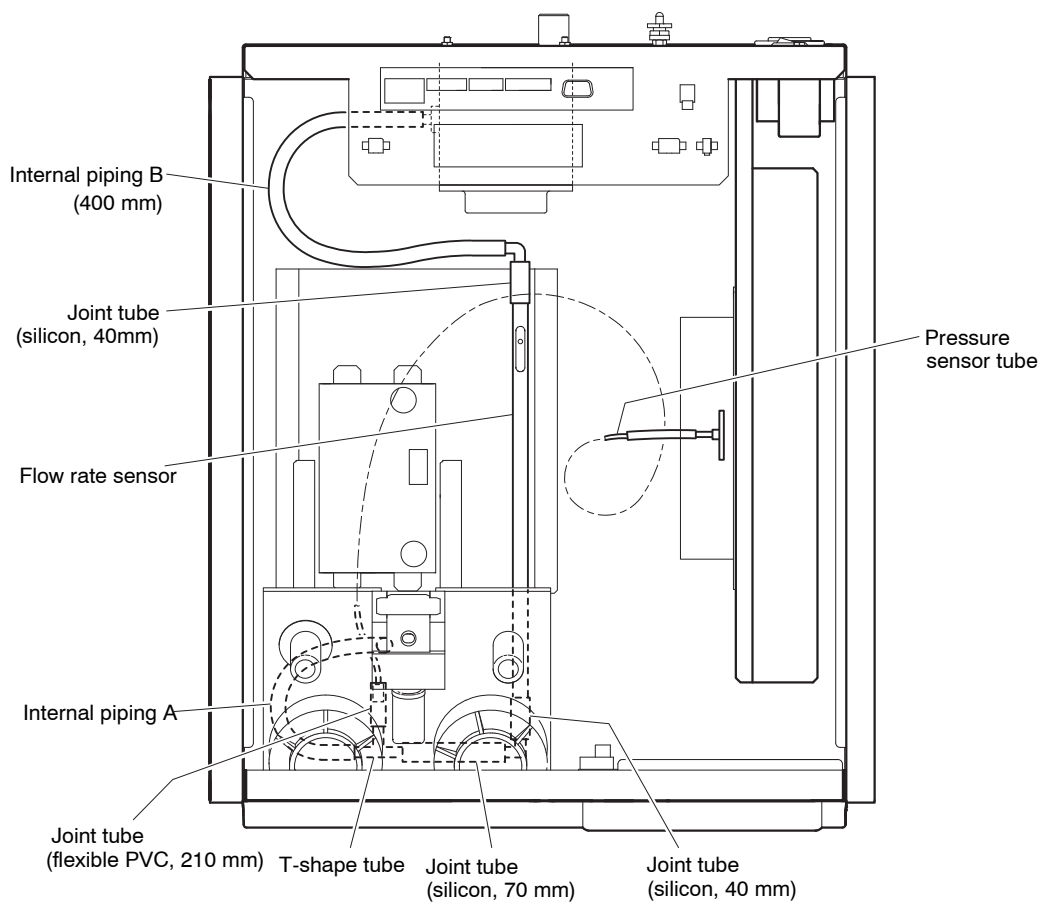
③ Replace the ZNR unit. Disconnect the used ZNR unit on the power supply board and connect a new ZNR unit.

④ Carry out the operation after maintenance. >> 6.3(2) "Operation after maintenance"

- 6.3(2) "① Attach the upper cover"
- 6.3(2) "② Attach the dehumidifier"
- 6.3(2) "③ Restore the measurement part"
- 6.3(2) "④ Fix the dehumidifier"
- 6.3(2) "⑤ Attach the front panel"
- 6.3(2) "⑥ Attach the window cover"
- 6.3(2) "⑦ Turn on the power"

6.14 Replacement of Internal Piping

The main piping of monitor is shown below. Replace the piping periodically.



Internal Piping

① Prepare a new piping.

[Replacement parts]

Internal piping A / B High plain tube ($\phi 7 \times \phi 11$), No. 116E022, necessary length: approx. 610mm

Joint tube Silicon tube ($\phi 8 \times \phi 14$), No. 116C011

② Carry out the operation before maintenance. >> 6.3(1) "Operation before maintenance"

- 6.3(1) "① Turn off the power"
- 6.3(1) "② Remove the window cover"
- 6.3(1) "③ Remove the retainer of dehumidifier"
- 6.3(1) "④ Remove the front panel"
- 6.3(1) "⑤ Check the O-ring"
- 6.3(1) "⑥ Pull out the measurement part"
- 6.3(1) "⑦ Pull out the dehumidifier"
- 6.3(1) "⑧ Remove the upper cover"

③ Disconnect the used piping. Disconnect internal piping A and internal piping B.

- If a piping cannot be disconnected even by using a flat-blade screwdriver or other tool, cut it by using a cutter knife, etc.

【IMPORTANT】 • Take care not to damage T-shape tube or joint.

④ Connect a new piping. Cut a new tube in the length of internal piping A and internal piping B, and then connect them surely not to cause leakage.

⑤ Carry out the operation after maintenance. > > 6.3(2) “Operation after maintenance”

6.3(2) “① Attach the upper cover”

6.3(2) “② Attach the dehumidifier”

6.3(2) “③ Restore the measurement part”

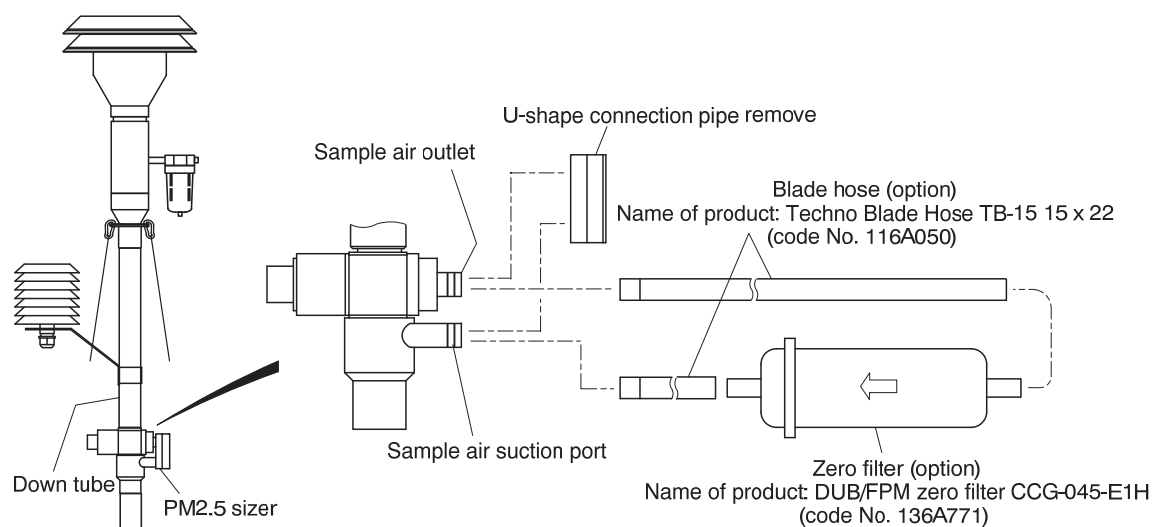
6.3(2) “④ Fix the dehumidifier”

6.3(2) “⑤ Attach the front panel”

6.3(2) “⑥ Attach the window cover”

6.3(2) “⑦ Turn on the power”

6.15 Blank Test



Connection of Zero Filter

WARNING

Fall Injury

- If there is a possibility of fall at high place such as roof of monitoring station, wear safety belt, etc. to prevent falling. In addition, to prevent injury, wear a helmet, life jacket, safety shoes, etc.

① Prepare the device.

Zero filter Name of product: DUB/FPM zero filter CCG-045-E1H No. 136A771

Blade hose $\phi 15 \times \phi 22$, length to which standard flowmeter can be connected

Name of product: Techno blade hose TB-15 15x22 code No. 116B050

② Clean up the duct of sample air. Carry out the operations related to cleaning in the following items.

6.4 “Cleaning of Impactor and Down Tube”

6.5 “Cleaning of PM2.5 Sizer”

6.6 “Cleaning of Dehumidifier and Replacement of O-ring”

6.8 “Cleaning of Radiation Source Window and Detection Window”

6.14 “Replacement of Internal Piping”

③ Carry out confirmation or calibration. Refer to the following items to carry out confirmation or calibration.

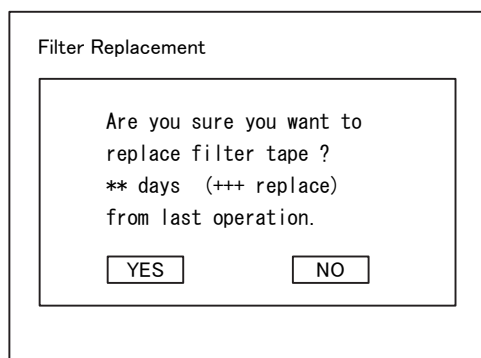
4.3 “Calibration of Flow Rate by Standard Flowmeter”

5.4(5) “Calibration of temperature, humidity, and pressure”

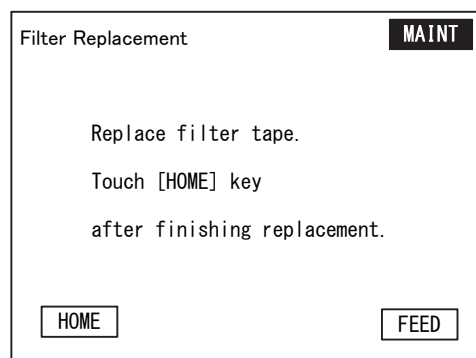
4.2 “Internal Calibration”

④ **Display “Filter Replacement Confirmation” screen.** Press **[FILTER]** on “Home” screen and press **[YES]** on the displayed “Filter Paper Replacement Confirmation” screen.

- If the pump stops and the lower measurement unit stops the downward movement, “Filter Replacement” screen is displayed and the temporary stop of measurement is continued.



Filter Confirmation Screen



Filter Replacement Screen

- ⑤ **Connect the zero filter.** Disconnect U-shape connection pipe of PM2.5 sizer unit and connect the zero filter (option) in between sample air discharge port and suction port using the blade hose.
- ⑥ **Check the measurement value.** Check the average value in 20 to 24 hours of 1-hour average value is within $\pm 5\mu\text{g}/\text{m}^3$.
- ⑦ **Return to sample air measurement.** Disconnect the blade hose with zero filter (option) from PM2.5 sizer unit and connect the U-shape connection pipe in the original state.
- ⑧ **Display “Home” screen.** Press **[HOME]** on “Data Display Menu” screen (2/3).
- The monitor goes into the normal measurement state.

6.16 Replacement of Lithium Battery

- (a) While the operation of monitor is stopped (turn off the main power), the battery is burned. It may be necessary to replace the lithium battery depending on the operation stop period.

Rough Indication of Lithium Battery Lifetime

Operation status of monitor	Rough indication of battery lifetime
Operation stop (power off)	Approx. 1 to 2 years
Intermittent operation (5000h/year)	Approx. 2 to 4 years
Continuous operation	Approx. 6 years

[NOTE] • The lithium battery is used for keeping the element in which date, time, and recorded data are memorized when the power supply of monitor is turned off.

- (b) If a battery error alarm (A704) is generated when the operation is restarted after the long-term operation stop, replace the lithium battery. After the replacement of battery, set date and time again.

[IMPORTANT] • If lithium battery is burned or pulled out, the setting of date and time is initialized and the log data (logging data) saved in the internal memory is cleared.

- (c) It is recommended to pull out the recorded measurement data by using a CF card before long-term operation stop.

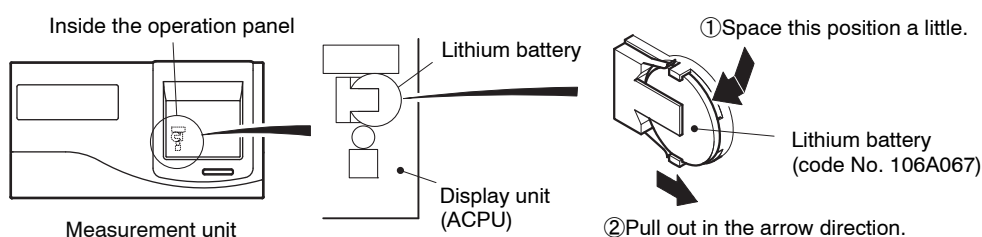
① Prepare the replacement parts.

[Replacement part] Lithium battery Name of product: Lithium battery CR2032, No.106B067

② Carry out the operation before maintenance. >> 6.3(1) “Operation before maintenance”

- 6.3(1) “① Turn off the power ”
 6.3(1) “② Remove the window cover”
 6.3(1) “③ Remove the retainer of dehumidifier”
 6.3(1) “④ Remove the front panel”
 6.3(1) “⑤ Check the O-Ring”
 6.3(1) “⑥ Pull out the measurement part”
 6.3(1) “⑦ Pull out the dehumidifier”
 6.3(1) “⑧ Remove the upper cover”

③ Replace the lithium battery. Pull out the used lithium battery in the display unit (ACPU) in the operation panel of measurement unit and attach a new lithium battery.



Removal of Lithium Battery

WARNING

Lithium Battery

- Do not charge, short, disassemble, deform, heat up lithium battery or put it in fire.
 - Discard the lithium battery as an industrial waste. When you discard it, insulate it by putting a tape on the terminal. If the battery contacts other metal or battery, it leads to heat generation, explosion, or ignition.
-

④ Carry out the operation after maintenance. >> 6.3(2) “Operation after maintenance”

6.3(2) “① Attach the upper cover”

6.3(2) “② Attach the dehumidifier”

6.3(2) “③ Restore the measurement part”

6.3(2) “④ Fix the dehumidifier”

6.3(2) “⑤ Attach the front panel”

6.3(2) “⑥ Attach the window cover”

6.3(2) “⑦ Turn on the power”

⑤ Reset the alarm. Reset the battery error alarm (A704). If an SRAM error alarm (A720) is generated, reset it too. >> 5.2 (2) “Confirmation and reset of alarm information”

[NOTE] • If the power is turned on again after the replacement of battery, an SRAM error alarm may be generated by clearance of data.

⑥ Set date and time. If date and time on “Home” screen are not correct, change them to the current date and time. >> 5.5(2) Change of date and time

- In case of indoor type, the step “⑦” shown below is not necessary.

7. Troubleshooting

7.1 Alarm Symbols and Measures

- (a) If an alarm is generated, refer to the table shown below to check the content corresponding to the alarm symbol, and then take measures. If an alarm is generated again after that, please contact the sales dealer.
- (b) For alarm screen and importance (alarm 1, alarm 2), refer to 5.2(2) “Confirmation and reset of alarm information”.
- (c) The “Content” in the following table is displayed on “Help” screen. For the arrangement of boards and connectors, refer to the figure “Arrangement and Connection of Connectors”.



Electric Shock

- Do not touch the terminals inside the product while power is applied. Touching the terminals can cause electric shock.

Alarm Symbols and Measures

Alarm symbol	Segment	Type	Content	Measures
Measurement operation error				
A725	AL2	Downward movement of measurement unit	The system is stopped because the downward movement of lower measurement unit did not stop and 5 seconds passed.	- Check the connection of connectors CN26 and CN23 of I/O board. - After taking measures, it is necessary to turn on and off the power.
A726	AL2	Upward movement of measurement unit	The system is stopped because the upward movement of lower measurement unit did not stop and 10 seconds passed.	- Check the connection of connectors CN26 and CN23 of I/O board. - After taking measures, it is necessary to turn on and off the power.
A727	AL2	Filter paper feeding operation	The system is stopped because the filter paper feeding operation did not stop and 10 seconds passed.	- Check the connection of connectors CN26 and CN23 of I/O board. - After taking measures, it is necessary to turn on and off the power.
A730	AL1	Temperature in measurement unit	The temperature in the measurement unit exceeded the range -20 to 50°C.	- To keep the temperature in measurement unit within the range, set the peripheral temperature within 0 to 40°C. - Check the connection of connector CN8 of I/O board.
A731	AL1	Temperature of measurement unit	The temperature of measurement unit exceeded the range -20 to 70°C.	Check the connection of connector CN7 of I/O board.

(To be continued)

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Alarm symbol	Segment	Type	Content	Measures
A732	AL1	Outdoor air temperature	The outdoor air temperature increased 70°C or more.	Check the connection of connector CN9 of I/O board and connector for outdoor air temperature of relay board.
A733	AL1	Temperature sensor	The temperature of temperature sensor exceeded the range -20 to 70°C.	Check the connection of connector CN7 of I/O board.
A734	AL2	Startup of pump	The pump was started up three times, however, the pressure of measurement unit when the pump was turned on does not become 5kPa or less for the atmosphere pressure.	Check the following items. 1. Connection of power cable for pump, control cable, and pump tube 2. Periodical replacement of diaphragm for pump 3. Connection of connector CN11 of I/O board
A735	AL1	Peripheral pressure	The peripheral pressure exceeded the range 50 to 130kPa.	Check the connection of connector CN12 of I/O board.
A736	AL1	Humidity of sample	The humidity of sample exceeded the range 0 to 100%.	Check the connection of connector CN13 of I/O board.
A737	AL1	Actual flow rate of sample	The actual flow rate of sample exceeded the range 16.7±1.5L/min.	Check the following items. 1. Connection of connector CN19 of I/O board 2. Connection of power cable for pump, control cable, and pump tube 3. Periodical replacement of diaphragm for pump 4. Disconnection or clogging of piping
A738	AL1	Noise voltage	The noise voltage increased to 1000mV or more.	Check the connection of connectors CN1 and CN14 of I/O board and the connection of connectors CN2 and CN3 of amplifier board.
A739	AL1	Movement of filter paper is frequently generated	- The movement of filter paper was generated three times during the measurement. - When the pressure of measurement unit decreased by 40kPa (large pump: 50kPa) or more than atmosphere pressure and 180 seconds passed, the movement of filter paper should be carried out.	Contact the sales dealer.
A740	AL2	Shortage of filter paper	The pressure of measurement unit increased than -5kPa pressure of atmosphere pressure.	Check the following items. 1. Damage of filter paper 2. Connection of power cable for pump, control cable, and pump tube 3. Periodical replacement of diaphragm for pump 4. Clogging of internal piping 5. Connection of connector CN11 of I/O board
A741	AL2	Clogging of filter paper	The pressure of measurement unit decreased by 50kPa (large pump: 60kPa) or more than atmosphere pressure.	Check the clogging of filter paper and the clogging of internal piping.

(To be continued)

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Alarm symbol	Segment	Type	Content	Measures
A743	AL2	Upper / lower limit of heater temperature	The heater temperature exceeded the range -20 to 80°C.	Check the connection of connectors CN6 and CN21 of I/O board
A750	AL2	DC power supply (+5V)	The voltage of 5V power supply for I/O board (SCPU) decreased.	Check the connection of connector CN18 of I/O board and CN4 of power supply board.
A760	AL1	Communication error of external terminal (A)	The communication between display unit (ACPU) and external terminal unit cannot be performed.	Check the following items. 1. Connection of external terminal unit and connector for external terminal on rear surface of measurement unit 2. Connection of connector CN16 of I/O board and CN7 of relay board 3. Connection of connector CN17 of I/O board and J10 of display unit
A761	AL1	Checksum error (A)	The communication between display unit (ACPU) and external terminal unit cannot be performed.	Contact the sales dealer.
A762	AL1	Response to error of external terminal (A)	The communication between display unit (ACPU) and external terminal unit cannot be performed.	Contact the sales dealer.
A766	AL1	Communication error of recorder	The communication between display unit (ACPU) and recorder cannot be performed.	Check the following items. 1. Connection of recorder and connector for recorder on rear surface of measurement unit 2. Connection of connector CN16 of I/O board and CN7 of relay board 3. Connection of connector CN17 of I/O board and J8 of display unit
A767	AL1	RS-232C communication error	The communication between display unit (ACPU) and external terminal unit cannot be performed.	Contact the sales dealer.

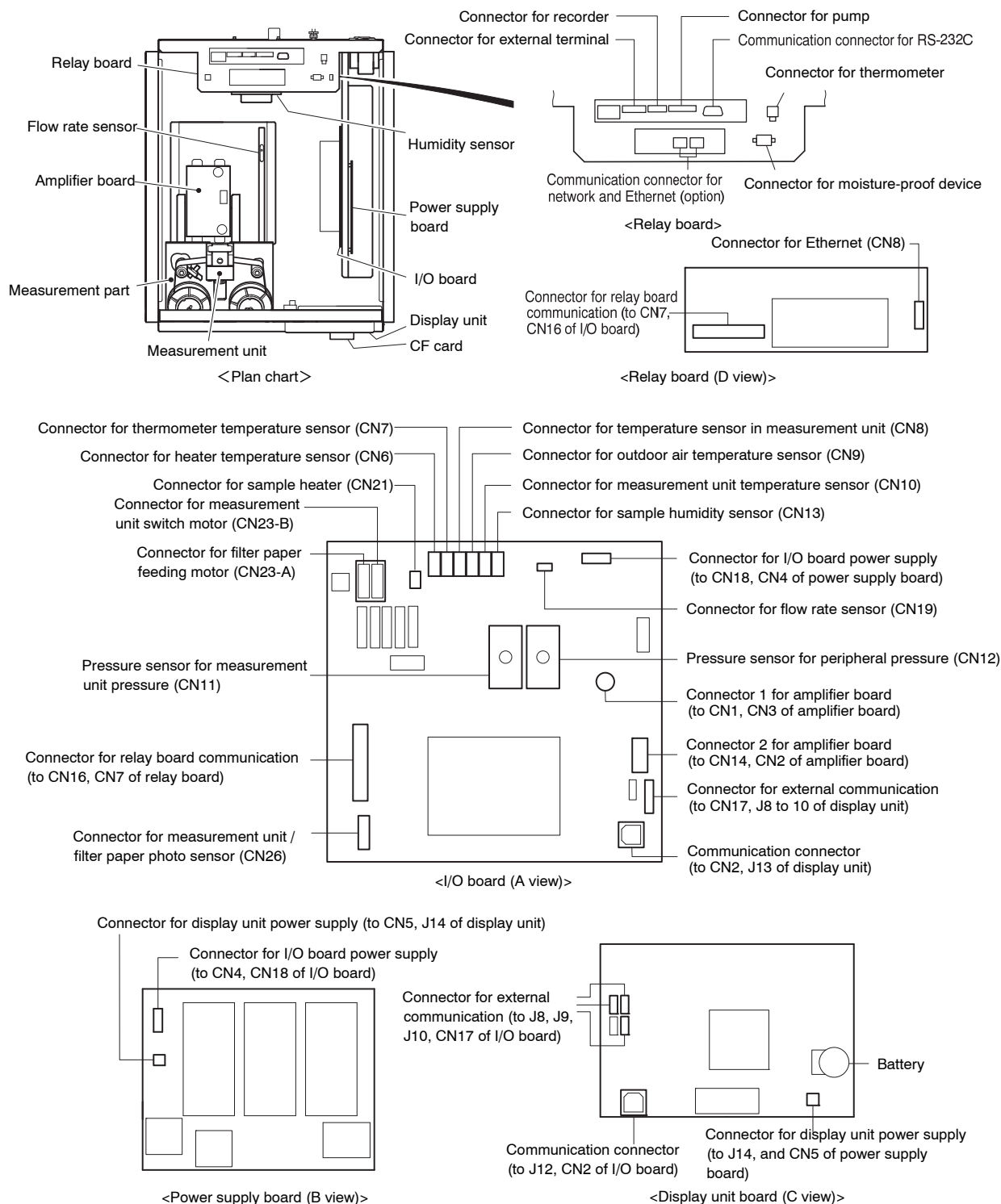
System error

A702	AL1	Initialization of parameter (A)	The parameter (variable) managed by display unit (ACPU) was reset to initial value.	The set value adjustment is necessary. Please contact the sales dealer.
A703	AL1	Initialization of parameter (S)	The parameter (variable) managed by I/O board (SCPU) was reset to initial value.	The set value adjustment is necessary. Please contact the sales dealer.
A704	AL1	Battery error	The battery of display unit (ACPU) was burned and the recorded data may be lost.	Insert a new battery properly. >> 6.16 “Replacement of Lithium Battery”
A710	AL1	Memory shortage of CF card	The remaining capacity of CF card became 1% or less.	Carry out the reduction of data in DATA folder of CF card.
A711	AL1	Writing error of CF card	The data cannot be written in the CF card.	- Use the CF card sold by our company. - Replace the CF card with a new one.
A712	AL1	EEPROM error (A)	EEPROM of display unit (ACPU) is defective and the parameter (variable) cannot be written in it.	Turn on the power again.

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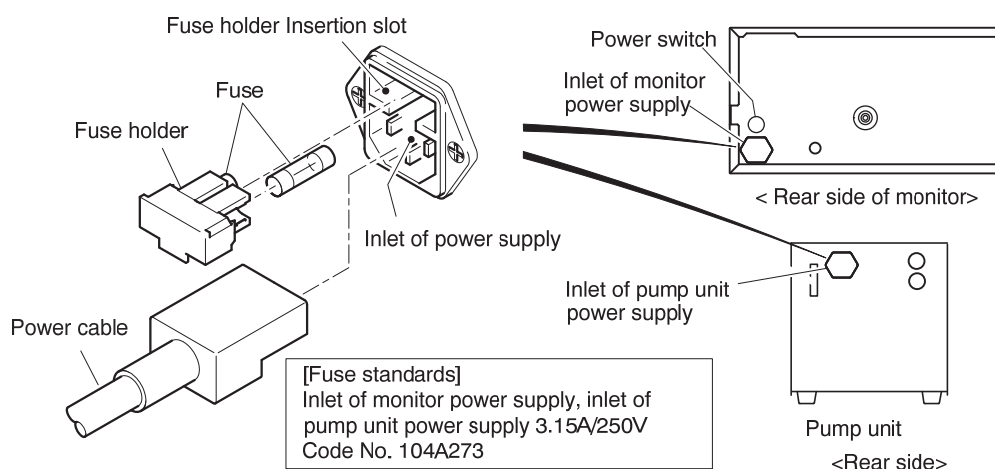
Alarm symbol	Segment	Type	Content	Measures
A713	AL1	EEPROM error (S)	• EEPROM of I/O board (SCPU) is defective and the parameter (variable) cannot be written in it.	• Turn on the power again.
A714	AL1	Writing error (A)	• The display unit (ACPU) is defective and the data cannot be sent to I/O board (SCPU).	• Turn on the power again.
A715	AL1	Writing error (S)	• I/O board (SCPU) is defective and the data cannot be sent to the display unit (ACPU).	• Turn on the power again.
A716	AL1	Mail error (A)	• The display unit (ACPU) is defective and the mail cannot be sent to I/O board (SCPU).	• Turn on the power again.
A717	AL1	Mail error (S)	• I/O board (SCPU) is defective and the mail cannot be sent to the display unit (ACPU)	• Turn on the power again.
A718	AL1	Link error (A)	• The communication between display unit (ACPU) and I/O board (SCPU) cannot be performed.	• Check the connection of connector J12 of display unit and connector CN2 of I/O board.
A720	AL1	SRAM error	• The parameter (variable) memorized in SRAM of display unit (ACPU) is abnormal.	• Turn on the power again.
A721	AL1	Parameter range error (A)	• The parameter (variable) memorized in EEPROM of display unit (ACPU) is abnormal.	• Turn on the power again.
A722	AL1	Parameter range error (S)	• The parameter (variable) memorized in EEPROM of I/O board (SCPU) is abnormal.	• Turn on the power again.



Arrangement and Connection of Connectors

7.2 Replacement of Fuses

There are fuses at the inlet of power supply of monitor and the inlet of power supply of pump unit. Check the fuse that seems to be blown and if it is blown, replace it with a new one.



Replacement of Fuses

① Prepare the replacement parts.

[Replacement part] Fuse Name of product: Fast acting fuse 217 3.15 3.15A 250V
Code No.: 103A273 Quantity: 1 to 4

② Disconnect the power cable. Disconnect the power cable from the inlet of power supply of monitor or the inlet of power supply of pump.

③ Pull out the fuse holder. Pull out the fuse holder at the inlet of power supply using a flat-blade screwdriver or other tool.

- Two fuses are inserted in the fuse holder. Two fuses function separately.

④ Check whether the fuse is blown. Disconnect two fuses from the fuse holder and check whether the wires in the fuses are broken.

- If both of two fuses are not blown, the steps “⑤ and ⑥” are not necessary. Restore the fuses in the fuse holder and go to the step “⑦ Restore the fuse holder”.

⑤ Eliminate the cause. If the fuse is blown, check the cause of blow-out, and then go to the next step.

⑥ Replace the fuse. Insert a new fuse in the fuse holder instead of blown fuse.

[IMPORTANT] • Two normal fuses need to be inserted in the fuse holder. Even if only one fuse is normal, the monitor does not operate.

⑦ Restore the fuse holder. Restore the fuse holder to the inlet of power supply again.

⑧ Connect the power cable. Connect the power cable to the inlet of power supply in the original state.

⑨ Turn on the power. To restart the operation, connect the power cable to the outlet and turn on the power switch (press I side).

- **Warm-up** and **MAINT** appear and they will disappear if the monitor enters measurement status (in 60 minutes at a maximum). >> “⑧ Wait for the measurement to be stabilized” in 3.1 “Procedure for Starting the Automatic Measurement”

●Repair contact

If a repair is required, please contact your sales representative or directly our sales office, or our service department. In this case, let us know the following information:

Model name (MODEL)

Serial number (SER. No.)

Manufacturing date (DATE)

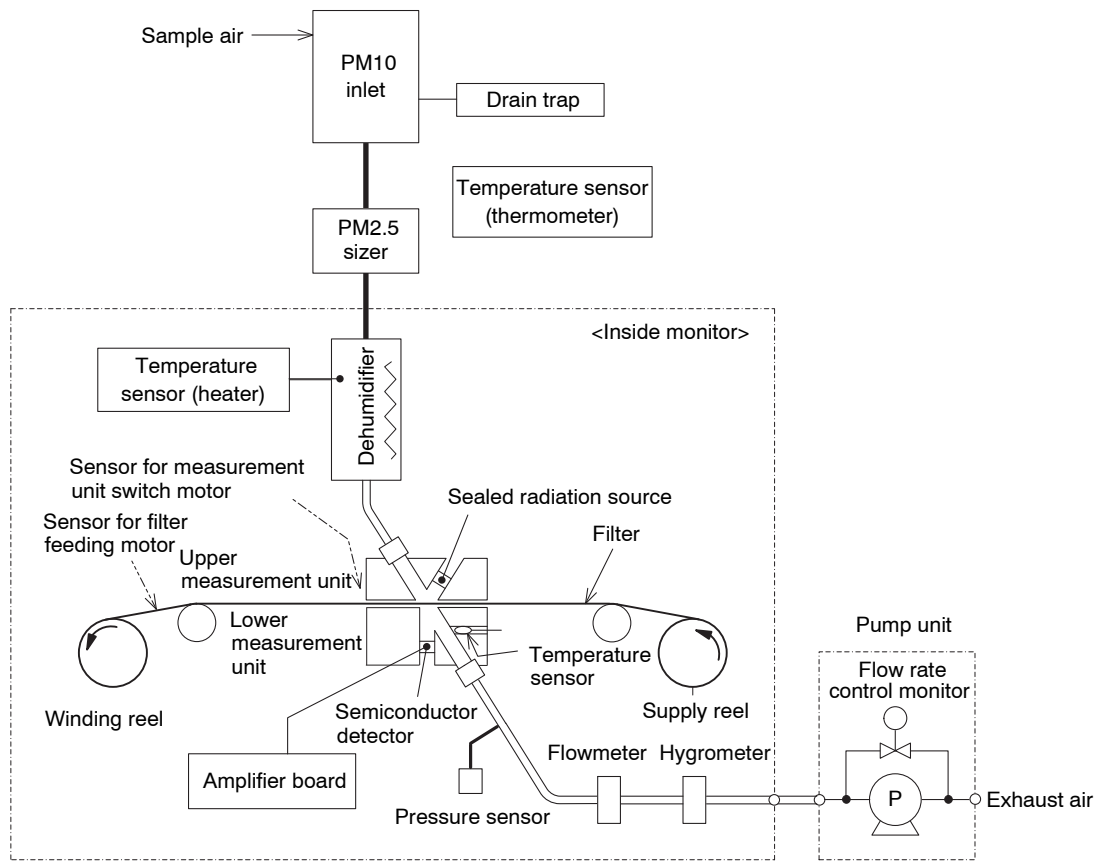
8. Specifications and Operational Explanation

8.1 Specifications

Product name	: PM2.5 monitor
Model name	: FPM-388
Element to be measured	: Fine particulate matter (PM2.5) in the air
Measurement method	: Beta (β) ray absorption method.
Measurement range	: 0 to 1000 $\mu\text{g}/\text{m}^3$
Display unit	: $\mu\text{g}/\text{m}^3$ (mg/m ³ at calibration of calibration film)
Material sampling	: Collection method Filtration method Sizing method Cyclone method (VSC cyclone) Suction flow rate 16.7L/min (actual flow rate) Used filter paper Filters made of polytetrafluoroethylene (rolled, replaced monthly) Material of contacting sample air ... Stainless steel, anodized aluminum Collection time 59 minutes
Performance	: Linearity (indication error) $\pm 3\%$ or less (depending on calibration film) Repeatability (zero) $\pm 2\%$ FS or less (depending on calibration film) Repeatability (span) $\pm 2\%$ or less (for calibration film value) Zero drift $\pm 2\%$ FS or less/24h (depending on calibration film) Span drift $\pm 3\%$ or less/24h (depending on calibration film) Temporal stability of flow rate $\pm 4\%$ /24 hours
Minimum display value	: 0.1 $\mu\text{g}/\text{m}^3$ when the display unit is set to $\mu\text{g}/\text{m}^3$
Recording capacity of CF card	: For 3 years (1-hour value, calibration log data, alarm log data, etc.) * Use the genuine product of our company (256MB).
Allowable ambient temperature and humidity (performance guarantee)	: 0 to 40°C 85%RH or less (no condensation)
Warm-up time	: 3 hours
Power supply	: 220VAC, 50/60Hz (same as power supply for pump)

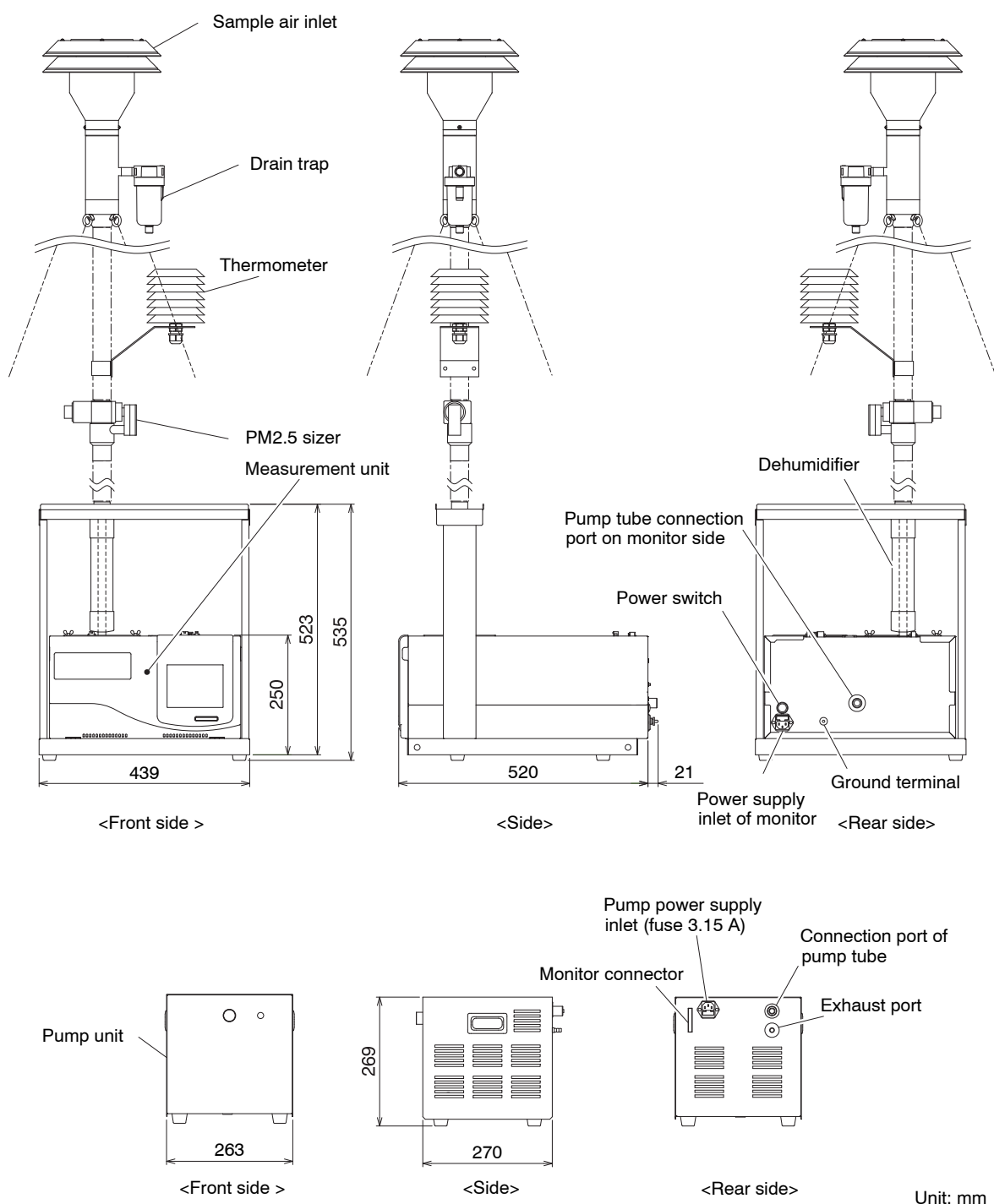
Power consumption	: Max. approx. 400VA, average 250W (including power supply for pump)
Mass	: Main unit 20kg Pump unit 12kg PM10 inlet + PM2.5 sizer 3.2kg (excluding down tube)

8.2 Measurement System



Measurement System Diagram

8.3 External Dimensions



External Dimensions

● Engineering Service Inspection Sheet

The “Engineering Service Inspection Sheet” at the end of the book is the recording sheet for periodical engineering service inspection. Please copy the sheet and submit it to the person in charge of engineering service.

Revision History

Instruction Manual No. FPM-AA819E	11/07/2017 (RG1)	New Version in English (RWG Kenmochi)
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Print size: A4 (Original size: A4), Perfect bind

YTD

Office building name	Production number	Program Main V sub.V	Special instruction
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Inspection item			Inspection date and time	Date:	Date:	Date:	Date:	Date:	Date:	Date:	Date:
			Unit	Criterion for judgment	Time:	Time	Time	Time	Time	Time	Time
Data		Integrated value	μg/m ³								
		β-ray count	—	8000 or more							
		Noise voltage	mV	800 or less							
		Temperature of measurement unit	°C	45 or less							
		Pressure of measurement unit	kPa	55 to 95							
		Sample flow rate	L/min								
	Reference value		Reference flowmeter	L/min							
			Reference thermometer	°C							
			Reference hygrometer	%							
			Reference barometer	kPa							
		Outdoor air temperature	°C								
		Outdoor air humidity	%								
		Ambient pressure	kPa								
		Temperature of heater	°C								
		Actual flow rate	L/min								
		Integrated flow rate	L								
	Internal temperature	°C									
	Span coefficient	—	0.9 to 1.3 and previous value±0.02								
Service		Correction of flow rate	L/min								
		Correction of outdoor air temperature	°C								
		Correction of humidity	%								
		Correction of ambient pressure	kPa								

Confirmation of record (check mark for OK)				☐	☐	☐	☐	☐	☐	☐	☐
Replacement of recording paper (check mark at replacement)				☐	☐	☐	☐	☐	☐	☐	☐
Replacement of filter paper (check mark at replacement)				☐	☐	☐	☐	☐	☐	☐	☐
Calibration film measurement		±0.02 of film value ①									
Film value ()		②									
		③									
Worker		Affiliation:	Name:								
Remark											



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